



FleetPulse

Standard Operating Procedures Manual Preventive Maintenance & Repair Operations for Petroleum Tanker Fleet

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Volume 1 — System Administration

2. Chapter 1: Roles & Access Matrix

This chapter defines every user role in the FleetPulse system, their permissions, and access boundaries. All personnel must understand which role they belong to and what actions they are authorized to perform.

1.1 Role Definitions

Role	System Label	Typical Personnel	Access Level
Super Admin	SUPER_ADMIN	IT Manager / System Owner	Full — including User Management & Django Admin
Admin	ADMIN	Operations Manager	All operations features except User Management
Staff	STAFF	Shop Supervisor / Lead Mechanic	Job Orders, PM views, Training supervision
Mechanic	MECHANIC	Diesel Mechanic / Technician	Assigned JOs only, limited read access
Contractor	CONTRACTOR	External Vendor Technician	Contractor JOs only, no access to other data
OJT / Trainee	TRAINEE	Apprentice / Under Training	Training dashboard & attendance only

1.2 Access Matrix by Feature

The following matrix shows which features are visible in the sidebar for each role. A check mark (✓) means the menu item is available.

Feature / Page	Super Admin	Admin	Staff	Mechanic	Contractor	OJT/Trainee
Dashboard	✓	✓	✓	✓	—	✓
Trucks	✓	✓	✓	✓	✓	✓
PMS Schedules	✓	✓	✓	—	—	—
PM Categories	✓	✓	—	—	—	—
PM Templates	✓	✓	—	—	—	—
Job Orders	✓	✓	✓	✓	✓	—
My Assignments	—	—	—	✓	✓	—

Feature / Page	Super Admin	Admin	Staff	Mechanic	Contractor	OJT/Trainee
Service Ledger	✓	✓	—	—	—	—
Contractors	✓	✓	—	—	—	—
KPI Reports	✓	✓	—	—	—	—
Predictive Analytics	✓	✓	—	—	—	—
User Management	✓	—	—	—	—	—
Training	✓	✓	✓	—	—	✓
Assign Training	✓	✓	—	—	—	—
Django Admin	✓	—	—	—	—	—

1.3 Role-Based Login Behavior

The system redirects users to different pages upon login depending on their role:

Role	Login Redirect	Rationale
Super Admin	Dashboard	Overview of all operations
Admin	Dashboard	Overview of all operations
Staff	Dashboard	Overview of shop activity
Mechanic	Dashboard	Quick view of assignments
Contractor	My Assignments	Immediate access to work orders
OJT / Trainee	Training Dashboard	Attendance & progress tracking

1.4 Security Principles

- **Least Privilege:** Each role has the minimum access required to perform its function. No role can access features outside its permission level.
- **Data Isolation:** Mechanics and Contractors can only see Job Orders assigned to them. They cannot browse all JOs in the system.
- **Immutable Audit Trail:** Once a Job Order is closed, all associated data is written to the Service Ledger and can never be modified or deleted.
- **Session Timeout:** Users are automatically logged out after 30 minutes of inactivity.
- **Password Policy:** All user accounts require passwords of at least 8 characters.

3. Chapter 2: User Account Lifecycle

2.1 Creating a New User

Responsible Role: Super Admin only

1. Log in as Super Admin.
2. Navigate to **Sidebar** → **User Management**.
3. Click the **"Add User"** button.
4. Complete the form with the following fields:
 - **Username:** Unique identifier (e.g., jdelacruz). Use first initial + last name in lowercase.
 - **Password:** Temporary password. Ensure at least 8 characters.
 - **Confirm Password:** Re-enter the same password.
 - **Role:** Select from the six roles defined in Chapter 1.
 - **First Name / Last Name:** Legal name as appears on company ID.
 - **Mobile Number:** Active cellular number for contact purposes.
 - **Specialization:** (Mechanics only) Primary area of expertise — e.g., Engine, Brakes, Electrical, Tank & Valves, General.
 - **Employee ID:** Company-assigned employee or contractor ID number.
5. Click **"Save"**.
6. Provide the new user their username and temporary password. Instruct them to change their password on first login.

2.2 Editing an Existing User

1. Navigate to **Sidebar** → **User Management**.
2. Locate the user in the list (use the role filter if needed).
3. Click the **user row** or **Edit button**.
4. Update the desired fields.
5. Click **"Save"**.

2.3 Resetting a User's Password

1. Navigate to **Sidebar** → **User Management**.
2. Find the user and click the **key icon** in the Actions column.
3. You will be redirected to the **Django Admin** password change form.
4. Enter the new password and confirm.
5. Click **"Change password"**.
6. Communicate the new temporary password to the user securely.

⚠ **Important:** Password resets redirect to the Django Admin interface, which is only accessible to Super Admin accounts. If you cannot access the Django Admin, check that you are logged in as Super Admin at /admin/.

2.4 Disabling / Deactivating a User

To prevent a user from logging in without deleting their account (preserving audit trail integrity):

1. Navigate to **Sidebar → User Management**.
2. Click the user to edit.
3. Uncheck the **"Active"** checkbox.
4. Click **"Save"**.
5. The user can no longer log in, but all their historical records (Job Orders, service logs, ratings) remain intact.

2.5 Creating an OJT/Trainee Account (Special Case)

Trainee accounts require an additional step beyond user creation:

1. **Step 1:** Create the user account with Role = "OJT / Trainee" (follow Section 2.1).
2. **Step 2:** Navigate to **Sidebar → Assign Training**.
3. Select the newly created OJT from the dropdown.
4. Assign a **Supervisor** (must be Staff, Admin, or Super Admin).
5. Set the **Start Date** (usually the OJT's first day).
6. Click **"Assign"**.
7. The OJT now has access to their Training Dashboard and the supervisor can begin ratings.

4. Chapter 3: System Navigation & UI Reference

3.1 Layout Overview

The FleetPulse system uses a consistent layout across all pages:

1. **Sidebar (Left):** Main navigation menu. Collapses on mobile screens. Shows menu items based on the user's role.
2. **Top Bar:** Displays current page title, a back button (where applicable), user name, and role badge. The back button (← arrow) returns to the previous logical page.
3. **Content Area:** Main body of the page. Contains lists, forms, detail views, and action buttons.
4. **Footer:** Company tagline and copyright information.

3.2 Page Loader Indicator

A thin blue progress bar animates across the top of the screen whenever a page is loading. This provides visual feedback during navigation. The bar fills from left to right and fades out once the page is ready.

3.3 Mobile Navigation

On small screens (phones, small tablets), the sidebar collapses. Tap the **≡ (hamburger icon)** in the top-left corner to expand the mobile menu.

3.4 Common UI Patterns

Back Buttons: The ← arrow in the top bar returns you to the previous page. This is context-aware — on a truck detail page, it returns to the truck list; on a Job Order detail, it returns to the JO list.

Search Boxes: Most list pages include a search box. Type any text to filter the list in real time. Searches match partial text against key fields (name, plate number, description, etc.).

Filters: Some pages (Job Orders, PMS Schedules) include dropdown filters to narrow results by status, type, truck, or other criteria. The filter and search work together.

Pagination: List pages display 50 items per page. Use the page controls at the bottom to navigate between pages.

CSV Export: The PMS Schedules list page includes a "CSV Export" button for downloading schedule data as a spreadsheet file.

Print: The PMS Schedules page includes a "Print" button that formats the table for browser printing (landscape orientation).

3.5 Logout Procedure

1. Click the **"Logout"** button in the sidebar (Account section).
2. A confirmation dialog appears: *"Are you sure you want to log out?"*
3. Click **OK** to confirm.
4. A full-screen overlay appears with a spinning favicon and "Logging out..." message.
5. After a brief delay, you are redirected to the login page.

3.6 Keyboard Shortcuts

Action	How
Navigate back	Alt + ← (browser back) or the ← top bar button
Submit form	Enter key on any input field
Cancel / close	Escape key on modal dialogs
Search	Click into any search box and start typing

6.7 Browser Support

The system is tested and supported on:

- **Google Chrome** (recommended — latest 2 major versions)
- **Mozilla Firefox** (latest 2 major versions)
- **Microsoft Edge** (Chromium-based, latest 2 major versions)

Mobile browsers are supported but the experience is optimized for desktop and tablet use.

Volume 2 — Fleet Asset Management

5. Chapter 4: Truck Registration & Documentation

4.1 Overview

Every truck in the fleet must be registered in the system with complete Certificate of Registration (CR) documentation. Proper registration ensures that all PM schedules are auto-generated, service history is tracked per vehicle, and regulatory compliance documents are accessible at all times.

4.2 Required Information for Truck Registration

Field	Required	Description
Unit Number	Yes	Internal fleet unit number (e.g., UNIT-001)
Plate Number	Yes	Government-issued license plate
Make	Yes	Manufacturer (e.g., Isuzu, Hino, Mitsubishi)
Model	Yes	Model name/number (e.g., Giga FVZ, Forward)
Year	Yes	Year of manufacture
Chassis Number	No	VIN / Frame serial number
Engine Number	No	Engine serial number
Tank Capacity	No	Capacity in liters (for fuel tanker trucks)
Fuel Type	No	Diesel, Gasoline, etc.

4.3 Certificate of Registration Fields

The following CR fields should be completed from the official LTO Certificate of Registration:

Field	Source Document
MV File Number	LTO MV File
Certificate of Registration (CR) Number	LTO CR
Official Receipt (OR) Number	LTO OR
Date of Registration	LTO CR
Date Expire	LTO CR

Field	Source Document
Make	LTO CR
Series/Body Type	LTO CR
Year Model	LTO CR
Engine Number	LTO CR (verify against engine block)
Chassis Number	LTO CR (verify against frame stamp)
Piston Displacement (cc)	LTO CR
Gross Vehicle Weight (GVW)	LTO CR
Net Capacity	LTO CR
Shipping Weight	LTO CR
Net Weight	LTO CR
LTO Registered Address	LTO CR
LTO Registered Owner	LTO CR

4.4 Registering a New Truck

Responsible Role: Admin or Super Admin

1. Navigate to **Sidebar** → **Trucks**.
2. Click the **"Add Truck"** button.
3. Complete all required fields (marked with an asterisk). Fill in optional fields where data is available.
4. Scan and attach CR documents if applicable.
5. Set the initial **Status** to **"Active"** if the truck is operational.
6. Enter the current **Mileage (km)** and **Engine Hours** from the truck's odometer and hour meter.
7. Click **"Save"**.
8. **Important:** The system automatically generates 154 PM schedules for the new truck based on the standard task templates. This process runs in the background after saving.
9. Verify the PM schedules were created by navigating to **Sidebar** → **PMS Schedules** and filtering by the new truck.

Note on Mileage and Engine Hours: Accurate initial values are critical. Every PM schedule's "next due" calculation starts from these baseline readings. If the truck has existing service records, enter the last-known PM completion data into the respective PM schedule records after creation.

4.5 Editing a Truck Record

1. Navigate to **Sidebar** → **Trucks**.
2. Click the truck you wish to edit (opens the Detail page).

3. Click the **"Edit"** button.
4. Update the desired fields. Note: Changing mileage or engine hours here will affect all PM schedule due-date calculations.
5. Click **"Save"**.

4.6 Truck Detail Page Layout

The truck detail page is organized into four sections displayed in a two-column layout:

Section	Content
Left Top: Truck Details	Unit number, plate, make, model, year, chassis/engine numbers, current mileage, engine hours, status, fuel type, tank capacity
Right Top: Certificate of Registration	All 17 CR fields from the LTO registration document
Left Bottom: PM Schedule	All PM items for this truck with colored status indicators
Right Bottom: Job Orders & Service History	Recent JOs and service log entries with links to full ledger

6. Chapter 5: Fleet Status Management

5.1 Truck Status Definitions

Status	Definition	Implications
Active	Truck is fully operational and available for dispatch	PM schedules are active. Job Orders can be created.
Down	Truck is currently out of service due to breakdown or major repair	Cannot be dispatched. PM schedules remain active (mechanical issues may be PM-related).
Retired	Truck has been permanently decommissioned, sold, or scrapped	No new Job Orders can be created. PM schedules are frozen but retained for historical reference.

5.2 Changing Truck Status

1. Navigate to the truck's Detail page.
2. Click **"Edit"**.
3. Change the **Status** field.
4. Optionally add a note in the description field explaining the reason for the status change.
5. Click **"Save"**.

Best Practice: When marking a truck as "Down," create a corresponding Job Order to track the repair work causing the downtime. This ensures the downtime is documented and can be analyzed in KPI reports.

5.3 Viewing Fleet Summary

The Dashboard shows a quick summary of fleet status:

- **Active Trucks:** Total count of trucks with Active status.
- **Overdue / Due PM:** List of trucks with PM items that are due or overdue.
- **Open Job Orders:** Count of all JOs currently Open or In Progress.

For a full fleet overview, navigate to **Sidebar → Trucks** to see all trucks in a sortable, searchable table.

5.4 Syncing New PM Templates to Existing Trucks

If new PM task templates are added to the system after trucks have already been registered, the new templates must be synced to existing trucks:

1. Navigate to **Sidebar → PMS Schedules**.
2. Click the **"Sync All Trucks"** button.
3. The system will create any missing PM schedule records for all trucks.
4. To sync a single truck: open the truck's Detail page and click the **Refresh icon** in the PM Schedule section.

Volume 3 — Preventive Maintenance

7. Chapter 6: PM Scheduling — Interval Types

6.1 Purpose of Preventive Maintenance

The Preventive Maintenance program ensures that every tanker truck in the fleet receives timely, scheduled service to prevent unexpected breakdowns, extend equipment life, maintain safety compliance, and comply with depot access requirements. A well-executed PM program reduces costly emergency repairs and keeps the fleet operational.

6.2 Interval Types

Each PM task in the system is assigned one of four interval types that determines when it becomes due:

Type	Icon	Basis	Example
MILEAGE		Odometer reading	"Oil change every 5,000 km" — due when <code>current_mileage</code> $\geq$ <code>last_mileage + 5,000</code>
HOURS		Engine hours meter	"Belt inspection every 500 hrs" — due when <code>current_hours</code> $\geq$ <code>last_hours + 500</code>
CALENDAR		Elapsed time in days	"Tank hydrostatic test every 365 days" — due when <code>today</code> $\geq$ <code>last_done + 365 days</code>
VISUAL		Manual judgment	"Pre-trip inspection" — always shows as "visual" status; no automatic calculation

6.3 Due Date Calculation

The system calculates PM status in real-time (no stored "due date" field). The formula for each interval type:

- **MILEAGE:** `current_mileage` $\geq$ `last_mileage_km + interval_value` $\rightarrow$ DUE. A 10% threshold applies: if within 10% of interval, status shows as "Due" (yellow) instead of "Overdue."
- **HOURS:** Same logic as mileage, using engine hours.
- **CALENDAR:** `today` $\geq$ `last_completed_at + timedelta(days=interval_value)` $\rightarrow$ DUE. Same 10% grace threshold.
- **VISUAL:** Always returns "visual" status — no auto-calculation.

6.4 Status Colors & Meanings

Status	Color	Meaning	Action Required
OK	Green	Within service interval	None — monitor at next interval
DUE	Yellow	Within 10% of interval threshold	Schedule service within the week
OVERDUE	Red	Past the interval threshold	Service immediately — do not dispatch until resolved
VISUAL	Blue	Manual inspection required	Inspect before next dispatch
NO DATA	Gray	Never completed, no baseline set	Perform initial service to establish baseline

6.5 How PM Schedules Update on Job Order Close

When a Job Order is closed, the system automatically:

1. Records the truck's **current mileage** and **engine hours** at the time of close.
2. Finds the relevant PM Schedule(s) that match the line items on the Job Order (matched by task template).
3. Updates those PM Schedules:
 - `last_completed_at` = JO close timestamp
 - `last_mileage_km` = mileage at close
 - `last_engine_hours` = engine hours at close
 - `status` recalculated automatically
4. Line items without a matching PM template do not update any schedule.

Important: PM schedules can also be updated manually. Click the pencil icon next to any PM schedule on the truck detail page to edit the last-completed values directly. This is useful when service was performed outside the Job Order system (e.g., third-party shop).

8. Chapter 7: PM Task Categories & Templates

7.1 System Structure

The PM system is organized into three levels:

1. **Task Categories (23):** High-level groups of related maintenance tasks. Each category represents a system or area of the truck (e.g., "Engine PM," "Brakes," "Tires").
2. **Task Templates (154):** Specific, pre-defined tasks within each category. Each template has a name, an interval value, an interval unit (km/hrs/days), and a category assignment.
3. **PM Schedules:** Per-truck instances of each template. Every truck gets one PM Schedule record for each template (154 per truck). The schedule tracks when it was last done and calculates the next due date.

7.2 The 23 PM Categories

#	Category Name	Total Templates	Focus Area
1	Engine PM	4	Oil & filter, coolant, radiator hoses, fan belt
2	Brakes	3	Lining, drum/disc, air brake system
3	Tires	2	Condition, tread depth & pressure
4	Tank & Valves	4	Overfill prevention, manlid, API adapters, camlocks
5	Vapor Recovery	2	Vapor recovery system, grounding/static
6	Hoses & Couplings	2	Product piping, hoses & couplings
7	Safety Equipment	1	Horn, reflective tape, labels, first aid, fall arrest
8	Electrical	3	Battery, alternator, starter, wiring/sensors
9	Lighting	1	Headlight, taillight, brake light, signal, clearance
10	Compliance & Regulatory	1	DOST stickers, HAZMAT placards, depot access, accreditation
11	Drivetrain	3	Differential, U-joints, wheel bearings
12	Water Brake System	4	Water brake components (test, valve, pump, line)
13	HVAC System	2	Air conditioning, heater
14	Hydraulics	2	Hydraulic pump, cylinder, lines
15	Suspension	2	Leaf springs, shock absorbers
16	Exhaust System	2	Exhaust pipe, muffler, mounting

#	Category Name	Total Templates	Focus Area
17	Body & Chassis	2	Body condition, chassis frame, mounting bolts
18	Fuel System	2	Fuel filter, fuel lines, injectors
19	Steering System	2	Steering gear, linkage, power steering
20	Transmission	2	Transmission oil, clutch adjustment
21	Cooling System (Aux)	2	Coolant level, radiator cap, recovery tank
22	Air Intake System	2	Air filter, intake piping, turbocharger
23	Miscellaneous	1	General visual inspection items

Note: The exact number of templates within each category may vary slightly from the table above as the task library evolves. The system administrator can add, modify, or archive templates at any time via **Sidebar → PM Templates** (Admin+ access).

7.3 Adding a New Task Template

Responsible Role: Admin or Super Admin

1. Navigate to **Sidebar → PM Templates**.
2. Click **"Add Template"**.
3. Complete the form:
 - **Task Name:** Clear, descriptive name (e.g., "Inspect and Adjust Air Brake Slack Adjuster").
 - **Category:** Select the appropriate category.
 - **Interval Value:** Numeric value of the interval (e.g., 5000).
 - **Interval Unit:** km, hrs, days, or "visual."
 - **Description:** Optional detailed notes about what this task entails.
4. Click **"Save"**.
5. **After saving:** Sync existing trucks to create PM Schedules for the new template (see Chapter 5.4).

9. Chapter 8: Step-by-Step PM Procedures

This chapter contains the complete step-by-step procedures for all 23 PM categories (34 total procedures). Every category now has a dedicated detailed procedure covering inspection steps, diagnostics, and acceptance criteria. Each procedure follows a standardized format with safety precautions, required tools, detailed steps, acceptance criteria, and documentation requirements.

All procedures assume the following **General Safety Rules** apply at all times:

- Wear appropriate Personal Protective Equipment (PPE): safety glasses, steel-toe boots, gloves, and high-visibility vest.
 - Ensure the truck is parked on level ground with the parking brake engaged.
 - Chock the wheels before working under or around the truck.
 - For tanker trucks: confirm the compartment is empty and gas-free before any hot work or entry.
 - Use lockout/tagout (LOTO) procedures when working on electrical, hydraulic, or pneumatic systems.
 - Do not work alone on hazardous tasks — always have a spotter or second mechanic present.
-

8.1 Engine Oil & Filter Change

Engine Oil and Filter Replacement

Purpose: Remove contaminated engine oil and replace with fresh oil of the correct grade. Replace oil filter to maintain proper lubrication and engine life.

Frequency: Every 5,000 km or 250 engine hours, whichever comes first

Tools & Materials: Socket wrench set, oil filter wrench (strap-type or cap-type), drain pan (15L+ capacity), funnel, torque wrench, rags, oil absorbent pads, new oil filter, correct grade engine oil (as specified by manufacturer, typically 15W-40 or 20W-50 for diesel), replacement drain plug gasket

Scope: All diesel-powered tanker trucks

PPE Required: Safety glasses, nitrile or latex gloves, steel-toe boots, coveralls

Step-by-Step Procedure:

1. Park truck on level ground, engage parking brake, chock wheels.
2. Warm up engine for 2-3 minutes to thin oil for better drainage (oil should be warm, not hot).
3. Turn off engine.
4. Position drain pan under the oil drain plug.
5. Using appropriate socket, loosen and remove drain plug.
6. Allow oil to drain completely (5-10 minutes).
7. Inspect drain plug gasket — replace if crushed or damaged.
8. While oil drains, move drain pan under oil filter.
9. Using oil filter wrench, loosen filter by turning counterclockwise.
10. Oil will spill — position rags to catch drips.
11. Remove filter completely.
12. Wipe the filter mounting surface on the engine block with a clean rag.
13. Apply a thin coat of fresh oil to the new filter's rubber gasket.
14. Screw new filter onto mounting stud by hand until gasket contacts the mounting surface.
15. Tighten an additional 3/4 turn by hand (do not overtighten).
16. Reinstall drain plug with new gasket.
17. Torque to manufacturer specification (typically 30-50 Nm for M14 plugs).
18. Refill engine with fresh oil through fill port.
19. Check capacity: typically 15-20 liters for medium-duty diesel trucks.
20. Pour 80% of expected capacity, then check dipstick.
21. Add remaining oil to reach FULL mark.
22. Start engine and run for 30 seconds.
23. Check for leaks around drain plug and filter.

24. Stop engine.
25. Wait 2 minutes for oil to settle.
26. Recheck dipstick and top off if necessary.
27. Record mileage, oil grade and quantity used, and date.

Acceptance Criteria: Oil level at FULL mark on dipstick after warm-up cycle. No leaks at drain plug or filter after 30-second run. Oil pressure gauge reads normal (typically 40-60 psi at idle when warm).

⚠ **SAFETY:** Hot oil can cause severe burns. Allow engine to cool slightly (2-3 minutes) before draining. Oil is an environmental hazard — collect all waste oil in approved containers for recycling. Never drain oil onto the ground or into drains.

Tip: Keep a log of oil consumption between changes. If the truck requires more than 1L top-up between changes, investigate for leaks or engine wear.

Documentation: Record in Job Order line item: mileage, oil grade/viscosity, quantity used, filter part number. Update PM schedule for Engine Oil & Filter.

Performing Mechanic → Lead Mechanic verification

8.2 Coolant System — Flush & Refill

Coolant System Flush and Refill

Purpose: Remove contaminated or degraded coolant, flush the cooling system, and refill with fresh coolant at the correct concentration to prevent overheating, cavitation, and freeze damage.

Frequency: Every 40,000 km or 2 years, whichever comes first (or per manufacturer specification)

Tools & Materials: Drain pan (20L+), radiator flush chemical (optional), distilled water, premixed coolant (typically 50/50 concentrate/distilled water), funnel, coolant hydrometer or refractometer, hose, rags, replacement radiator cap, replacement drain plug gasket if applicable

Scope: All liquid-cooled diesel truck engines

PPE Required: Safety glasses, heat-resistant gloves, coveralls, face shield (when opening hot system)

Step-by-Step Procedure:

1. Ensure engine is **cold** (coolant temperature below 40°C).
2. Place drain pan under radiator drain plug (typically at bottom corner of radiator).
3. Slowly open radiator cap to release any residual pressure.
4. Remove drain plug and allow coolant to drain completely.
5. Close drain plug.
6. If using flush chemical: fill system with water + flush chemical per manufacturer instructions.
7. Run engine with heater on HIGH for 10-15 minutes.
8. Drain again.
9. Refill with clean distilled water, run engine 5 minutes, drain again (repeat until draining water is clear).
10. Close drain plug.
11. If not using chemical flush: fill with distilled water, run engine 10 min, drain.
12. Repeat until draining water is clear.
13. Close drain plug.
14. Inspect all coolant hoses for cracks, soft spots, or swelling.
15. Replace if compromised.
16. Inspect radiator cap gasket and pressure seal.
17. Replace if worn.
18. Mix coolant concentrate with distilled water to achieve 50/50 ratio (or as specified for your climate — 60/40 for extreme cold).
19. Fill radiator through cap opening until full.
20. Fill recovery tank to COLD FILL line.
21. Install radiator cap tightly.
22. Run engine with heater on HIGH until thermostat opens (upper radiator hose becomes hot).

23. Check for leaks at hoses, water pump, drain plug, and radiator seams.
24. Stop engine and allow to cool.
25. Recheck coolant level and top off as needed.

Acceptance Criteria: Coolant free of contamination (no rust, oil, or debris). Coolant concentration 50% $\pm$ 5% (tested with refractometer — reading 1.10-1.14 SG for ethylene glycol). No leaks at any connection. Radiator cap holds pressure (test with pressure tester if available). Recovery tank at correct level.

⚠ SAFETY: Coolant is toxic (ethylene glycol). Do not discharge into drains or onto ground. Collect all waste coolant in sealed containers for proper disposal. Never open a hot cooling system — severe scalding will occur. Wait until engine is cold (at least 1 hour after shutdown).

Tip: Use distilled water, not tap water. Tap water contains minerals that cause scale buildup and reduce cooling efficiency. For trucks with Cummins or Caterpillar engines, use SCA (Supplemental Coolant Additive) test strips to verify cavitation protection.

Documentation: Record mileage, coolant type/concentration, quantity used, date performed.

Performing Mechanic

8.3 Radiator & Hose Inspection

Radiator and Coolant Hose Inspection

Purpose: Inspect radiator core, tanks, fins, and all coolant hoses for leaks, damage, blockages, or deterioration that could lead to overheating.

Scope: All trucks

Frequency: Every 10,000 km or 6 months

PPE Required: Safety glasses, gloves

Tools & Materials: Flashlight, pressure tester (cooling system), inspection mirror, rag, coolant (for top-off)

Step-by-Step Procedure:

1. With engine cold, remove radiator cap and inspect cap gasket and seal.
2. Visually inspect radiator core for bent or damaged fins.
3. Use flashlight to check for debris blockages between fins (bugs, mud, leaves).
4. Inspect radiator seams (where plastic tank meets aluminum core) for white or pink deposits indicating seepage.
5. Check upper and lower radiator hoses for: cracks, soft spots (collapsible under squeeze), bulging, chafing on fan shroud, oil or fuel contamination (swollen hose).
6. Check heater hoses (firewall connections) for same defects.
7. Check all hose clamp connections.
8. Confirm clamps are tight, not over-tightened (cutting into hose), and positioned past the hose barb.
9. Inspect coolant recovery tank: check for cracks, verify hose connection to radiator overflow nipple is secure and clear.
10. Pressure test: install pressure tester on radiator neck, pump to system pressure rating (typically 12-16 psi for trucks).
11. Watch gauge for 2 minutes — pressure should hold steady.
12. If pressure drops, locate leak by listening for hissing and inspecting all connections with a mirror.
13. Relieve pressure and remove tester.
14. Top off coolant if needed.
15. Install radiator cap.

Acceptance Criteria: No leaks detected under pressure test (pressure holds for 2 minutes). Hoses firm but pliable — no cracks, soft spots, or swelling. Clamps secure and correctly positioned. Radiator fins clean and undamaged (at least 90% open).

⚠ **SAFETY:** Do not pressure test a hot cooling system. Ensure engine is cold.

Tip: Squeeze-test hoses: a good hose feels firm but springy. A hose that feels mushy or collapses easily is showing internal degradation and should be replaced preemptively.

Documentation: Record inspection findings, pressure test result (PSI held / duration), any hoses replaced.

Performing Mechanic

8.4 Fan Belt & Tensioner Inspection

Fan Belt (Serpentine/V-Belt) and Tensioner Inspection

Purpose: Inspect condition and tension of all drive belts and the automatic tensioner to prevent belt failure, loss of cooling, alternator output, and power steering.

Frequency: Every 10,000 km or 6 months

Tools & Materials: Belt tension gauge (optional), 12mm/14mm socket, pry bar (for manual tensioners), flashlight

Scope: All trucks with belt-driven accessories

PPE Required: Safety glasses, gloves (avoid loose clothing near moving parts)

Step-by-Step Procedure:

1. Turn off engine and remove key.
2. Visually inspect belt(s) along entire length: look for cracks on the ribbed surface, glazing (shiny, hardened surface), fraying on edges, missing chunks of rubber, separation between layers.
3. Check belt routing against diagram on fan shroud or owner manual.
4. Confirm belt is correctly seated in all pulley grooves.
5. Check automatic tensioner: inspect for wobble or misalignment (indicates bearing wear).
6. Check tensioner pulley for smooth rotation — spin by hand; should turn freely without grinding or wobble.
7. Measure belt deflection: push down on belt at longest unsupported span.
8. Acceptable deflection: 10-15mm (3/8 to 5/8 inch) for typical truck belts.
9. For vehicles with automatic tensioner: verify indicator mark is within the acceptable range on the tensioner body.
10. For manual tensioner systems: adjust as needed by loosening mounting bolt and pivoting alternator or accessory.
11. Check accessory pulleys (alternator, water pump, AC compressor, power steering): spin each by hand.
12. Feel for roughness, looseness, or wobble.
13. Check pulley alignment: lay a straightedge across pulleys — all pulley grooves should align within 1mm.

Acceptance Criteria: Belt shows no cracks, glazing, fraying, or chunking. Tensioner indicator in acceptable range (or deflection within 10-15mm). All pulleys spin smoothly without wobble. Belt properly seated in all grooves.

⚠ SAFETY: Never inspect belts with engine running. Loose clothing, hair, or tools can be pulled into rotating components. Ensure engine is off and key is removed.

Tip: If one belt shows significant wear, replace all belts in the system simultaneously. A failing tensioner often damages the new belt prematurely.

Documentation: Record belt condition, tension measurement, any replacements (belt part numbers).

Performing Mechanic

8.5 Brake Lining Inspection

Brake Lining (Brake Shoe Pad) Inspection

Purpose: Inspect brake lining thickness, condition, and contact pattern on all wheels. Determine if replacement is needed before lining material is critically low.

Frequency: Every 10,000 km or 3 months for severe service

Tools & Materials: Tire chocks, jack stands (minimum 6-ton capacity), wheel chocks, lug wrench, torque wrench, brake spoon or adjustment tool, ruler or brake lining thickness gauge, flashlight, penetrating oil, air gun (for dust removal — use with dust collection)

Scope: All trucks — inspect all wheels (typically 6-10 wheels per truck)

PPE Required: Safety glasses, gloves, steel-toe boots, hearing protection (if using air tools)

Step-by-Step Procedure:

1. Chock wheels on axles that will remain on ground.
2. Loosen lug nuts slightly while truck is on ground.
3. Jack up truck and place jack stands under frame rails at specified lifting points — **never work under a truck supported only by a jack.**
4. Remove wheels and set aside.
5. Remove brake drum inspection plug (rubber plug on backing plate) or remove drum entirely.
6. For drum brakes: Use brake spoon to back off the adjuster (turn in direction of arrow or counterclockwise) to release brake shoes from drum.
7. Pull drum straight off — may require penetrating oil on hub/drum mating surface if seized.
8. Inspect drum inner surface (see Section 8.6 for drum inspection).
9. Measure lining thickness at both ends of each shoe (leading and trailing).
10. Compare thickest and thinnest points.
11. Replace if thickness is $\leq 3\text{mm}$ (service limit) at any point.
12. For disc brakes (if equipped): Remove caliper retaining pins or bolts.
13. Swing caliper up or remove entirely (suspend with wire — do not let hang by hose).
14. Remove brake pads from caliper.
15. Measure pad friction material thickness — replace if $\leq 3\text{mm}$.
16. Check for uneven wear (tapered pads indicate caliper guide pin binding).
17. Inspect rotor surface (see Section 8.6).
18. Check brake hardware: return springs (drum), retaining clips, anti-rattle springs, caliper slide pins.
19. Replace any corroded or weak hardware.
20. Clean brake dust from backing plate and hub surface (use approved vacuum — avoid breathing brake dust).

21. Reassemble: Install new or existing shoes/pads.
22. For drum brakes: pre-adjust shoes before drum installation.
23. For disc brakes: compress caliper piston using appropriate tool before reinstalling.
24. Install wheel, tighten lug nuts in star pattern to manufacturer torque (typically 450-600 Nm for heavy trucks).
25. Lower truck.
26. Re-torque lug nuts after 50 km of driving.

Acceptance Criteria: Lining thickness $\geq$ 6mm minimum on friction material (replace at 3mm). Even contact pattern across entire lining surface (at least 80% contact). No oil or grease contamination on lining. All hardware present and functional.

⚠ **SAFETY:** Brake dust may contain asbestos. Never use compressed air to clean brake dust. Use HEPA vacuum or wet rag method. Always use jack stands rated for the vehicle weight. Never rely solely on a hydraulic jack.

Tip: Measure lining thickness at the thinnest point. If one wheel requires replacement, inspect both sides of that axle — uneven braking can cause dangerous pulling. Always replace brake shoes/pads in axle sets (both sides).

Documentation: Record lining thickness for each wheel position (LF, RF, LR, RR, etc.). Note any drums/rotors replaced. Record remaining thickness of replaced components.

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8.6 Brake Drum / Disc Inspection

Brake Drum and Brake Disc (Rotor) Inspection

Purpose: Inspect braking surfaces for wear, scoring, cracks, heat damage, and out-of-round condition that affects braking performance.

Frequency: Every brake service (lining replacement) or every 30,000 km

Tools & Materials: Vernier caliper or drum micrometer, dial indicator with magnetic base (for runout), flashlight, 150-grit sandpaper (for minor surface rust)

Scope: All trucks — inspect each brake drum or disc during lining inspection

PPE Required: Safety glasses, gloves, steel-toe boots

Step-by-Step Procedure:

- 1. Drum Inspection:** Clean drum interior surface with brake cleaner and rag.
- Visually inspect friction surface for severe scoring (grooves deeper than 1mm), heat checking (hairline cracks on surface), hard spots (discolored, shiny areas — "glazing"), and cracks extending to edge or bolt holes.
- Measure drum inside diameter at several points around circumference using drum micrometer or inside caliper.
- Compare to maximum allowable diameter (stamped on drum outer surface or manufacturer spec).
- If worn past maximum + 0.060 inch (1.5mm), replace drum.
- Check for out-of-round: measure diameter at multiple points around circumference; variation should be ≤ 0.005 inch (0.13mm).
- 7. Disc (Rotor) Inspection:** Clean rotor friction surface with brake cleaner.
- Visually inspect for scoring, cracks, heat discoloration (blue/purple indicates overheating), hard spots, and rust.
- Measure rotor thickness using micrometer at several points around circumference, about 10mm from outer edge.
- Minimum thickness is stamped on rotor edge — replace if at or below that value.
- Thickness variation between measurements should be ≤ 0.001 inch (0.025mm).
- Check rotor runout: mount dial indicator against rotor friction surface, rotate hub.
- Runout should be ≤ 0.004 inch (0.10mm).
- If excessive, check hub bearing play first (bearing wear can cause false runout).

Acceptance Criteria: Drum: inside diameter within manufacturer limit, no cracks, no severe scoring. Maximum out-of-round 0.005 inch (0.13mm). Disc: thickness above minimum, no cracks, no hard spots. Thickness variation ≤ 0.001 inch (0.025mm). Runout ≤ 0.004 inch (0.10mm).

⚠ **SAFETY:** Wear particulate respirator if cleaning with abrasives. Rotors can develop hard spots from overheating that cause braking vibration and reduced friction.

Tip: When replacing drums or rotors, always replace in axle pairs. New drums/rotors with old linings can cause uneven braking. Resurfacing is rarely cost-effective for heavy truck components — replacement is preferred.

Documentation: Record drum/rotor condition, measurements (diameter, thickness, runout), replacement decisions with part numbers.

Performing Mechanic

8.7 Air Brake System Inspection

Air Brake System Inspection (DOT-compliant)

Purpose: Inspect the complete compressed air brake system including compressor, governor, air tanks, lines, chambers, slack adjusters, and brake valve to ensure safe stopping performance.

Scope: All trucks equipped with air brake systems

Frequency: Weekly or every 5,000 km

PPE Required: Safety glasses, gloves, hearing protection

Tools & Materials: Tire thumper or mallet, soapy water spray bottle, air pressure gauge, ruler, flashlight, crows foot wrench set (for slack adjuster), chocks

Step-by-Step Procedure:

1. **Pre-test:** Chock wheels.
2. Ensure air system fully charged (governor cut-out at ~120-135 psi).
3. Turn off engine.
4. **Leak Test:** With engine off and full air pressure, note pressure reading.
5. Observe pressure drop over 1 minute.
6. With brakes released: maximum 2 psi drop per minute.
7. With brakes applied (push pedal fully): maximum 3 psi drop per minute.
8. If exceeds, locate leaks using soapy water at all fittings, chambers, and valve connections.
9. **Low Pressure Warning:** With engine off, slowly drain air by depressing brake pedal repeatedly.
10. Low pressure warning (buzzer + light) must activate before pressure drops below 60 psi (typically at 70-75 psi).
11. **Spring Brake Activation:** Continue draining air.
12. Spring brakes should mechanically apply (truck will not roll) when pressure drops to approximately 30-40 psi.
13. **Compressor & Governor:** Start engine and let air pressure build.
14. Governor cut-in should be approximately 100-110 psi.
15. Governor cut-out at approximately 120-135 psi.
16. Listen for abnormal compressor noise (knocking, continuous running).
17. **Air Tank Drain:** Open each air tank drain valve (wet tank, primary, secondary) fully.
18. Observe discharge for excessive moisture, oil contamination, or rust particles.
19. Close valves once drained.
20. **Slack Adjuster Inspection:** With brakes released, measure pushrod stroke at each brake chamber using ruler.
21. Apply brakes fully (100 psi) and measure again.
22. Stroke should be within limits: Type 20/24 chamber: 1.75 inch max.

23. Type 30 chamber: 2.0 inch max.
24. Adjust if stroke exceeds limit: hold adjusting nut (usually 9/16 or 5/8 inch), turn adjusting bolt counterclockwise until lining contacts drum, then back off 1/2 turn.
25. **Brake Chamber Inspection:** Inspect chamber housing for cracks, corrosion holes, loose mounting bolts.
26. Check pushrod for bent condition.
27. Verify chamber type is same on both sides of axle.
28. **Hose & Tubing:** Inspect all air hoses for chafing, cracks, bulges, or weather checking.
29. Ensure hoses are properly supported and not contacting hot surfaces or moving parts.
30. Check nylon tubing for embrittlement near connections.

Acceptance Criteria: Air pressure drop within limits (2 psi released, 3 psi applied). Low pressure warning activates above 60 psi. Governor cuts in at 100-110 psi, cuts out at 120-135 psi. No air leaks detected. Pushrod stroke within limits — automatic adjusters not over-extended. Air tanks drained — minimal moisture/oil present.

⚠ **SAFETY:** Never work on a air brake system without fully draining air pressure first. The energy stored in air tanks can cause sudden, violent movement of components. Chock wheels before any brake work. Do not attempt to disassemble spring brake chambers — they contain a powerful compressed spring that can cause fatal injuries. Spring brake disassembly requires a certified technician with specialized equipment.

Tip: If automatic slack adjusters consistently require manual adjustment, they may be at the end of their service life. Replace them rather than continuing to adjust. Check pushrod stroke on both sides of each axle — uneven stroke indicates a problem with the automatic adjuster on the longer-stroke side.

Documentation: Record: cut-in/cut-out pressures, pushrod stroke for each wheel position, any adjustments made, air dryer service interval status.

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8.8 Tire Condition Inspection

Tire Condition Visual Inspection

Purpose: Inspect all tires for damage, irregular wear, proper inflation, and safety defects to prevent blowouts and ensure road safety.

Frequency: Daily (pre-trip) and every 5,000 km (detailed)

Tools & Materials: Flashlight, tread depth gauge, tire pressure gauge (dual-foot type for steering and drive tires), hammer or tire thumper

Scope: All trucks — all tire positions including spare

PPE Required: Safety glasses, gloves, high-visibility vest

Step-by-Step Procedure:

1. **Visual Walk-Around:** Walk completely around the truck inspecting each tire.
2. Check for: under-inflation (bulging sidewalls, excessive sidewall flex), over-inflation (center tread wear, crowning).
3. Use tire thumper to strike each tire — a properly inflated tire produces a solid, resonant sound; an under-inflated tire sounds dull.
4. **Sidewall Inspection:** Inspect both sidewalls of each tire for: cuts, snags, bulges, blisters, exposed cords or belts, weather checking (fine cracks), abrasion damage.
5. **Tread Inspection:** Look for: irregular wear patterns (cupping, feathering, shoulder wear, center wear), stones or debris embedded in tread (remove with scraper), foreign object penetration (nails, screws, metal), chunking or tearing of tread rubber.
6. **Valve & Cap:** Check valve stem for cracks at the base.
7. Confirm valve cap is present and tight (caps are secondary seals).
8. **Rim & Wheel Hardware:** Inspect rim for cracks (especially around lug holes and weld joints).
9. Check for bent or damaged rim flanges.
10. Inspect all lug nuts — none should be loose, missing, or show signs of movement (rust ring around nut indicates loosening).
11. Check for wheel cracks (especially on aluminum wheels around lug holes).
12. **Spare Tire:** Verify spare tire is present, properly mounted, and inflated to recommended pressure.

Acceptance Criteria: No cuts, bulges, blisters, or exposed cords on any tire. Tread depth $\geq 2/32$ inch (1.6mm) on all tires — $4/32$ inch (3.2mm) minimum for steering axle tires in most jurisdictions. No irregular wear exceeding $2/32$ inch (1.6mm) difference across the tread. Valve caps present on all tires. All lug nuts tight and properly torqued. No rim cracks or damage.

⚠ **SAFETY:** A tire blowout at highway speed can cause loss of vehicle control. Never attempt to repair a tire with sidewall damage, exposed cords, or bulges — replace immediately. Do not inflate a tire past the maximum pressure stamped on the sidewall.

Tip: Steering axle tires should always have the deepest tread and be the newest tires in the fleet. When replacing, always use matched tires on the same axle (same brand, model, tread depth within 2/32 inch). Never mix radial and bias-ply tires on the same axle.

Documentation: Record tire positions with significant wear or damage, tread depth measurements (steering and outer drive positions). Note any replacement tire serial numbers.

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8.9 Tread Depth & Pressure Measurement

Tire Tread Depth and Pressure Measurement

Purpose: Quantitatively measure tread depth and inflation pressure on all tires to ensure compliance with legal minimums and optimal tire life.

Frequency: Weekly or every 5,000 km

Scope: All trucks — weekly for long-haul, daily for local operations

PPE Required: Safety glasses, gloves, high-visibility vest

Tools & Materials: Tread depth gauge (dial-type or digital), tire pressure gauge (calibrated), air compressor with tire chuck, valve core tool (for replacing cores)

Step-by-Step Procedure:

1. **Pressure Check:** Check tire pressures when tires are **cold** (driven less than 1 mile at moderate speed).
2. Remove valve cap.
3. Press gauge firmly onto valve stem and read pressure.
4. Compare to the manufacturer's recommended pressure (stamped on sidewall or in vehicle placard).
5. Steering tires: typically 100-120 psi.
6. Drive tires: typically 90-110 psi.
7. Add air if below recommended pressure.
8. If a tire requires more than 10% of its pressure to fill, inspect for leaks (soapy water at valve core, bead area, and tread).
9. Reinstall valve cap.
10. **Tread Depth Check:** Insert tread depth gauge probe into the tread groove at a shallow area (usually near center or outer edge).
11. Push gauge base flat against tread surface.
12. Read depth in 32nds of an inch or millimeters.
13. Measure at three positions across the tread (outer shoulder, center, inner shoulder) and at two points around the circumference (180 degrees apart).
14. Record the lowest reading.
15. **Legal Minimums:** Steer axle: 4/32 inch (3.2mm) minimum in most commercial regulations.
16. Drive and trailer axles: 2/32 inch (1.6mm) minimum.
17. Flag any tire below these limits for immediate replacement.

Acceptance Criteria: All tires within 5 psi of recommended cold pressure. Tread depth above legal minimums. No more than 2/32 inch (1.6mm) difference between tires on same axle. Spare tire at maximum recommended pressure.

⚠ **SAFETY:** Over-inflating a tire beyond its rated pressure can cause explosive failure. Use a tire cage when inflating a tire that has been run flat or shows sidewall damage. Split rim wheels require special handling — deflate completely before removing any hardware.

Tip: Tire pressure should be checked and logged consistently. A tire losing more than 2 psi per week has a slow leak that should be investigated. Under-inflation by just 20% reduces tire life by 25% and increases fuel consumption by 2-3%.

Documentation: Record pressures and tread depths for each tire position. Use consistent format: LF (left front), RF (right front), LDO (left drive outer), LDI (left drive inner), etc.

Performing Mechanic or Driver

8.10 Overfill Prevention System Inspection

Overfill Prevention System (Vapor Recovery Compatible) Inspection

Purpose: Inspect and test the overfill prevention system to ensure it prevents tank compartment overfilling during loading, which could result in spillage, environmental damage, and safety hazards.

Frequency: Monthly or per depot access requirements

Tools & Materials: Multimeter (for electrical sensors), soft brush, rag, approved solvent (for cleaning), airline for drying

Scope: All petroleum tanker trucks with overfill prevention sensors/valves

PPE Required: Safety glasses, gloves, chemical-resistant gloves (if fuel contact possible)

Step-by-Step Procedure:

1. **Visual Inspection:** Access each compartment's overfill prevention sensor/valve (typically located on top of each compartment or at the vapor recovery adaptor).
2. Check for physical damage: cracked housing, broken wires, loose mounting.
3. Check for debris or contamination around sensor tip.
4. Clean sensor tips gently with soft brush and approved solvent.
5. **Mechanical Float Test (if float-type):** Manually actuate the float mechanism by lifting the float to its upper limit.
6. Verify that the valve closes completely when the float is in the raised position (simulating a full compartment).
7. Release float — valve should open freely.
8. Check float arm pivot for free movement.
9. **Electrical Sensor Test (if electronic):** Use multimeter to check sensor resistance across terminals.
10. Compare to manufacturer specification (normally open or normally closed configuration).
11. Cycle sensor by simulating full condition — multimeter reading should change state.
12. **Optical Sensor Test (if optical type):** Clean optical window.
13. Use reflective strip or manufacturer test tool to simulate product presence.
14. Observe indicator light change on control board (if accessible).
15. **Cable & Connector:** Inspect cable from sensor to truck's vapor recovery adaptor or interrupter.
16. Check for cuts, abrasion, or crushed areas.
17. Verify connector pins are clean, straight, and fully seated.
18. **Functional Test at Loading Rack (if possible):** Coordinate with depot for a test loading cycle.
19. Observe that overfill prevention initiates a slow-down and shut-down signal when triggered.

Acceptance Criteria: Float-type: valve closes fully when float raised, opens freely on release. Electronic: sensor state changes correctly when triggered, resistance within spec. Optical: indicator

confirms detection. No physical damage, corrosion, or contamination on any sensor. Cable fully intact with good connector.

⚠ **SAFETY:** Petroleum products are flammable and toxic. Purge compartments before accessing internal sensors. No hot work within 10 meters of compartment openings. Use only non-sparking tools when working near tank openings.

Tip: The overfill prevention system is the most critical safety device on a tanker truck. Many depots will refuse access if the overfill prevention is not functioning correctly. Test compatibility with different depot systems during inspection.

Documentation: Record sensor type per compartment, test method, result (pass/fail), cleaning performed, any replacement parts.

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8.11 Manlid & Gasket Inspection

Manlid Cover and Gasket Inspection

Purpose: Inspect each compartment's manlid cover, hinge, clamping mechanism, and gasket to ensure vapor-tight sealing and prevent product loss or contamination during transport.

Frequency: Weekly or before every loading

Tools & Materials: Flashlight, rag, gasket replacement kit (as needed), torque wrench

Scope: All tanker trucks — inspect each compartment manlid individually

PPE Required: Safety glasses, gloves, steel-toe boots

Step-by-Step Procedure:

1. Open each manlid fully and secure with safety prop (if equipped).
2. Inspect gasket on lid underside: check for cuts, compression set (flattened permanently), brittleness, cracks, separation from lid channel.
3. Inspect gasket channel for rust or debris buildup.
4. Clean channel with rag if needed.
5. Inspect lid sealing surface: must be clean, flat, free of dents or distortion.
6. Check hinge pin for wear and lubrication.
7. Verify hinge allows full opening (90+ degrees).
8. Inspect clamping mechanism: check over-center latches or T-bolts for smooth operation.
9. Check latch rollers for flat spots or wear.
10. Check latch strike plates on tank body for deformation.
11. Verify all clamps engage fully and hold lid firmly closed.
12. Close manlid and engage clamping mechanism.
13. Verify that the lid closes with moderate resistance (compressing the gasket).
14. Check for gaps between lid and tank body — use feeler gauge if available.
15. Maximum acceptable gap: 0.5mm.
16. **Seal Test (if time permits):** Close all manlids.
17. Pressurize tank vapor system to 1 psi using vapor recovery connection.
18. Apply soapy water around each manlid perimeter.
19. Look for bubbles indicating leaks.

Acceptance Criteria: Gasket pliable (not rock-hard or crumbling), no cuts or compression set exceeding 50% of original thickness. Lid closes with consistent compression across all edges. No gap exceeds 0.5mm between lid and body. All clamping mechanisms engage fully and hold. No leakage detected in seal test.

⚠ **SAFETY:** When opening manlids on compartments that have carried product, vapors may escape. Stand upwind and allow vapors to dissipate before leaning over the opening. Never open a manlid that has internal pressure — relieve pressure through vapor recovery system first.

Tip: Gaskets should be replaced on a preventive schedule (every 12 months or when inspection shows compression set). A leaking manlid gasket causes product loss through evaporation, contamination from water ingress, and vapor leaks at depots. It also creates a foul odor around the truck.

Documentation: Record gasket condition per compartment (1-5 scale), any gaskets replaced with date and material type (Buna-N, Viton, EPDM).

Performing Mechanic

8.12 API Adapters & Camlock Couplings Inspection

API Bottom-Loading Adapter and Camlock Coupling Inspection

Purpose: Inspect the API bottom-loading adapters, camlock couplings, dust caps, and associated fittings for wear, damage, leakage, and correct operation to ensure safe loading/unloading.

Frequency: Weekly or before every loading shift

Tools & Materials: Flashlight, small pick or screwdriver (for o-ring inspection), lubricant (silicone grease), rag

Scope: All tanker trucks with bottom-loading and vapor recovery adapters

PPE Required: Safety glasses, chemical-resistant gloves, steel-toe boots

Step-by-Step Procedure:

1. **API Bottom-Loading Adapter:** Inspect adapter body for cracks, dents, or impact damage.
2. Check sealing mechanism: manually open and close the internal valve (if accessible).
3. The valve should open smoothly and close with a positive seal.
4. Inspect the poppet and seat for debris, scoring, or wear.
5. Check return spring action — valve should close automatically when released.
6. Inspect the interlock mechanism (if equipped) that prevents opening unless vapor system is connected.
7. Check all o-rings on the adapter face: inspect for cuts, flattening, or extrusion.
8. Measure o-ring protrusion — should be approximately 1-2mm above the face surface.
9. **Camlock Couplings (Top Loading / Unloading):** Inspect camlock body for cracks or distortion (especially at the cam ear pivot points).
10. Check cam arms for free movement — should pivot smoothly through full travel.
11. Inspect cam arm mating surfaces for wear (rounded-off edges reduce locking effectiveness).
12. Check gasket inside female end — must be present, pliable, and not swollen or deformed.
13. Replace if any o-ring is flat, cut, or leaking.
14. **Dust Caps:** Verify dust cap is present on each adapter.
15. Check hinge and latch for proper operation.
16. Cap must close and latch securely.
17. **Threaded Connections:** Inspect any threaded caps, plugs, or adapters for thread damage (cross-threading, galling, stripped threads).

Acceptance Criteria: API adapter: internal valve opens/closes fully, o-rings protrude 1-2mm with no cuts, no impact damage, interlock functional. Camlock: free-moving cam arms, rounded edges not excessive, gasket present and pliable, no cracks in body. Dust caps present and close securely.

⚠ **SAFETY:** A failed API adapter during bottom-loading can release thousands of liters of product. If any adapter shows significant damage or wear, tag the compartment out of service until repaired. Depots may reject a truck with faulty adapters.

Tip: Apply a thin film of silicone grease to all o-rings and camlock gaskets during inspection to extend service life. Do not use petroleum-based lubricants — they cause o-ring swelling and degradation. Buna-N o-rings (black) are standard for petroleum products. Viton (brown) for higher temperature or chemical resistance.

Documentation: Record adapter condition per position (compartment 1-6), any o-rings replaced, any seal failures.

Performing Mechanic

8.13 Vapor Recovery System Inspection

Vapor Recovery System Inspection

Purpose: Inspect the vapor recovery system that captures fuel vapors during loading and returns them to the terminal, ensuring the system is vapor-tight and functioning correctly for depot compliance.

Frequency: Monthly or per depot access requirements

Tools & Materials: Manometer or low-pressure gauge, soapy water spray bottle, flashlight, plug/blank for vapor opening

Scope: All tanker trucks equipped with vapor recovery systems

PPE Required: Safety glasses, gloves, hearing protection

Step-by-Step Procedure:

1. **Vapor Hose Inspection:** Inspect vapor return hose (typically 4-inch diameter, attached to vapor recovery adaptor).
2. Check for cuts, abrasion, weather checking, and soft spots.
3. Verify hose is properly supported and not dragging on the ground.
4. Check hose end fitting for damage.
5. **Vapor Adaptor:** Inspect the vapor recovery adaptor (located at rear of truck or side cabinet).
6. Check the poppet valve for free movement.
7. Inspect sealing surfaces and o-rings — replace if worn.
8. **Vapor Plumbed Lines:** Trace vapor lines from each compartment (connected to each manlid or internal vapor valve) to the main vapor header.
9. Inspect for corrosion, damage, or loose connections.
10. Check all flexible vapor connections at tank ends for deterioration.
11. **Vapor Valve (Pressure/Vacuum Vent):** Locate the P/V vent on top of the tank (usually highest point).
12. Inspect for correct orientation (must be upright).
13. Check for free movement of the vent valve.
14. Clean vent screen of debris or insect nests.
15. **Seal Test:** Close all manlids and vapor valves.
16. Attach low-pressure source (regulated shop air at 1-2 psi maximum) to vapor adaptor.
17. Apply soapy water solution to all vapor connections, gaskets, and manlid seals.
18. Look for bubbles indicating leaks.
19. Hold pressure for 5 minutes — system should hold within 10% of initial pressure.

Acceptance Criteria: Vapor hose intact with no cuts, bulges, or soft spots. Adaptor poppet moves freely. All vapor line connections tight and undamaged. P/V vent upright and functional, screen

clear. Seal test: holds pressure for 5 minutes with minimal drop ($\leq 10\%$). No bubbles at any connection.

⚠ **SAFETY:** Do not pressurize vapor system beyond 3 psi — overpressure can damage tank structure and vapor components. Some depots have specific pressure test requirements for vapor system certification. Petroleum vapors are flammable — no smoking or open flames within 10 meters during testing.

Tip: A failed vapor recovery system is one of the most common reasons for depot access refusal. Many terminals perform a pressure test before every loading. If the truck fails depot vapor test, document the psi reading observed at the terminal to help diagnose the leak location.

Documentation: Record pressure test psi, hold time, leak locations (if any), repairs performed, vapor hose replacement date.

Performing Mechanic → Lead Mechanic verification

8.14 Grounding & Static Discharge System Inspection

Grounding Cable and Static Discharge System Inspection

Purpose: Inspect the truck's grounding system to ensure a reliable electrical connection between the truck chassis and the depot grounding point, preventing static electricity buildup during loading operations.

Scope: All tanker trucks

Frequency: Weekly

PPE Required: Safety glasses, gloves

Tools & Materials: Multimeter (continuity function), flashlight, rag, file or emery cloth, dielectric grease

Step-by-Step Procedure:

1. **Cable Inspection:** Locate grounding cable(s) — typically a green/yellow striped cable with a clamp at one end, attached near the rear of the chassis or in a side cabinet.
2. Inspect cable jacket for cuts, abrasion, corrosion.
3. Check for broken strands at attachment points.
4. **Cable Route:** Verify cable is routed without sharp bends, kinks, or contact with hot surfaces or moving components.
5. Ensure cable length is sufficient to reach typical depot grounding points (minimum 3 meters).
6. **Grounding Clamp (Spring Clamp):** Inspect clamp jaws for clean, bare metal contact surfaces.
7. Remove any paint, rust, or contamination using file or emery cloth.
8. Test spring tension — clamp should hold firmly on a 10mm steel test piece.
9. Check clamp pivot for smooth operation.
10. Inspect clamp-to-cable connection: verify it is mechanically secure and electrically continuous.
11. **Chassis Ground Connection:** Trace cable to chassis attachment point.
12. Verify bolt or stud is tight.
13. Inspect connection for corrosion.
14. Clean and apply dielectric grease if needed.
15. **Continuity Test:** Set multimeter to continuity or resistance mode (200 ohms range).
16. Place one probe on the end of the grounding clamp contact surface.
17. Place the other probe on a clean, bare metal chassis point (scrape paint if needed).
18. Resistance reading should be ≤ 5 ohms.
19. If higher, check all connections and cable for damage.
20. **Stored Energy Test (optional but recommended):** After truck has been driven, use a static discharge meter (if available) to measure static charge on the tank body before grounding.
21. Reading should rapidly drop to near zero when clamp is attached.

Acceptance Criteria: Grounding cable: intact jacket, no broken strands, length sufficient. Clamp: clean bare-metal contact surface, strong spring tension. Chassis connection: tight, corrosion-free. Continuity: ≤ 5 ohms from clamp to chassis ground.

⚠ SAFETY: Static discharge is a leading cause of tanker truck fires during loading. Never begin loading without the grounding clamp securely attached to a verified grounding point. If the grounding system fails continuity test, tag the truck out of service immediately for repair.

Tip: The depot grounding system also requires inspection — but that is the depot's responsibility. Your job is to ensure the truck-side system is 100% reliable. Apply a thin coat of dielectric grease to the clamp contact surface after cleaning to prevent rapid re-corrosion.

Documentation: Record continuity reading (ohms), clamp condition, cable inspection result, any cleaning or repairs performed.

Performing Mechanic

8.15 Product Piping & Valves Inspection

Product Piping, Internal Valves, and Discharge System Inspection

Purpose: Inspect all product-carrying piping, internal safety valves, discharge valves, and associated fittings for leaks, corrosion, and mechanical integrity.

Frequency: Monthly

Tools & Materials: Flashlight, small mirror, rag, leak detection spray (for non-product sections), torque wrench

Scope: All tanker trucks — inspect per compartment and common discharge line

PPE Required: Safety glasses, chemical-resistant gloves, steel-toe boots

Step-by-Step Procedure:

1. **Internal Safety Valve (Foot Valve):** Located at bottom of each compartment.
2. Verify valve operator linkage (cable or rod) is intact and properly adjusted.
3. Operate the emergency/pneumatic release handle at the rear of the truck — confirm each compartment's internal valve opens and closes correspondingly.
4. Check for leaks at valve body (product seepage past valve seat).
5. **Product Piping (Bottom-Loading / Discharge):** Visually inspect all product piping from each compartment's internal valve to the discharge outlet.
6. Check for corrosion, dents, rubbing marks, weld cracks, and joint leaks.
7. Pay special attention to pipe supports and areas where pipes pass through frame members.
8. Check threaded connections at flanges and unions for seepage.
9. **Discharge Valve (Butterfly / Ball Valve):** Operate each discharge valve fully open and closed.
10. Valve should turn smoothly with moderate resistance (not too tight, not sloppy).
11. Inspect valve body for cracks or corrosion.
12. Verify valve position indicator is accurate (open/closed markings).
13. **Emergency Shutoff System:** Inspect pneumatic or mechanical linkage from the emergency release at the rear of the truck to each internal valve.
14. Confirm all connecting rods, cables, and levers are secure and properly adjusted.
15. Test emergency shutoff by pulling the handle — all internal valves should close.
16. **Piping Supports:** Verify all pipe clamps and brackets are present and tight.
17. Look for rubber or plastic chafing strips where pipes contact frame or other components.
18. Replace missing chafing protection.

Acceptance Criteria: Internal valves: open and close fully, no leakage past seat. Piping: no corrosion deeper than 10% of wall thickness, no dents restricting flow, no leaks at any joint. Discharge valves: smooth operation through full travel. Emergency shutoff: all internal valves close when handle is pulled. Piping properly supported with chafing protection intact.

⚠ **SAFETY:** Product piping contains flammable liquids. Any leak is a fire hazard and environmental violation. If you detect any leak, tag the compartment out of service immediately. Conduct repair only after compartment is purged and gas-free certified.

Tip: Corrosion often starts at the bottom of pipes where water accumulates. Pay extra attention to pipe low points and drain plugs. If a pipe shows significant corrosion, use an ultrasonic thickness gauge (if available) to measure remaining wall thickness.

Documentation: Record per-compartment inspection results (pass/fail), any leaks found and locations, valve operation condition, emergency shutoff test result.

Performing Mechanic → Lead Mechanic

8.16 Transfer Hoses & Couplings Inspection

Product Transfer Hoses and End Couplings Inspection

Purpose: Inspect all flexible product transfer hoses (loading/unloading hoses) and their end couplings for signs of wear, damage, deterioration, and leakage to ensure safe product transfer.

Frequency: Weekly visually; monthly detailed inspection

Tools & Materials: Flashlight, mirror, rag, hose gauge or caliper (for measuring cover thickness)

Scope: All tanker trucks with on-board hose racks

PPE Required: Safety glasses, chemical-resistant gloves

Step-by-Step Procedure:

- 1. Hose Cover Inspection:** Uncoil or lay out hose full length on clean surface.
2. Visually inspect the full circumference of the hose.
3. Look for: cuts or gouges penetrating through the cover to reinforcement layers, abrasion wear (flat spots), blisters or bubbles (cover separation from reinforcement), chemical attack (soft or sticky spots, swelling), heat damage (hardened, cracked areas), weather checking (fine surface cracks), kinking or crushing damage.
- 4. Hose Reinforcement:** If cover damage is found, inspect the underlying reinforcement.
5. Exposed fabric or wire braid requires immediate hose replacement.
- 6. End Couplings:** Inspect each coupling (male and female) for corrosion, cracks, or distortion.
7. Check coupling retention — the coupling must be securely crimped or swaged to the hose.
8. Look for evidence of coupling movement (gap between coupling ferrule and hose cover).
9. Inspect coupling threads for damage (cross-threading, galling).
10. Clean threads with wire brush if needed.
- 11. Hose Markings:** Verify hose is marked with: manufacturer name, date of manufacture, rated working pressure, and any applicable standards (API, EN, etc.).
12. Confirm hose has not exceeded its recommended service life (typically 5-7 years from manufacture date, regardless of condition).
- 13. Hose Storage:** Inspect hose rack or storage area.
14. Verify hose is stored on a clean surface or rack — never on bare ground.
15. Check that hose is not draped over sharp edges or in contact with chemicals, oil, or heat sources.
- 16. Pressure Test (annual):** Hydrostatically test each hose at 1.5x rated working pressure.
17. Hold pressure for 5 minutes.
18. No leakage or hose movement at fitting.

Acceptance Criteria: No cuts exposing reinforcement. No blisters, soft spots, or swelling.
Couplings: securely attached, clean threads, no cracks. Working pressure rating meets or exceeds

system pressure (typically 150-300 psi for gravity/unloading hoses). Hose not past service life.
Annual pressure test passed.

⚠ **SAFETY:** A hose failure during loading under pressure can cause a massive product spill with fire and environmental consequences. A hose failure while unloading at a customer site can cause serious injury. Never use a damaged hose — replace immediately regardless of cost.

Tip: Rotate hoses periodically (every 6 months) to distribute wear. Label each hose with an ID number and installation date. The leading cause of hose failure is abrasion from dragging on concrete. Use hose ramps or always lift rather than drag.

Documentation: Record hose ID, condition rating, manufacture date, test pressure, test result, date of next test due.

Performing Mechanic

8.17 Safety Equipment Inspection

Truck-Mounted Safety Equipment Inspection

Purpose: Inspect all safety equipment carried on the truck including fire extinguisher, first aid kit, reflective triangles, wheel chocks, HAZMAT placards, and other safety devices to ensure they are present, accessible, and serviceable.

Scope: All trucks

Frequency: Weekly (pre-trip)

PPE Required: Safety glasses, gloves

Tools & Materials: Flashlight, inspection tag (for fire extinguisher)

Step-by-Step Procedure:

1. **Fire Extinguisher:** Locate extinguisher mounting bracket — verify extinguisher is securely mounted and accessible.
2. Check gauge needle: must be in green (operable) range.
3. Inspect cylinder for corrosion, dents, or damage.
4. Check plastic seal/tamper indicator — must be intact.
5. Verify extinguisher has not passed its inspection date (monthly visual, annual maintenance, 6-year hydrostatic test).
6. **Reflective Triangles / Warning Devices:** Verify three reflective triangles are present in their storage location.
7. Check that reflective surfaces are clean and unbroken.
8. Check triangle hinges/folding mechanism — must open and stand stably.
9. **Wheel Chocks:** Confirm two wheel chocks are present (typically stored under passenger seat or in side compartment).
10. Must be heavy-duty rubber or aluminum — check for excessive wear, cracks, or damage.
11. **First Aid Kit:** Verify first aid kit is present and fully stocked.
12. Minimum contents: adhesive bandages (assorted sizes), sterile gauze pads, adhesive tape, antiseptic wipes, disposable gloves (nitrile), CPR mask, trauma shears, emergency blanket.
13. Check expiration dates on sealed items.
14. **Reflective Tape:** Inspect all reflective tape on truck body (sides, front, rear).
15. Must be present per DOT/B+P/TES regulations.
16. Check for peeling, fading, or damage.
17. Replace any missing or degraded sections (at least 50% reflective area must be intact per side).
18. **HAZMAT Placards:** Verify correct placards are present for the product being carried.
19. Most petroleum tankers use UN 1203 (Gasoline) or UN 1993 (Diesel/Fuel Oil) placards.
20. Placards must be securely mounted, clean, and legible.
21. Minimum two placards per side, one front, one rear.
22. Verify the placard holder or bracket is functional.

- 23. **Compartment Labels:** Each compartment must be labeled with product identification (if carrying multiple products).
- 24. Labels must be legible and securely attached.
- 25. **Horn & Backup Alarm:** Test horn — must be clearly audible (minimum 90 dB).
- 26. Test backup alarm (if equipped) — engage reverse gear and verify alarm sounds audibly.
- 27. **Fall Arrest / Top Rail:** If truck is equipped with fall arrest system or handrails on top deck, inspect for secure mounting, corrosion, and functionality.

Acceptance Criteria: Fire extinguisher: gauge in green, no damage, current inspection tag. Triangles: three present, reflective surface intact. Wheel chocks: present. First aid kit: stocked, items within expiration. Reflective tape: ≥50% reflective area per side, no large missing sections. Placards: correct, clean, legible. Horn: functional. Labels: present and legible. Fall arrest: secure.

⚠ **SAFETY:** In an emergency, every second counts. Safety equipment stored under cargo or behind obstacles is useless — ensure all items are easily accessible from outside the cab. Fire extinguishers must be accessible without entering the cab or reaching over the hood.

Tip: Use a pre-trip checklist card to ensure no safety item is missed. Log the inspection in the system under the Safety Equipment PM schedule. If any item is missing or expired, tag the truck and do not dispatch until resolved.

Documentation: Record all safety equipment status (present/serviceable/expired/missing). Note fire extinguisher last inspection date. Record any placards replaced.

Performing Mechanic or Driver

8.18 Battery & Alternator Inspection

Battery System and Alternator Charging Test

Purpose: Inspect batteries, cables, terminals, and alternator output to ensure reliable engine starting and electrical system operation.

Scope: All trucks

Frequency: Monthly or every 10,000 km

PPE Required: Safety glasses, acid-resistant gloves (for wet-cell batteries)

Tools & Materials: Digital multimeter, battery load tester (optional), terminal brush, baking soda solution (for cleaning), dielectric grease, flashlight, wrench set

Step-by-Step Procedure:

1. **Battery Visual Inspection:** Inspect battery case for cracks, bulging, or leaks.
2. Check battery hold-down bracket — must be tight to prevent vibration damage.
3. Check battery terminal posts for corrosion (white/green/blue powder) — clean if present using terminal brush and baking soda solution.
4. **Cable Inspection:** Inspect both positive and negative battery cables from terminal to chassis/ground and to starter.
5. Check for: frayed or broken strands at terminals, cracked insulation, corrosion under the insulation (green substance), loose terminal clamps.
6. **Battery Voltage Test (Resting):** Engine off, all loads off.
7. Set multimeter to DC volts (20V range).
8. Measure voltage across battery terminals.
9. 12.6V+ = fully charged.
10. 12.4V = 75% charged.
11. 12.2V = 50% charged.
12. Below 12.0V = discharged — recharge and investigate cause.
13. **Battery Load Test:** If battery voltage is low, perform load test.
14. Apply load tester per manufacturer instructions (typically half the CCA rating for 15 seconds).
15. Minimum voltage during load: 9.6V at room temperature (higher in cold).
16. **Alternator Output Test:** Start engine.
17. Measure voltage across battery terminals at idle: should read 13.8-14.5V (12V system).
18. Rev engine to 1500 RPM — voltage should rise to 14.0-14.7V (regulation point).
19. Turn on all electrical loads (headlights, AC, hazards) — voltage should remain above 13.0V.
20. If voltage stays below 13.0V at idle with loads, alternator is undercharging.
21. If voltage exceeds 15.0V, voltage regulator is failing (overcharging).
22. **Belt Check:** Inspect alternator drive belt condition and tension.
23. See Section 8.4 for belt procedure.

24. **Battery Water Level (Serviceable Batteries):** If batteries have vent caps, remove caps and check electrolyte level in each cell.
25. Level should cover the plates (typically 10-15mm above plates).
26. Top off with **distilled water only** — never add acid.
27. **Parasitic Draw Test (if battery drains):** With engine off, set multimeter to amps (10A range).
28. Disconnect negative battery terminal and connect meter in series (meter between terminal and cable).
29. Reading should be below 50mA.
30. Higher indicates parasitic draw that needs diagnosis.

Acceptance Criteria: Resting voltage $\geq 12.4V$ (75%+ charge). Load test: maintains $\geq 9.6V$ under load. Alternator: 13.8-14.5V at idle, 14.0-14.7V at 1500 RPM. No corrosion on terminals. Cables intact, no fraying. Battery water covers plates (if serviceable). Parasitic draw $\leq 50mA$.

⚠ **SAFETY:** Batteries contain explosive hydrogen gas. No smoking, sparks, or flames near batteries. Disconnect negative terminal first, connect positive first. When jump-starting, connect dead positive first, then good positive, then good negative (to chassis ground, not dead battery), then dead negative last. Reverse order for removal. Acid can cause severe burns — flush immediately with water if contact occurs.

Tip: For trucks with dual batteries (common for diesel trucks), test each battery individually. Replace batteries in pairs — mixing old and new batteries reduces the life of both. Log the installation date on each battery to track service life (typically 3-5 years).

Documentation: Record battery resting voltage, alternator output voltage at idle and 1500 RPM, any cleaning or topping off performed, battery replacement dates and CCA rating.

Performing Mechanic

8.19 Starter & Starting Circuit Inspection

Starter Motor and Starting Circuit Wiring Inspection

Purpose: Inspect the starter motor, solenoid, and all associated wiring for conditions that could cause starting failure, high current draw, or electrical fire.

Scope: All trucks

Frequency: Every 20,000 km or when starting issues are observed

PPE Required: Safety glasses, gloves, insulated tools (when working near battery)

Tools & Materials: Digital multimeter, wrench set, flashlight, contact cleaner

Step-by-Step Procedure:

1. **Visual Inspection:** Locate starter motor on engine bell housing.
2. Inspect starter body for cracks, oil leaks, or impact damage.
3. Check mounting bolts — must be tight.
4. Inspect starter solenoid: check for loose or corroded small-wire connections at the "S" (start signal) and "I" (ignition bypass) terminals.
5. **Main Power Cable:** Trace battery positive cable to starter solenoid large terminal.
6. Check for: corrosion, loose nut, frayed cable at terminal, melted insulation (indicates high resistance connection causing heat), chafing along cable route.
7. **Ground Circuit:** Check battery negative cable at engine block or chassis ground.
8. Must be tight and corrosion-free.
9. Also check secondary ground strap between engine and chassis (engine-to-body ground) — this is critical for starter current return path.
10. **Voltage Drop Test (Cranking Circuit):** Disable ignition or fuel system to prevent start (for safety and isolation).
11. Set multimeter to DC volts.
12. **Positive side drop:** Place positive probe on battery positive post, negative probe on starter solenoid terminal.
13. Crank engine and read voltage drop: should be $\leq 0.5V$.
14. **Negative side drop:** Place positive probe on starter housing (clean metal surface), negative probe on battery negative post.
15. Crank engine: drop should be $\leq 0.2V$.
16. Higher drops indicate high resistance connections that need cleaning or cable replacement.
17. **Starter Current Draw Test (if available):** Use inductive ammeter clamp.
18. Crank engine and read current.
19. Typical diesel starter: 200-400 amps initial, 100-200 amps cranking.
20. Excessively high current may indicate bad starter, seized engine, or failing battery.
21. **Pinion Gear Check:** If starter is removed for service, inspect the Bendix drive gear teeth for chipping or wear.

22. Inspect ring gear teeth (access through starter opening) for damage.

23. Lightly lubricate Bendix drive mechanism with high-temp grease (do not over-lubricate).

Acceptance Criteria: Starter: securely mounted, no physical damage. Cables: no corrosion, fraying, or melted insulation. Voltage drop: positive $\leq 0.5V$, negative $\leq 0.2V$. Engine cranks briskly. Current draw within normal range.

⚠ **SAFETY:** Disable the ignition system or fuel shutoff before performing cranking tests to prevent accidental engine start. The starter circuit carries several hundred amps — short circuits can cause arc flashes, burns, and fire. Use insulated tools. Never bypass the neutral safety switch on automatic transmissions.

Tip: A slow crank is most often caused by a bad connection (corrosion or loose terminal), not a bad starter. Always check voltage drops before replacing the starter. If the starter solenoid clicks but the engine does not crank, suspect the battery or main cable connection first.

Documentation: Record: visual condition, voltage drop measurements (V+ and V-), current draw reading (if measured).

Performing Mechanic

8.20 Lighting Systems Inspection

Full Lighting System Inspection

Purpose: Inspect all exterior lighting to ensure full compliance with road safety regulations — all lights must be functional, properly aimed, and with correct color lenses.

Frequency: Daily (pre-trip) via walk-around

Scope: All trucks

PPE Required: High-visibility vest, safety glasses (for night inspections)

Tools & Materials: Assistant or reflective surface (for checking brake/tail lights), flashlight, bulb replacement kit (assorted 12V/24V bulbs), electrical contact cleaner

Step-by-Step Procedure:

1. **Set up:** Park vehicle, engage parking brake.
2. Turn headlights ON (low beam).
3. Walk completely around the vehicle.
4. **Front Lighting:** Verify headlights (low beam and high beam) are both operational.
5. Verify parking lights (front position lights) are illuminated.
6. Verify turn signals (left and right) flash at correct rate (60-120 flashes per minute).
7. No hyper-flash (rapid flashing) — indicates a burned-out bulb on that circuit.
8. **Side Lighting:** Verify side marker lights (amber front, red rear) on both sides.
9. Check clearance lights on cab roof and tank top (if equipped).
10. **Rear Lighting:** Tail lights: both must be illuminated with headlights on.
11. Brake lights: have assistant press brake pedal or use reflective surface.
12. Both brake lights must illuminate brighter than tail lights.
13. Turn signals: confirm rear indicators flash correctly (amber or red depending on regulation).
14. License plate light: must illuminate plate.
15. Reverse light(s): engage reverse and verify.
16. Hazard warning lights: activate hazard switch and confirm all four turn signals flash simultaneously.
17. **Instrument Panel:** Verify all dashboard warning lights function on bulb check (key-on, engine-off position).
18. Check dome/cab interior light.
19. Check all switches are illuminated (if backlit switches are equipped).
20. **Light Condition:** Check lenses for cracks, moisture ingress (fogging or water inside the housing), hazing or yellowing (reduces light output — polish or replace).
21. Check housing mounting for loose or broken brackets.
22. Check wiring and connectors at each light assembly for corrosion or damage.

23. **Aiming (as needed):** Park 25 feet from a wall on level ground.
24. Mark the center of each headlight beam on the wall.
25. Low beams should be aimed slightly downward and to the right (driver side slightly lower).
26. Adjust using aiming screws on headlight housing.

Acceptance Criteria: All lights operational: headlights (low + high), tail lights, brake lights, turn signals (front and rear), hazard lights, marker/clearance lights, license plate light, reverse light. Flash rate 60-120/min. No burned-out bulbs. No cracked lenses. No water inside housings. Headlights properly aimed.

⚠ **SAFETY:** Driving with defective lighting is a traffic violation and safety hazard. A burned-out taillight or brake light can lead to rear-end collision. If any required light is non-functional, the truck should not be dispatched until repaired.

Tip: Keep a stock of common bulb types on hand: 1157 (dual-filament tail/brake), 1156 (single-filament turn), 168/194 (marker lights), H4 or 9004 (headlight bulbs). For LED-equipped trucks, carry replacement LED modules. Check trailer connector (if equipped) for pin corrosion.

Documentation: Record any burned-out bulbs replaced, any lenses replaced, any aiming adjustments made.

Performing Mechanic or Driver

8.21 Compliance & Regulatory Inspection

Regulatory Compliance and Documentation Inspection

Purpose: Verify the truck carries all required legal and regulatory documentation, stickers, permits, and markings necessary for operation on public roads and access to petroleum depots.

Scope: All trucks

Frequency: Monthly or upon any registration change

PPE Required: None required

Tools & Materials: Checklist, flashlight (for hard-to-read stickers)

Step-by-Step Procedure:

1. **LTO Registration (Official Receipt / Certificate of Registration):** Confirm the original OR/CR documents or authorized copies are in the truck's document pouch.
2. Verify registration expiry date — must not be expired.
3. Confirm plate number on documents matches the truck.
4. **Insurance:** Verify Certificate of Insurance (COI) or insurance policy is in the document pouch.
5. Check coverage dates — must be current.
6. Verify insurance type: comprehensive or third-party liability (TPL) as required.
7. Check insurance provider's name, policy number, and vehicle details.
8. **Vehicle Inspection Report (VIR) / Private Motor Vehicle Inspection (PMVI):** Verify current PMVI sticker is displayed on the windshield.
9. Check inspection expiry date.
10. If the truck has been inspected by LTO-accredited inspection center, ensure the report is in the document pouch.
11. **DOST / PSTC Stickers:** Check for DOST (Department of Science and Technology) or PSTC stickers for calibration.
12. Verify calibration validity dates for: tank compartment capacity (certified tank volume), meter calibration (if meter-equipped), and weighbridge certificate (if applicable).
13. **Depot Access Requirements:** Verify required documents specific to petroleum depots (typically: valid insurance, valid registration, HAZMAT compliance, vapor recovery certificate, fire extinguisher tag, speed limiter certificate if required).
14. Check that the truck's depot access card or transponder is present and functional.
15. **Speed Limiter:** If the truck is equipped with a speed limiter, verify the seal or tamper indicator is intact.
16. Note the set speed on the verification tag.
17. Typical setting: 80 km/h for heavy trucks.
18. **Tollway RFID / Autosweep / Easytrip:** Verify RFID tags are properly mounted and functional.
19. Check for loose or detached tags.

20. **HAZMAT / Dangerous Goods:** Verify driver has valid HAZMAT training certificate (if required for the product).
21. Confirm HAZMAT registration and permit documents are current.
22. Ensure UN marking panels are present and correct for the product.
23. **Compliance Plate:** Check the vehicle's compliance plate or data plate on the chassis — must be present and legible with correct GVW, GAWR, VIN.

Acceptance Criteria: All documents present, valid (not expired), and in good condition. All required stickers displayed. Depot access requirements met. HAZMAT documentation current. Toll tags functional.

⚠ **SAFETY:** Operating a truck with expired registration, insurance, or permits is a serious legal violation that can result in impoundment, fines, and liability exposure. Maintenance supervisors must maintain a compliance calendar to track all expiry dates.

Tip: Create a compliance calendar with 30-day advance alerts for each document expiry. Use the truck detail page in the system to record document numbers and expiry dates for quick reference. Take photos of documents and store them digitally for backup.

Documentation: Record each document's expiry date, any documents replaced or renewed, depot access card/VT number, speed limiter seal condition.

Lead Mechanic or Fleet Supervisor

8.22 Drivetrain — Differential, U-Joints, Wheel Bearings

Drivetrain System Inspection (Differential, U-Joints, Wheel Bearings)

Purpose: Inspect the complete drivetrain from transmission output to wheel hubs, checking for wear, lubrication condition, and mechanical integrity that could lead to catastrophic failure.

Scope: All trucks

Frequency: Every 20,000 km or 6 months

PPE Required: Safety glasses, gloves, hearing protection (road test)

Tools & Materials: Flashlight, pry bar, grease gun (with lithium complex or moly grease), torque wrench, 3/8-inch drive ratchet with extensions

Step-by-Step Procedure:

- 1. U-Joint Inspection:** With truck on level ground, transmission in neutral, parking brake OFF, chock all wheels.
2. Crawl under truck.
3. Inspect each U-joint (typically 2-3 on driveline: transmission output, carrier bearing, differential input).
4. Grasp driveline near each U-joint and attempt to twist — there should be ZERO rotational play.
5. Attempt to move driveline up and down sideways near U-joint — maximum acceptable play: 0.010 inch (0.25mm).
6. Check U-joint retaining clips or strap bolts — must be tight.
7. Look for rust-colored dust around U-joint cross (indicates wear).
8. Grease each U-joint (if equipped with grease fitting) until fresh grease purges from all four seals.
- 9. Center / Carrier Bearing:** Inspect rubber cushion for cracking or deterioration.
10. Check bearing for smooth rotation — spin by hand; should feel smooth with no roughness.
11. Inspect mounting bracket for cracks.
- 12. Differential Inspection:** Check differential oil level: remove fill plug (top of differential housing).
13. Oil should be level with the bottom of the fill hole.
14. If low, top off with correct grade gear oil (typically 80W-90 or 85W-140 for severe service).
15. Check oil condition: dip a clean finger into fill hole — oil should be clear to amber-brown.
16. Milky, white, or grey oil indicates water contamination.
17. Black, burnt-smelling oil indicates overheating.
18. Inspect differential housing for leaks at: pinion seal (front, where driveline enters), axle seals (inner wheel ends), cover gasket, vent tube (must be clear and routed upward).
- 19. Axle U-joints:** If equipped with independent suspension, inspect half-shaft U-joints or CV joints.

20. Check boots for tears and grease leakage.
21. **Road Test:** Drive truck in a straight line on a smooth road at 30-40 km/h.
22. Listen for: whining or howling from differential (bearing or gear wear), clicking from U-joints (worn cross bearings), clunking on acceleration/deceleration (excessive driveline backlash), vibration at specific speeds (driveline balance or worn carrier bearing).
23. While driving, gently turn left and right — listen for clicking from front CV joints (if equipped).

Acceptance Criteria: U-joints: zero rotational play, minimal lateral play (<0.010 inch). Carrier bearing: smooth rotation, no cracked rubber cushion. Differential: oil level correct, oil clean, no leaks. Road test: no unusual noises or vibrations.

⚠ **SAFETY:** A driveline failure at highway speed can separate from the vehicle, causing loss of control and potential injury. If any U-joint shows significant play or roughness, replace immediately. A driveline vibration that suddenly worsens may indicate imminent bearing failure.

Tip: Most driveline vibration issues are caused by worn U-joints rather than out-of-balance shafts. Check U-joints first. When replacing U-joints, replace all on the same driveline (transmission output and both differential ends). Use only grease-able U-joints with a proper grease schedule.

Documentation: Record: U-joint condition, differential oil level and condition, any leakage found, grease points serviced, road test results.

Performing Mechanic → Lead Mechanic verification

8.23 Water Brake System Inspection

Water Brake (Retarder) System Inspection

Purpose: Inspect the water brake hydraulic retarder system, including pump, valve, lines, and cooler, to ensure it provides supplemental braking power for heavy loads on grades.

Frequency: Monthly or every 10,000 km

Tools & Materials: Flashlight, wrenches, rag, coolant refractometer (if water brake shares engine coolant)

Scope: Trucks equipped with water brake / hydraulic retarder systems

PPE Required: Safety glasses, gloves, hearing protection

Step-by-Step Procedure:

1. **System Overview:** Identify the water brake components: hydraulic pump (usually on front of engine or PTO-driven), control valve (cab-mounted or chassis-mounted), lines/hoses, heat exchanger (radiator or shell-and-tube type), and actuation cylinder (driveline or transmission-mounted).
2. **Fluid Level Check:** Locate the hydraulic fluid reservoir.
3. Check fluid level — should be between MIN and MAX marks.
4. Fluid should be clear amber (hydraulic oil) or green (approved coolant, depending on system type).
5. If fluid is milky, dark, or smells burnt, a complete fluid change is needed.
6. Note the fluid type required (typically AW-32 or AW-46 hydraulic oil, or proprietary fluid).
7. **Pump Inspection:** Inspect pump mounting bolts — must be tight.
8. Check pump drive coupling for wear (if accessible).
9. Inspect pump shaft seal for leaks.
10. Listen for pump operation with engine at idle — should produce a steady, smooth sound without cavitation (rattling/gravelly sound indicates low fluid or air in system).
11. **Hose and Line Inspection:** Trace all hydraulic lines from pump to control valve to heat exchanger to actuation cylinder.
12. Inspect for: chafing, cracks, bulges, leaks at fittings, collapsed sections (restricted flow).
13. Pay special attention to areas near exhaust or other heat sources.
14. **Heat Exchanger / Cooler:** Inspect for debris blockage on fins (if air-cooled).
15. Check for coolant leaks (if water-cooled via engine system).
16. Verify cooler mounting brackets are intact.
17. **Control Valve:** Test control lever or switch in cab — must move through full range smoothly.
18. Verify valve linkage is properly adjusted (cable or rod).
19. Inspect valve body for external leaks.
20. **Actuator / Caliper:** Inspect actuation cylinder — check for leaks at rod seal, check for smooth rod movement.

21. If water brake acts on a driveline disc, inspect the disc surface for scoring, heat checking, or warpage.
22. Check disc thickness and compare to minimum specification.
23. **Test Operation:** In a safe area with adequate stopping distance, drive truck to 30 km/h and apply water brake (50% engagement).
24. Truck should decelerate smoothly without pulling or vibration.
25. Overheating after 1 minute of continuous use at 50% engagement is normal — full engagement should not be held for more than 30 seconds.
26. Verify that the system disengages fully when control is released.

Acceptance Criteria: Fluid level correct and fluid condition good. No leaks at any connection. Pump operates smoothly without unusual noise. Hoses intact with no chafing or bulges. Control linkage operates through full range smoothly. Braking: smooth deceleration without pulling or vibration. System disengages fully on release.

⚠ **SAFETY:** Water brake systems can generate extreme heat during sustained use. If the red warning light or temperature gauge indicates overheating, release the brake and allow system to cool. Do not touch hydraulic components immediately after heavy use — they can be hot enough to cause burns.

Tip: The water brake is a supplemental braking system only. It is not a parking brake. Always use the spring parking brake for parking. For trucks hauling heavy loads in mountainous terrain, the water brake should be applied continuously to maintain speed on downgrades, preserving the service brakes for emergency stops.

Documentation: Record fluid level, condition, any leaks repaired, heat exchanger condition, road test result.

Performing Mechanic

8.24 Transmission & Clutch Inspection

Transmission Oil, Clutch System, and Clutch Booster Inspection

Purpose: Inspect the transmission oil level and condition, clutch adjustment, and the clutch booster (air/hydraulic servo) system. Identify early warning signs of booster failure such as air leaks, hydraulic leaks, and pedal feel changes before catastrophic failure occurs.

Frequency: Every 10,000 km or 3 months

Tools & Materials: Flashlight, 3/8-inch ratchet set, clutch adjustment gauge or ruler, soapy water spray bottle, air pressure gauge, brake fluid (DOT 3/4), transmission oil (correct grade), drain pan, rag

Scope: All trucks with manual transmissions and air/hydraulic clutch boosters

PPE Required: Safety glasses, gloves, hearing protection (road test), steel-toe boots

Step-by-Step Procedure:

- 1. Transmission Oil Level Check:** Park truck on level ground, engine off. Locate the transmission fill plug (typically on the side of the transmission case). Clean around the plug before removing. Remove fill plug and check oil level — oil should be level with the bottom of the fill hole. If low, add correct grade oil (typically 80W-90 gear oil or 10W-30/15W-40 engine oil, depending on transmission type). Check oil condition: dip a clean finger into the fill hole. Oil should be clear amber to light brown. Milky or white oil indicates water contamination. Burnt smell or dark/black oil indicates overheating or wear. If contaminated, schedule transmission oil change. Reinstall fill plug and torque to specification.
- 2. Clutch Pedal Free Play Check:** With engine off, gently press the clutch pedal by hand until you feel the first resistance (the point where the release bearing contacts the diaphragm spring fingers). Measure the distance the pedal travels from its resting position to this contact point. Standard free play: 25-40mm (1 to 1.5 inches) for most heavy truck clutches. If free play is excessive (>40mm), the clutch may not fully disengage (hard shifting, grinding). If free play is insufficient (<25mm), the release bearing may be pre-loaded, causing premature bearing failure and partial clutch engagement. Adjust the clutch linkage or slave cylinder pushrod to bring free play within specification.
- 3. Clutch Booster (Air/Hydraulic Servo) — Visual Inspection:** Locate the clutch booster — typically mounted on the bell housing or frame rail near the transmission. Inspect for: oil or hydraulic fluid leakage around the booster body and connections, air leakage around the booster (listen for hissing with engine running and clutch depressed), rust or corrosion on the housing, cracks or dents in the housing, loose mounting bolts, damaged air hoses, damaged hydraulic line, torn rubber boots (if equipped). If any leakage, cracks, or severe corrosion are present, replace the booster before failure occurs.
- 4. Clutch Booster — Listen for Air Leaks:** Start the engine and let air pressure build to full governor cut-out (120-135 psi). With engine running, press and release the clutch pedal several times. Listen for continuous hissing after the pedal is released. A brief hiss during pedal movement is normal (air being admitted to the booster). Continuous hissing indicates: internal seal failure, diaphragm leak, or air valve leakage. These are early warning signs that the

booster will fail soon. Use soapy water spray at the booster air inlet, exhaust port, and hose connections to pinpoint the leak. Bubbles indicate the leak location.

5. **Clutch Pedal Feel Test:** With engine running, operate the clutch through a full cycle. Note any changes in feel: pedal becoming harder than normal (indicates booster not providing full assist — possible air pressure issue or internal restriction), pedal becoming softer than normal (indicates internal bypass or hydraulic leak), delayed clutch release (pedal comes up slowly or hesitates — indicates sticky booster valve or restricted exhaust), jerky engagement (chattering — indicates uneven pressure plate release or worn clutch disc), pedal not returning fully to rest position (indicates weak return spring or sticky booster). Any of these symptoms should trigger immediate further inspection and booster replacement if confirmed.
6. **Air System Health Check:** Since the clutch booster depends on compressed air, inspect: air pressure regulator setting (should match manufacturer specification, typically 100-120 psi to booster), compressor output, air dryer operation (service per manufacturer interval), drain air tanks fully — check for excessive moisture or oil contamination in the air system. Water inside the air system is one of the biggest contributors to premature booster failure (causes internal corrosion and seal degradation).
7. **Hydraulic System Check:** Trace the hydraulic line from the clutch master cylinder (mounted on the firewall, connected to the pedal) to the clutch booster. Check: master cylinder for external leakage at the pushrod seal and hose fitting, fluid level in the master cylinder reservoir, fluid condition (should be clear amber — if dark or contaminated, flush and replace), flexible hoses for chafing, cracks, or bulging, slave cylinder (if separate from booster) for leaks and pushrod travel. Low fluid or contaminated fluid shortens booster life and can cause sudden failure.
8. **Clutch Adjustment Verification:** A clutch that is out of adjustment forces the booster to travel farther and operate under higher loads than designed. During PM: measure clutch free play at the pedal (as above), verify release bearing clearance (should be 2-3mm gap between bearing and diaphragm spring fingers when clutch is engaged), check clutch brake operation (if equipped — pushes against input shaft to stop gear rotation for easier shifting). Adjust according to the manufacturer's specification.
9. **Transmission Mounts:** Inspect all transmission mounts for cracks, deterioration, or looseness. Tighten any loose mounting bolts. Worn mounts cause misalignment that stresses the clutch linkage and booster.
10. **Road Test:** Drive the truck in a safe area. Test: clutch engagement smoothness in all forward gears, easy gear engagement without grinding (especially reverse and first gear — difficult engagement suggests incomplete clutch release), clutch chatter or vibration during take-off, clutch slip (engine RPM increases without corresponding speed increase, especially under load in higher gears), unusual transmission noises (whining, howling, clunking). Record any abnormalities for further investigation.

Acceptance Criteria: Transmission oil at correct level, condition clear amber, no burnt smell. Clutch free play 25-40mm. Clutch booster: no external leaks, no continuous air hissing, no corrosion or damage. Pedal feel smooth and consistent through full travel. Air system dry (no

excessive moisture). Hydraulic fluid clean and at correct level. Road test: smooth engagement, no slipping, no unusual noises, all gears engage easily.

⚠ **SAFETY:** A failed clutch booster can rupture explosively, sending metal fragments and hydraulic fluid at high velocity. If the booster shows signs of severe corrosion, a cracked housing, or a torn diaphragm, replace immediately. Do not attempt to repair a failed booster — replace as a complete unit. Wear safety glasses when working near the booster during system pressure tests. The air system can store high pressure even with the engine off — depressurize before disconnecting any air lines.

Tip: If one clutch booster has failed in the fleet, investigate the root cause: is air pressure exceeding specification? Is the air dryer overdue for service? Are air tanks accumulating water? Are clutches improperly adjusted? Are the boosters all of similar age? If multiple trucks have boosters of similar age, proactive replacement may be more cost-effective than waiting for in-service failures.

Documentation: Initial and transmission oil level at condition, clutch free play measurement, booster visual condition (pass/fail and any notes), air system dryness observation, clutch engagement smoothness, road test observations.

Mechanic → Lead Mechanic verification

8.25 HVAC System Inspection

Heating, Ventilation, and Air Conditioning System Inspection

Purpose: Inspect the HVAC system including air conditioning compressor, condenser, evaporator, heater core, blower motor, and controls to ensure cab comfort and demisting/defrosting capability for safe operation.

Scope: All trucks

Frequency: Every 20,000 km or 6 months (pre-season AC check before summer)

PPE Required: Safety glasses, gloves (refrigerant-safe when handling AC)

Tools & Materials: AC manifold gauge set (R-134a or R-1234yf), thermometer (vent probe type), multimeter, flashlight, UV leak detection kit (if available), belt tension gauge

Step-by-Step Procedure:

- 1. AC Compressor Inspection:** With engine off, inspect compressor body for oil leaks (refrigerant oil leakage appears as dark oily residue around compressor seals and fittings). Check compressor clutch operation: have an assistant turn AC ON and OFF while you observe the clutch plate. The clutch should engage with a crisp click and the plate should spin with the pulley when engaged. Listen for abnormal compressor noises when engaged: rattling, screeching, or continuous cycling (short cycling) indicate problems. Check compressor mounting bolts — must be tight. Check compressor belt condition and tension (see Section 8.4 for belt inspection).
- 2. Condenser Inspection:** Inspect the condenser (mounted in front of the radiator) for: bent or damaged fins, debris blockage between fins (bugs, leaves, mud), corrosion or refrigerant oil residue (indicates leak), physical damage from road debris. Clean the condenser fins gently with compressed air (blowing from the back toward the front) or low-pressure water. Do not use high pressure — this can bend the fins and reduce efficiency.
- 3. AC Performance Test:** Start engine, set to fast idle (~1500 RPM). Turn AC to MAX cold, blower to HIGH, and close all windows. Place a vent thermometer in the center dash vent. Let the system stabilize for 5 minutes. Measure the vent temperature. Center vent temperature should be 4-10°C (40-50°F) in moderate ambient conditions (25-35°C / 77-95°F). If vent temperature is above 15°C (60°F), the system likely needs service. If equipped, connect manifold gauges: low side should read 15-40 psi, high side 150-250 psi (varies with ambient temperature). Compare readings to manufacturer specifications.
- 4. Cabin Air Filter:** Locate the cabin air filter (typically behind the glove box or under the dashboard). Remove and inspect the filter. Replace if dirty, clogged, or has a musty odor. Recommend replacement every 20,000 km or annually.
- 5. Heater Operation:** Test heater by setting temperature to MAX hot, fan to HIGH. Verify that warm air flows from the vents within 5 minutes of engine start (on a cold engine). If insufficient heat, check: coolant level in the cooling system, heater control valve operation (cable or vacuum-actuated), blend door operation (listen for movement when changing temperature), and coolant flow through the heater core — feel both heater hoses; both should be hot when the heater is on.

6. **Blower Motor Inspection:** Run the blower fan at all speed positions (1, 2, 3, 4, MAX). Verify airflow at all speeds. If one or more speeds are missing, the blower motor resistor may be faulty. Listen for: bearing noise (whining or grinding) indicates failing blower motor, dirt or debris on the blower wheel causing imbalance (vibration or wobble sound), foreign objects inside the duct (rattling). Ensure air flows correctly from defrost, dash, and floor positions when selected.
7. **AC Refrigerant Leak Check:** If system performance is low, check for refrigerant leaks using: electronic leak detector (sniffer), UV dye (inject with approved dye kit and inspect with UV light), soapy water spray at all service ports and fittings (bubbles indicate leak). Common leak points: compressor shaft seal, hose connections at condenser and evaporator, Schrader valves at service ports, and O-rings at line connections.
8. **Control and Switch Operation:** Operate all HVAC controls: temperature slider/knob — smooth operation throughout range, mode selector (defrost/panel/floor/bi-level) — correct airflow at each position, recirculation button — should hear blend door move, AC button — compressor clutch engages/disengages, rear defrost (if equipped) — indicator light and grid operation.
9. **AC System Pressure Test (if available):** Connect manifold gauges and record static pressure (engine off, system equalized). Static pressure should be proportional to ambient temperature (approximately 75-100 psi at 30°C / 85°F). If static pressure is very low (<50 psi), the system is low on refrigerant and needs service. Record high and low side operating pressures for diagnosis.

Acceptance Criteria: AC compressor clutch engages and disengages cleanly. No abnormal noises from compressor or blower. Center vent temperature 4-10°C (40-50°F) at max cooling. Heater produces warm air within 5 minutes. Blower fan operates at all speeds. Airflow correct at each mode position. No unusual odors from vents. Cabin air filter clean or replaced. No refrigerant leaks detected.

⚠ **SAFETY:** Refrigerant systems operate at high pressure (150-250+ psi on the high side). Never open the AC system to the atmosphere. Refrigerant can cause frostbite on contact with skin or eyes. Only certified technicians should recover, evacuate, and recharge refrigerant. Use approved recovery equipment.

Tip: The most common cause of poor AC performance on trucks is a dirty condenser (blocked by road debris and bugs). Clean the condenser before assuming refrigerant is low. Second most common is a failing compressor clutch that slips under load.

Documentation: Initial center vent temperature at ambient temperature, AC pressures (static at operating), visual condition of compressor at condenser, cabin air filter condition, blower motor operation at all speeds, heater operation.

Mechanic na Gumawa

8.26 Hydraulics System Inspection

Hydraulic System Inspection (Pumps, Cylinders, Valves, Lines)

Purpose: Inspect all hydraulic system components on the truck including pumps, cylinders, valves, hoses, and fittings for leaks, wear, and correct operation to prevent system failure and potential safety hazards.

Frequency: Monthly or every 10,000 km

Tools & Materials: Flashlight, rag, hydraulic fluid (correct grade), pressure gauge (if test port available), multimeter (for electric-over-hydraulic systems), wrenches

Scope: Trucks equipped with hydraulic systems (lift gates, PTO-driven pumps, hydraulic brakes, hydraulic fans)

PPE Required: Safety glasses, chemical-resistant gloves, steel-toe boots

Step-by-Step Procedure:

- 1. Fluid Level and Condition Check:** Park on level ground with engine off and all hydraulic functions in the OFF/NEUTRAL position. Locate the hydraulic fluid reservoir. Clean the cap area before opening. Check fluid level: should be between MIN and MAX marks (or within sight glass range). Fluid condition: new hydraulic fluid is clear amber (AW-32, AW-46, or 10W depending on system). Milky, foamy, or gray fluid indicates water or air contamination — causes pump cavitation and component wear. Dark or black fluid with burnt smell indicates overheating and oxidation — reduces lubrication and causes varnish buildup on valves. If fluid is contaminated, schedule a complete fluid change and filter replacement. Note the correct fluid type required (stamped on reservoir or in the manual).
- 2. Hydraulic Pump Inspection:** Inspect pump mounting bolts — must be tight. Check pump shaft seal for leakage (drips or wetness around the shaft). Check pump drive coupling (if accessible): look for wear, cracks, or misalignment, if PTO-driven — check PTO mounting bolts and driveline angle. Listen for pump operation: with engine at idle, cycle the hydraulic function. The pump should produce a steady, smooth sound. Cavitation (rattling/gravelly sound) indicates low fluid, restricted inlet, or air ingestion. Excessive whining or screeching indicates worn internal components or low fluid. High-pitched squealing on startup may indicate cold, thick fluid or a restricted inlet filter.
- 3. Hydraulic Cylinder Inspection:** Inspect each hydraulic cylinder externally: check the rod for scoring, pitting, rust, or chrome flaking (damaged rod seals will fail quickly). Check the rod seal for leakage (oil film or drips at the rod wiper seal area). Minor sweaty film may be acceptable; active dripping requires seal replacement. Inspect the cylinder barrel for dents, corrosion, or weld cracks. Check cylinder pivot pins and bushings for wear. Grease all cylinder pivot points. Verify that cylinders fully extend and retract without binding or hesitation.
- 4. Valve Inspection:** Operate each control valve through its full range. The valve lever/spool should move smoothly with moderate resistance — not too tight, not sloppy. Check for external leakage at valve spool seals and hose connections. If applicable: check solenoid valve electrical connections for corrosion and loose terminals. Verify that the valve returns to NEUTRAL when released (spring-centered valves). If the valve spool sticks or requires excessive force, the valve may need disassembly, cleaning, or replacement.

- 5. Hose and Line Inspection:** Trace all hydraulic hoses and rigid lines from the pump to the control valve to the cylinder and back to the reservoir. Inspect for: chafing or abrasion wear (especially where hoses rub against frame members or other hoses), cuts, cracks, or bulges in the hose cover, soft spots (indicate internal ply separation), leaks at fittings (tighten or replace as needed), corrosion on rigid tubing, collapsed or kinked sections (restricted flow). Pay special attention to areas near exhaust components, moving parts, and sharp edges. Replace any hose showing signs of wear, damage, or deterioration.
- 6. Filter and Breather Check:** Locate the hydraulic return filter and/or pressure filter. Check filter condition indicator (if equipped) — replace if in the red zone. If no indicator, follow the manufacturer's recommended replacement interval (typically 500-1000 hours or annually). Clean or replace the reservoir breather cap/filter to prevent contamination ingress.
- 7. System Pressure Test (if applicable):** Connect a pressure gauge to the system test port (if available). Operate the hydraulic function at idle and at governed RPM. Compare the pressure reading to manufacturer specification. Low pressure indicates: pump wear, pressure relief valve set too low (adjust if adjustable), internal leakage past cylinder seals or control valve spools, or a slipping pump drive. High pressure or pressure spikes may indicate: relief valve stuck closed or restricted return line.
- 8. Leak-down Test:** Fully extend all cylinders, stop the engine, and observe. Cylinders should hold position without drifting for at least 5 minutes. Rapid drift indicates internal seal leakage past the piston seals or control valve spool bypassing. Note which circuit drifts for diagnosis.

Acceptance Criteria: Fluid level between MIN and MAX, fluid clear amber (no milkiness, foaming, or burnt smell). Pump operates smoothly without cavitation or excessive noise. Cylinder rods undamaged, rod seals not actively leaking. Control valves operate smoothly through full range. No hose chafing, cracks, bulges, or leaks. Filters within service interval. System holds pressure without excessive leak-down.

⚠ SAFETY: Hydraulic systems can store high pressure even with the engine off. Always relieve system pressure before disconnecting any line or fitting. Escaping hydraulic fluid under pressure can penetrate the skin and cause severe injury or death. Use a piece of cardboard (not your hand) to check for leaks. Wear chemical-resistant gloves when handling hydraulic fluid — it is an irritant and can cause dermatitis with prolonged exposure.

Tip: Contamination is the #1 cause of hydraulic system failure. Keep the reservoir cap clean, change filters on schedule, and use clean fluid from a sealed container. When adding fluid, always pour through a filter funnel (100 mesh minimum). A 10-micron absolute filter on the return line is the best insurance for system longevity.

Documentation: Initial fluid level at condition, pump operation observation, cylinder condition per location, valve operation, hose condition (especially chafed hoses requiring replacement), filter service hours, pressure test readings, leak-down test results.

Mechanic → Lead Mechanic verification

8.27 Suspension System Inspection

Suspension System Inspection (Leaf Springs, Shock Absorbers, Bushings)

Purpose: Inspect the suspension system including leaf springs, shock absorbers, spring bushings, U-bolts, and suspension brackets for wear, damage, and loose components that affect ride quality, stability, and tire wear.

Scope: All trucks

Frequency: Every 20,000 km or 6 months

PPE Required: Safety glasses, gloves, steel-toe boots

Tools & Materials: Flashlight, pry bar, torque wrench (high-range), wire brush, hammer, ruler or tape measure

Step-by-Step Procedure:

- 1. Visual Walk-Around:** Park on level ground with full fuel and no payload (curb weight). Walk alongside the truck and visually inspect both sides of the suspension. Note any obvious sagging, leaning, or misalignment. Compare left to right ride height at the frame rail above each axle — height should be equal within 10mm.
- 2. Leaf Spring Inspection:** Inspect each leaf spring pack carefully: check for broken or cracked leaves — look for gaps between leaves that are wider at one end than the other (indicates a broken leaf hidden between leaves). Check for shifted leaves (a leaf that has moved sideways out of alignment with the rest of the pack). Inspect the main (longest) leaf at the spring eye — cracks here can lead to complete spring failure. Check for interleaf wear (polished spots between leaves indicate excessive friction). Check for rust pitting on spring surfaces — deep pitting reduces spring life. Measure ride height at each axle: compare to manufacturer specification. Sagging springs reduce load capacity and cause bottoming out.
- 3. Spring Bushings and Pins:** Check the spring eye bushings (front and rear of each spring pack) for wear: insert a pry bar between the spring eye and the frame bracket and attempt to move the spring sideways — excessive movement (>3mm / 1/8 inch) indicates worn bushings. Check the spring pivot pin for wear and secure retention. Grease the spring pins (if equipped with grease fittings) until fresh grease purges.
- 4. U-Bolt Inspection:** Inspect U-bolts on each axle: check for looseness (use a torque wrench — typical torque: 300-500 Nm depending on axle capacity). Check for corrosion damage — deeply pitted or rusted U-bolts should be replaced. Check for thread damage — replace if threads are stripped or cross-threaded. Verify that U-bolt nuts are present and torqued evenly on both sides. Re-torque U-bolts after the first 500 km following any suspension work or spring replacement.
- 5. Shock Absorber Inspection:** Inspect each shock absorber: check for external oil leakage (dampness or dripping at the shaft seal area — light oil film is acceptable, active dripping is not). Check the shock body for dents or damage. Check the mounting bushings at both ends — worn bushings cause knocking noises and reduced damping effectiveness. Check mounting bolts — must be tight. Perform a bounce test: push down firmly on the bumper or frame rail at each corner and release. The truck should settle within 1-2 oscillations. Excessive bouncing (3+ cycles) indicates worn shocks. For cab-mounted shock absorbers (on cab suspension), check

the same items. Cab shock wear causes excessive cab movement, driver discomfort, and possible cab mount damage.

- 6. Suspension Bracket and Frame Inspection:** Inspect the spring hanger brackets (frame-mounted brackets that hold the spring eyes). Check for cracks at the welds — especially at the rear spring hanger where loads are highest. Check for elongated bolt holes (indicates loose U-bolts or hanger bolts). Check the equalizer bracket (on tandem axle suspensions) for wear at the pivot point and bushing condition. Inspect the frame rail near suspension attachment points for cracks (especially at the crossmember connections adjacent to spring hangers).
- 7. Stabilizer Bar (Anti-sway Bar) Inspection:** If equipped, inspect the stabilizer bar and link assemblies: check stabilizer bar bushings for wear or deterioration (cracked or missing rubber). Check stabilizer end links: move by hand to check for play — worn links cause clunking noises during turns and uneven braking. Check link mounting bolts for tightness.
- 8. Air Suspension (if equipped):** For trucks with air suspension, also inspect: air springs (air bags) — look for cracks, rubbing wear, or bulges on the bag. Check air lines for chafing and secure routing. Check the height control valve linkage — should move freely and maintain the correct ride height. Check the leveling valve for air leakage. Measure ride height from the axle to frame — compare to manufacturer specification. Check the air dryer that supplies the suspension air system.

Acceptance Criteria: No broken or cracked leaves in any spring pack. Ride height equal left-to-right within 10mm. Spring eye bushings show no excessive play (<3mm lateral movement). U-bolts torqued to specification. Shock absorbers not actively leaking, mounting bushings intact. No cracks at suspension bracket welds. For air suspension: air bags intact, correct ride height, no air leaks.

⚠ **SAFETY:** A broken leaf spring can cause sudden loss of ride height, tire contact with the body, or complete axle separation. Replace cracked or broken springs immediately. Never mix spring packs from different manufacturers on the same axle. When replacing U-bolts, always use new hardware — never reuse U-bolts that have been torqued.

Tip: Leaf springs should be replaced in pairs on the same axle. Mixing a new spring with an old one on the same axle will cause uneven load distribution, handling issues, and premature wear of the new spring. If one spring in a pack is broken, the entire pack should be replaced.

Documentation: Itala ang ride height measurements, leaf spring condition per position, bushing wear observations, shock absorber condition, U-bolt torque readings, suspension bracket crack check results.

Mechanic na Gumawa

8.28 Exhaust System Inspection

Exhaust System Inspection (Pipes, Muffler, DPF, Hangers)

Purpose: Inspect the exhaust system from the engine manifold to the exhaust outlet for leaks, damage, corrosion, and secure mounting. Exhaust leaks can cause asphyxiation risk in the cab, reduce engine performance, and are a noise and emissions violation.

Scope: All trucks

Frequency: Every 20,000 km or 6 months

PPE Required: Safety glasses, gloves (heat-resistant for hot components), hearing protection

Tools & Materials: Flashlight, rubber mallet, 14mm/16mm wrenches, penetrating oil, wire brush, mirror on a stick

Step-by-Step Procedure:

- 1. Preparation:** Park the truck and allow the exhaust system to cool completely — exhaust temperatures can exceed 300°C (570°F) and cause severe burns. Perform the inspection on a cool engine.
- 2. Visual Walk-Under:** Crawl under the truck and inspect the entire exhaust system from the exhaust manifold flange to the tailpipe. Look for: soot marks around joints and gaskets (indicate exhaust leaks), blue/black soot on the ground underneath the truck, physical damage (dents, crushed sections), corrosion (especially at pipe welds and clamp areas), green/white corrosion on stainless steel components (stress cracking indicator).
- 3. Exhaust Manifold and Gaskets:** Inspect the exhaust manifold for cracks — pay special attention to the area between cylinders where thermal stress is highest. Check manifold bolts for looseness or breakage. Look for soot around manifold gaskets — black carbon traces indicate a blown gasket. Tighten manifold bolts to manufacturer torque specification (typically 40-60 Nm for medium-duty). Replace broken bolts as needed.
- 4. Flexible Coupling (Flex Pipe):** If the truck has a flexible exhaust coupling, inspect it for: cracks in the braided section (if exposed type), separation of the flex pipe from its end fittings, exhaust soot indicating a leak through the flex section. A failed flex coupling must be replaced — it cannot be repaired.
- 5. Muffler and Exhaust Pipe:** Inspect the muffler body for corrosion — deep pitting or holes require replacement. Check the muffler internal baffles by tapping with a rubber mallet — a solid sound indicates intact baffles; a rattling sound indicates broken internal baffles (reduces noise suppression). Inspect all exhaust pipe connections: check U-bolt clamps for tightness, check slip joints for leaks (soot marks), ensure pipes are properly aligned and not rubbing against frame or suspension components.
- 6. Diesel Particulate Filter (DPF) — if equipped:** If the truck has a DPF, check the differential pressure sensor lines for blockages or damage. Check the DPF housing for physical damage. Verify the DPF regeneration indicator light functionality (if equipped). Note: DPF service intervals and regeneration are typically managed by the engine ECU — consult manufacturer for specific DPF maintenance requirements. If the DPF soot load indicator shows high or the

dash displays a DPF warning, perform a stationary regeneration or take the truck for a regeneration drive per manufacturer instructions.

- 7. Exhaust Gas Recirculation (EGR) — if visible:** Inspect EGR valve and cooler for external leaks (soot/carbon deposits around gaskets). Check EGR cooler coolant lines for leaks. Note heavy carbon buildup or wet soot that may indicate a failing EGR cooler.
- 8. Exhaust Hangers and Brackets:** Inspect all exhaust hangers and mounting brackets. Rubber hangers: check for cracks, stretching, or heat damage (hardened rubber). Metal brackets: check for cracks at weld points. Hanger bolts: check for tightness. The exhaust system should be securely supported and able to move slightly (thermal expansion) without contacting other components. Excessive movement (>20mm in any direction when pushed by hand) indicates a failed hanger.
- 9. Heat Shields:** Inspect all heat shields for: looseness (rattling), corrosion (especially at mounting points), contact with exhaust pipes (causes vibration and noise). Heat shields protect nearby components (wiring, hoses, brake lines, fuel lines) from exhaust heat. Any missing or damaged heat shield must be replaced.
- 10. Tailpipe:** Inspect the tailpipe outlet for obstructions. Check for excessive carbon buildup (black soot) — indicates incomplete combustion or over-fueling. Check for oil residue in the tailpipe (blue-black, wet soot) — indicates oil burning (worn turbo seals, piston rings, or valve guides).
- 11. Exhaust Leak Test (Final):** With the engine running at idle, listen for exhaust ticking or hissing sounds (indicating leaks at gaskets or joints). Feel for pulses of exhaust gas at each joint (use caution — move your hand near, not on, the joint to feel for air movement). If a leak is suspected but cannot be located, seal the tailpipe temporarily with a rag — pressure will increase and the leak location will become audible.

Acceptance Criteria: No exhaust leaks (no soot marks at any joint or gasket). Exhaust manifold bolts tight, no cracks. Muffler body intact without holes or deep corrosion. All hangers present and secure — exhaust system does not contact frame or suspension. Heat shields present and secure. No rattling or unusual exhaust noises. DPF (if equipped): no damage, no warning lights. Tailpipe: no excessive soot or oil residue.

⚠ SAFETY: Exhaust gases contain carbon monoxide (CO) — a colorless, odorless, deadly gas. Never run the engine in an enclosed space without proper ventilation. A small exhaust leak in the engine bay can allow CO to enter the cab through the HVAC system. Check for CO leaks if the driver reports headaches or drowsiness while driving. Hot exhaust components can cause severe burns — allow the system to cool completely before inspection.

Tip: A persistently leaking exhaust manifold gasket on a diesel engine can be identified by a sharp, ticking sound that speeds up with RPM. This is often mistaken for a valve train issue. Use a piece of rubber hose as a stethoscope to pinpoint the leak location. The sound will be loudest at the leaking gasket.

Documentation: Itala ang exhaust system condition, leaks found at joints, manifold bolt torque check, hanger condition, heat shield presence at condition, DPF/EGR observations, tailpipe soot condition.

Mechanic na Gumawa

8.29 Body & Chassis Inspection

Body and Chassis Frame Inspection

Purpose: Inspect the chassis frame, body mounting, cab mounts, steps, and grab handles for cracks, corrosion, loose fasteners, and structural integrity to prevent frame failure and body separation during operation.

Scope: All trucks

Frequency: Every 30,000 km or annually

PPE Required: Safety glasses, gloves, steel-toe boots

Tools & Materials: Flashlight, wire brush, inspection mirror, torque wrench, hammer, ruler, penetrating oil

Step-by-Step Procedure:

- 1. Chassis Frame Rail Inspection:** Park on level ground, chock wheels. Clean the frame rails of mud and debris using a wire brush or pressure washer (avoid electrical components). Visually inspect both frame rails from front to rear. Look for: cracks — pay close attention to: areas around spring hanger brackets, crossmember connections, steering gear mounting area, fifth wheel mounting area (for tractor-trailers), and PTO mounting locations. Cracks often start at holes, weld toes, and sharp corners. Inspect frame flanges (the top and bottom horizontal surfaces) for cracks perpendicular to the flange edge. Check frame web (vertical surface) for buckling or deformation — indicates overstress or previous impact damage. Check for elongated or worn bolt holes at suspension and body mounting points. Measure frame width at several points — should be consistent within 3mm. Frame rail twisting or distortion indicates significant damage.
- 2. Crossmembers:** Inspect all crossmembers for: cracks at the welds where crossmembers attach to frame rails, loose or missing bolts at crossmember connections, corrosion damage — especially at lower flanges where water and debris accumulate. Bent or damaged crossmembers from impact (forklifts, loading docks). Replace or repair any cracked or damaged crossmembers.
- 3. Body Mounting Bolts and Brackets:** Inspect all body-to-frame mounting bolts (tank body mounting on tanker trucks, dump body mounting on dump trucks). Verify all bolts are present and tight. Check for cracked or broken mounting brackets — especially at the front body mounts where torsional loads are highest. Check for corrosion under body mount pads — corrosion between the body and frame can create dangerous stress concentrations. Inspect rubber body mount cushions for deterioration or compression set (flattened rubber).
- 4. Cab Mounts:** Inspect cab mount rubber cushions (typically 4 mounts: 2 front, 2 rear). Check for: cracking or splitting of the rubber, compression set (excessive sag), separation of rubber from metal bonding plates. Inspect cab mount bracket at the frame for cracks. Check cab mount bolts for tightness. Worn cab mounts cause excessive cab movement, vibration, and poor door alignment.
- 5. Steps, Grab Handles, and Deck Plate:** Inspect all cab entry steps and grab handles. Check: mounting bolts for tightness, welds for cracks, anti-slip surface on steps for wear. Inspect tank top deck plate (on tanker trucks) for: corrosion perforation, cracked weld joints, missing or damaged anti-slip coating. Check all handrails and grab handles on the tank top — must be

secure and able to support a person's full weight. Inspect rear bumper step and tow hooks for secure mounting.

- 6. Mud Flaps and Splash Guards:** Check all mud flaps and splash guards. Must be present behind all wheels (if required by local regulations). Check for: torn or missing flaps, bracket damage, flaps dragging on the ground. Replace any damaged or missing mud flaps.
- 7. Fender and Body Panel Inspection:** Inspect fenders (front and rear) for: cracks, corrosion, loose mounting. Check body panels for damage, loose fasteners, or corrosion. On fiberglass bodies (some trucks), check for cracks, delamination, or UV damage. Inspect the hood and hood hinges/latches for proper operation and secure latching.
- 8. Undercoating and Corrosion Protection:** Inspect frame and body areas for rust-through, especially in regions where road salt is used. Check for corrosion at: frame sections where water and mud accumulate (top of frame flange behind cab), around battery tray area (acid corrosion), at the back of cab and under body mounts. Treat surface rust with rust converter and undercoating. Heavy corrosion requiring structural repair should be flagged for a specialist assessment.

Acceptance Criteria: No cracks in frame rails or crossmembers. All body mounting bolts present and tight. Cab mount rubber cushions intact — no cracking or compression set. Steps and grab handles secure, anti-slip surfaces in good condition. No corrosion perforation in frame or body structure. Mud flaps present and serviceable.

⚠ **SAFETY:** A cracked frame rail can lead to catastrophic frame failure and loss of vehicle control. If cracks are found, do not dispatch the truck. Consult the manufacturer or a qualified frame repair specialist. Never weld on a frame rail without manufacturer approval — improper welding can create stress risers that cause further cracking.

Tip: Frame cracks most commonly start at the top flange of the frame rail, near the rear spring hanger on the drive axle. This is the highest stress point on a truck frame. Inspect this area carefully on all trucks, especially those that operate on rough roads or are frequently overloaded.

Documentation: Itala ang frame condition (no cracks o crack locations), crossmember condition, body mount bolt torque check, cab mount condition, step/grab handle security, corrosion observations.

Lead Mechanic o Fleet Supervisor

8.30 Fuel System Inspection

Fuel System Inspection (Filters, Lines, Injectors, Pump)

Purpose: Inspect the complete fuel system including fuel filters, water separator, fuel lines, injection pump, and injectors for leaks, contamination, and proper operation to prevent engine performance issues and fuel system damage.

Scope: All diesel-powered trucks

Frequency: Every 10,000 km or at every oil change

PPE Required: Safety glasses, fuel-resistant gloves, steel-toe boots

Tools & Materials: Flashlight, clean rag, fuel filter wrench, diesel fuel container, water indicating paste (for water separator test), fuel pressure gauge (for common rail systems), multimeter (for electric fuel pump)

Step-by-Step Procedure:

- 1. Fuel Filter — Primary (Water Separator):** Locate the primary fuel filter/water separator (typically mounted on the frame rail with a clear or metal bowl). Inspect the filter housing for leaks. If it has a clear bowl: check for water at the bottom (water settles below fuel — look for a visible separation line). If equipped with a drain valve: open the drain briefly into a container to check for water. Drain any accumulated water until clean fuel appears. If no drain valve, the filter must be replaced if water contamination is suspected. Check the filter condition indicator if equipped. Replace the primary fuel filter per the maintenance schedule (typically every 10,000-20,000 km). Use water indicating paste on the bowl if there is any doubt about water presence.
- 2. Fuel Filter — Secondary (Fine Filter):** Locate the secondary fuel filter (typically on the engine, near the injection pump or high-pressure pump on common rail engines). Inspect for leaks at the filter housing seal and at the fuel line connections. Replace per the maintenance schedule (typically every 20,000-30,000 km). Note: On common rail diesel engines, always use the manufacturer-specified fuel filter — aftermarket filters may not provide adequate filtration for the high-pressure injection system and can cause injector failure.
- 3. Fuel Lines — Supply and Return:** Visually trace the fuel supply line from the fuel tank to the primary filter, to the secondary filter, to the injection pump. Trace the return line from the injectors/injection pump back to the tank. Inspect for: chafing or rubbing wear (especially where lines pass through frame holes or near moving parts), cracks or brittleness in rubber hose sections, loose connections at banjo fittings and hose clamps, corrosion on metal fuel lines, soft or bulging sections on rubber hoses (indicates internal deterioration). Pay special attention to areas near the exhaust system, turbocharger, and engine block where heat can degrade fuel hoses.
- 4. Fuel Tank Inspection:** Visually inspect the fuel tank (if accessible) for: dents, cracks, or leaks, loose or missing mounting straps, corrosion on the tank surface, fuel cap condition — gasket should seal properly, cap should vent correctly. Check the fuel pickup tube inside the tank (if accessible through the sender opening) for blockage or debris. Verify that the fuel return line

is connected and not clogged — a restricted return line causes excessive fuel pressure and can damage injectors.

- 5. Fuel Injection Pump (Low-Pressure Side):** Inspect the injection pump for external fuel leaks (especially at the pump shaft seal, inlet/outlet fittings, and governor cover). Check pump mounting bolts for tightness. Listen for unusual pump noises with the engine running — a steady whirring is normal; knocking, screeching, or irregular sounds indicate internal wear. On common rail systems: check the high-pressure pump for leaks at the high-pressure outlet fitting — a leak here sprays fuel at extremely high pressure (20,000+ psi) and can cause fires.
- 6. Fuel Injectors:** With the engine running at idle, listen to each injector using a mechanic's stethoscope or a long screwdriver placed against the injector body (handle to ear). A healthy injector produces a consistent, sharp clicking sound at idle speed. Irregular clicking, missing clicks, or metallic knocking indicates an injector problem. Check for external fuel leakage at: injector fuel return lines (small diameter hoses connecting injectors), injector hold-down clamps — must be tight, injector seal (combustion seal) — black soot around the injector base indicates a blown injector copper seal (compression leak). For common rail engines: injector return flow should be minimal — excessive return flow indicates a worn injector that needs replacement.
- 7. Fuel System Priming:** After any fuel filter replacement or fuel line work: fill the new filter with clean diesel before installation (to reduce air in the system). Use the hand primer pump (typically on top of the primary filter or on the injection pump) to purge air from the system. Pump until resistance is felt and the primer plunger becomes firm. Start the engine and check for leaks. If the engine starts but runs rough or stalls, there may still be air in the system — allow it to idle until it smooths out, or bleed air at the injector line connections.

Acceptance Criteria: No fuel leaks at any filter, line, pump, or injector. Fuel filters replaced within the maintenance schedule. No water in the fuel system. Fuel lines secure, no chafing or cracking. Injection pump and injectors operate with normal sound — no knocking or irregular clicking. Fuel cap seals properly. No check engine light for fuel system codes.

⚠ SAFETY: Diesel fuel is flammable. Clean up any fuel spills immediately. Do not smoke or have open flames near the fuel system. On common rail diesel engines, the high-pressure fuel system operates at 20,000+ psi — fuel spray from a pinhole leak can penetrate skin and cause blood poisoning. Use a piece of cardboard (not your hand) to check for high-pressure fuel leaks. Always relieve fuel pressure before disconnecting any high-pressure fuel line.

Tip: The most common cause of fuel system problems in diesel trucks is water in the fuel. Water promotes microbial growth (diesel bug) that clogs filters and corrodes injectors. Drain the water separator at every PM, and if you find water, treat the fuel tank with a biocide and recommend more frequent filter changes until the contamination is cleared.

Documentation: Itala ang primary at secondary filter replacement (date at mileage), water presence (yes/no at amount), fuel line condition, injection pump at injector sound observations, any fuel leaks found at repaired.

8.31 Steering System Inspection

Steering System Inspection (Gear Box, Linkage, Power Steering)

Purpose: Inspect the complete steering system including steering gear box, pitman arm, drag link, tie rod ends, kingpins, power steering pump, and hoses for wear, looseness, and leaks to ensure safe and precise vehicle control.

Scope: All trucks

Frequency: Every 10,000 km or 3 months

PPE Required: Safety glasses, gloves, steel-toe boots

Tools & Materials: Flashlight, pry bar, ruler or tape measure, steering free play gauge (or helper), torque wrench, power steering fluid (correct type), rag

Step-by-Step Procedure:

- 1. Power Steering Fluid Level and Condition:** With the engine off and truck on level ground, locate the power steering reservoir (typically mounted on the power steering pump or remote-mounted near the firewall). Clean the cap area before opening. Check fluid level: should be between MIN and MAX marks (or within the sight glass range). Fluid should be clear amber to light brown (ATF or power steering fluid depending on manufacturer). Milky, foamy fluid indicates air or water contamination. Dark or burnt-smelling fluid indicates overheating (pump wear, system restriction, or overloading). If the fluid is low, check for leaks before topping off. Use the correct fluid type specified by the manufacturer.
- 2. Power Steering Pump Inspection:** Start the engine and listen for pump operation. A healthy pump produces a steady, low whirring sound. Abnormal noises: whining or screeching (low fluid or air in the system), groaning (low fluid or mechanical wear), rattling (internal wear or loose mounting). Inspect the pump for external leakage at: the shaft seal (drips at the front of the pump behind the pulley), the high-pressure outlet fitting, the low-pressure return line connection. Check pump mounting bolts for tightness. Check pump belt condition and tension (see Section 8.4).
- 3. Power Steering Hoses and Lines:** Visually trace the high-pressure hose from the pump to the steering gear, and the low-pressure return line from the gear back to the pump or reservoir. Inspect for: chafing or abrasion wear (especially at frame contact points), cracks, bulges, or soft spots on rubber hoses, loose or leaking fittings at the pump, gear, and cooler (if equipped), corrosion on metal lines. A leak at the high-pressure hose can empty the system quickly and cause sudden loss of power steering while driving — replace any suspect hose immediately.
- 4. Steering Gear Box Inspection:** Inspect the steering gear box (typically mounted on the left frame rail connected to the steering column). Check for: external oil leakage at the input shaft seal (where the steering column connects), the output shaft seal (pitman arm shaft), and the top cover/end plug. Check the gear box mounting bolts — must be tight. Check for play: with wheels on the ground and engine running, have an assistant slightly rock the steering wheel back and forth while you observe the pitman arm shaft. There should be minimal delay between steering wheel movement and pitman arm movement. Excessive play before the pitman arm moves indicates worn internal steering gear components (sector shaft adjustment or worn ball nuts). Inspect the steering column universal joints for play or binding.

- 5. Steering Linkage Inspection:** Chock all wheels and raise the front axle with jack stands (ensure the suspension is at normal ride height). Crawl under the truck and inspect each steering linkage component: pitman arm — check for cracks and secure connection to steering gear output shaft. Drag link (center link) — check both ends for ball joint wear: grasp the linkage near each joint and attempt to move it up/down and in/out. Any noticeable play indicates a worn ball joint. Tie rod ends (left and right) — check for play the same way. Also check the tie rod end boots for cracks or grease leakage. Grease all steering linkage fittings until fresh grease purges from seals. Include the kingpin grease fittings (typically two per side on the front axle) — pump grease until fresh grease purges from the seals. Check tie rod and drag link for bending damage. Check locking nuts and cotter pins at all tie rod ends and ball joints.
- 6. Steering Free Play Measurement:** With the engine running and wheels pointed straight ahead, center the steering wheel. Mark the steering wheel position with tape (or use the straight-ahead mark on the steering wheel). Slowly turn the steering wheel in one direction until you feel resistance (the wheels begin to move). Measure the distance the steering wheel rim travels. Repeat in the other direction. Total free play (both directions combined) should not exceed: 50mm (2 inches) for a 500mm diameter steering wheel (or manufacturer specification, typically 10-15 degrees of steering wheel rotation). In terms of steering wheel angle: no more than 10-15 degrees of rotation before road wheels respond. Excessive free play is a safety violation and indicates worn steering components. If free play is excessive, isolate the source: column U-joints, steering gear, or linkage.
- 7. Front Wheel Bearings and Kingpins:** With the front axle raised: grasp the tire at the top and bottom and attempt to rock it. Movement indicates kingpin/wheel bearing wear. Grasp at front and back and rock — movement indicates tie rod end or wheel bearing wear. Spin the wheel and listen for bearing noise (grinding or rumbling). Check kingpin play: place a pry bar under the tire and lift — measure the vertical movement at the kingpin area. Excessive kingpin play (>3mm) requires kingpin and bushing replacement.
- 8. Steering Damper (If Equipped):** Inspect the steering damper (shock absorber connected to steering linkage). Check for: oil leakage, damaged body, loose mounting brackets, worn bushings at both ends. A failed steering damper allows excessive steering wheel movement on rough roads.
- 9. Road Test:** Drive the truck in a safe area and evaluate steering feel: steering should track straight with minimal correction on a flat road, no excessive steering wheel play at highway speeds, no wandering or pulling to one side, steering returns to center after a turn (self-centering action), no noise (groaning or whining) during turning at low speeds (indicates pump or fluid issue).

Acceptance Criteria: Power steering fluid at correct level, fluid condition good (clear amber). No power steering leaks at pump, hoses, or gear box. Steering gear box securely mounted, no external oil leaks. Steering linkage: no perceptible play at any ball joint, all grease fittings lubricated. Steering free play within specification ($\leq 50\text{mm}$ or ≤ 15 degrees). Power steering pump operates quietly. Road test: steering tracks straight, self-centers, no pulling or wandering.

⚠ **SAFETY:** Steering system failure while driving causes complete loss of vehicle control, which can result in serious accidents. Any steering component with excessive play must be replaced before the truck is dispatched. Never adjust steering free play by modifying tie rod ends — replacement of worn components is the only safe repair.

Tip: Steering free play increases gradually as components wear, making it difficult for drivers to notice the change. Track steering free play measurements over time in the PM records. A sudden increase in free play (e.g., 5mm at one PM to 20mm at the next) indicates rapid component wear that needs immediate investigation.

Documentation: Initial and power steering fluid level at condition, pump noise observation, steering gear external condition, linkage play measurements (pass/fail per joint), steering free play measurement (mm), any leaks found at repaired, road test observations.

Mechanic → Lead Mechanic verification

8.32 Cooling System — Auxiliary Components Inspection

Auxiliary Cooling System Components Inspection

Purpose: Inspect auxiliary cooling system components including the engine cooling fan, fan clutch, coolant recovery tank, radiator cap, and coolant level/condition to supplement the main coolant system procedures in Sections 8.2 and 8.3.

Scope: All trucks

Frequency: Every 10,000 km or 6 months

PPE Required: Safety glasses, gloves, hearing protection (for fan operation test)

Tools & Materials: Flashlight, coolant refractometer or hydrometer, radiator cap tester, pressure tester (for cap), rag, coolant (correct type)

Step-by-Step Procedure:

- 1. Engine Cooling Fan and Fan Clutch Inspection:** With the engine OFF, visually inspect the cooling fan blades for: cracks (especially at the base of blades where stress is highest), bent or missing blades, unbalanced rotation (wobble when spun by hand), debris wrapped around the fan hub. Check the fan shroud for cracks or missing sections — the shroud directs airflow through the radiator and must be intact. For viscous fan clutches: inspect the fan clutch body for oil leakage (wetness around the clutch bearing/seal area). Check the fan clutch for play: grasp the fan blades and attempt to rock the fan front-to-back and side-to-side — excessive movement indicates worn fan clutch bearings. For electric fans: check the fan motor operation by turning the AC on and monitoring fan engagement. Listen for bearing noise. Check the electric fan wiring and connectors for damage. Test the fan by manually bridging the thermal switch or relay (if safe to do so). For on/off fan clutches (air or electrically actuated): cycle the clutch by applying the fan control (engine brake or HVAC demand) and verify that the fan engages with a whoosh of air. The fan should spin freely when disengaged and lock solidly when engaged.
- 2. Coolant Recovery Tank:** Visually inspect the coolant recovery tank (plastic tank connected to the radiator overflow). Check for: cracks or leaks (especially at the seam and hose barb), correct coolant level — should be at the COLD FILL line when the engine is cold, between COLD FILL and HOT FILL when warm, coolant condition — should be clear and the correct color (green, red/orange, or blue depending on coolant type). If coolant appears rusty, muddy, or has particles suspended, the system needs flushing. Check the overflow hose from the radiator neck to the recovery tank — must be clear and securely connected.
- 3. Radiator Cap Inspection:** Remove the radiator cap (engine cold). Inspect the cap gasket for: compression set (flattened or misshapen), cuts, cracks, or debris. Inspect the cap spring and sealing diaphragm for damage. Pressure test the cap using a radiator cap tester: install the cap on the tester and pump to the cap's rated pressure (typically 10-16 psi for trucks, stamped on the cap). The cap should hold pressure and release at the rated pressure (you will hear a pop as the release valve opens). Replace the cap if: it fails to hold pressure within 2 psi of the rated pressure, the gasket is damaged, or if the release mechanism does not operate smoothly. A failed radiator cap causes coolant loss (pressurized system loses coolant through overflow) or allows air into the system (causes overheating).

4. **Coolant Condition and Freeze Point:** Using a refractometer (preferred) or coolant hydrometer: draw a sample of coolant from the radiator (with engine cold). Measure the specific gravity or refractive index. Calculate the freeze point and boil point. For most climates: 50/50 mix provides freeze protection to -37°C (-34°F) and boil protection to 108°C (226°F). For extreme climates: 60/40 (coolant/water) provides freeze protection to -52°C. Never exceed 70% coolant concentration — higher concentrations reduce heat transfer and freeze protection. Check coolant color: green (conventional IAT), orange/red (OAT — extended life), blue/red (HOAT — hybrid). Do not mix coolant types unless specified as compatible by the manufacturer. If the coolant is discolored, has floating particles, or smells burnt, schedule a coolant system flush and refill.
5. **Auxiliary Coolant Pump (if equipped):** Some trucks have an auxiliary electric coolant pump for the heater or auxiliary equipment. Check for: leaks at pump seals, pump operation (should run when the heater is on or when engine is cold — varies by vehicle), electrical connections — secure and corrosion-free.
6. **Fan Belt Re-check:** Verify the condition and tension of the fan drive belt (already covered in Section 8.4) — the fan cannot provide adequate cooling if the belt slips. Re-check belt deflection at the longest span between the fan pulley and the crankshaft pulley.

Acceptance Criteria: Cooling fan: no cracked or bent blades, fan clutch operates correctly (viscous: no oil leakage, correct engagement; electric: runs when commanded; on/off: engages/disengages cleanly). Fan shroud intact. Coolant recovery tank: correct level, no cracks, hose connected. Radiator cap: holds rated pressure within 2 psi, gasket intact. Coolant freeze point adequate for climate (-37°C for 50/50 mix). Coolant color correct — not rusty or contaminated.

⚠ **SAFETY:** The engine cooling fan can engage suddenly without warning, even with the engine off (on some systems with thermal fan clutches). Keep hands, tools, and loose clothing clear of the fan at all times. Do not run the engine with the fan shroud removed — the shroud protects against fan blade contact. A failed fan clutch can allow the fan to spin freely, causing immediate engine overheating.

Tip: The radiator cap is the most overlooked cooling system component. A failed cap is often the root cause of coolant loss and mysterious overheating that does not respond to other repairs. Replace the cap proactively every 2 years.

Documentation: Itala ang fan blade condition, fan clutch type at operation test (pass/fail), recovery tank level at condition, radiator cap pressure test result (psi at pass/fail), coolant freeze point at concentration, coolant color at condition.

Mechanic na Gumawa

8.33 Air Intake System Inspection

Air Intake System Inspection (Filter, Piping, Turbocharger, Intercooler)

Purpose: Inspect the complete engine air intake system from the air filter intake to the intake manifold, including the turbocharger and charge air cooler, to ensure the engine receives clean, unrestricted air for proper combustion and long engine life.

Frequency: Every 10,000 km or at every oil change

Scope: All diesel-powered trucks with turbochargers

PPE Required: Safety glasses, gloves, hearing protection (turbocharger inspection)

Tools & Materials: Flashlight, air filter restriction gauge (or vacuum gauge), screwdrivers, mirror on a stick, rag, boost pressure tester (if available), pry bar (for turbo shaft play check)

Step-by-Step Procedure:

- 1. Air Filter Restriction Indicator:** Locate the air filter restriction indicator (typically mounted on the air cleaner housing or on the air intake piping near the filter). Check the indicator window: green zone = filter OK, red zone or colored band visible = filter restricted and needs replacement. If the vehicle has a dashboard warning light for air filter restriction, verify that it self-tests (lights on key-on/bulb check) and is not illuminated while running. If no restriction indicator is installed, use a vacuum gauge on the intake piping downstream of the filter — restriction should not exceed manufacturer specification (typically 635mm / 25 inches of water column for heavy truck air filters).
- 2. Air Filter Element Inspection:** Remove the air filter element from the housing (follow manufacturer procedure — some require removing the end cap, others have quick-release clamps). Inspect the element for: excessive dirt loading (compare to a new element — if heavily clogged, replace), oil saturation (on oiled foam pre-filters — clean and re-oil per manufacturer spec), damage to the pleated paper (folds, tears, or holes), damage to the sealing gasket (the rubber gasket at both ends of the element must be intact and flexible). Replace the element if: the restriction indicator shows red, the element is physically damaged, it has been in service for more than 1 year, or it has been removed and reinstalled more than 3 times (paper elements degrade with handling). Never attempt to clean a dry-type air filter element by tapping or blowing compressed air from the inside — this damages the paper pleats and creates holes that pass unfiltered air. Use a new element only.
- 3. Air Cleaner Housing:** Inspect the air cleaner housing for: cracks or damage to the plastic or metal housing, missing or damaged clips/latches that secure the cover, missing or damaged gasket between the housing and the cover. A leaking air cleaner housing allows unfiltered air to enter the engine, causing rapid engine wear. Check the dust ejector valve (if equipped) — this rubber flap on the housing allows dust to be ejected during engine operation. It should be flexible and not clogged. Clean or replace if blocked.
- 4. Intake Piping — Filter to Turbo:** Inspect all intake piping from the air cleaner outlet to the turbocharger inlet. Check for: loose or disconnected hose clamps, cracks or splits in rubber or silicone hoses, collapsed sections (restricted airflow), chafing on frame or engine components,

oil mist inside the piping (normal for turbocharged engines with crankcase ventilation — excessive oil indicates worn turbo seals or excessive crankcase pressure). Tighten all clamps. Replace any damaged hoses.

- 5. Turbocharger Inspection:** With the engine OFF and cold: check the turbocharger for shaft play: remove the air intake pipe at the turbo inlet. Reach inside and grasp the compressor wheel (the small fan blade visible at the turbo inlet). Attempt to move the shaft up/down and in/out. Acceptable radial (up/down) play: very slight movement (approximately 0.5mm / 0.020 inch). Acceptable axial (in/out) play: virtually zero — slight movement (less than 0.1mm / 0.004 inch) may be acceptable. Any contact between the compressor wheel and the housing (wheel-to-housing rub) indicates turbo failure. Spin the compressor wheel by hand — it should spin freely without binding or scraping sounds. Inspect the compressor wheel blades for damage: bent, chipped, or eroded blades reduce turbo efficiency and can unbalance the rotating assembly. Check the compressor housing for oil deposits (indicates leaking turbo seals). Check for oil in the intake pipe downstream of the turbo. Inspect the turbo center housing for oil leaks at the oil supply line and oil drain line connections. Check the wastegate actuator linkage (if equipped) for free movement and secure connections.
- 6. Charge Air Cooler (Intercooler) Inspection:** Inspect the intercooler (mounted in front of the radiator or in a separate location) for: physical damage (bent or crushed fins, impacted by road debris), corrosion on the intercooler core or end tanks, oil deposits at intercooler connections (indicates leaking turbo seals or crankcase ventilation bypass), cracks or leaks at the intercooler end tank welds. Check all charge air cooler hoses (from turbo outlet to intercooler inlet, and from intercooler outlet to intake manifold). These hoses carry pressurized air (20-40 psi on modern diesels) and must be: securely clamped, free of cracks, bulges, or soft spots (internal ply separation causes hose failure under boost), not rubbing against other components. Tighten all charge air hose clamps. If a boost leak is suspected: pressurize the system with a boost leak tester (an adapter that seals the intake system and allows shop air at 20-30 psi) — listen for hissing to locate leaks.
- 7. Intake Manifold:** Inspect the intake manifold for: external damage or cracks, loose mounting bolts or nuts, evidence of leaks at gaskets (black soot or oil residue around gasket surfaces). If the truck has an intake manifold pressure sensor, check the sensor for secure connection and clean the sensor port if accessible.
- 8. Cold Start Aids (If Equipped):** Check the operation of cold start aids: glow plugs — measure resistance at each glow plug (typically 0.5-2.0 ohms for modern metal sheathed plugs). Open circuit indicates a failed glow plug. Intake heater (grid heater) — visually inspect the grid for damage; check the relay operation. Ether injection system — verify the presence of the ether canister, check that the injection nozzle is clear.

Acceptance Criteria: Air filter restriction indicator in green zone. Air filter element clean and undamaged (or replaced within interval). Air cleaner housing sealed and intact. No cracks or leaks in intake piping from filter to turbo. Turbocharger: minimal shaft play (radial <0.5mm, axial <0.1mm), compressor wheel undamaged, spins freely. Intercooler: no damage, no boost leaks. All charge air hoses secure and undamaged. Intake manifold: no leaks.

⚠ **SAFETY:** A turbocharger with excessive shaft play can fail catastrophically, sending metal fragments into the engine intake and causing severe engine damage. If the turbo shaft has significant play (compressor wheel contacts housing), do not start the engine. The turbocharger operates at speeds up to 150,000 RPM and temperatures up to 750°C (1,380°F) — even a small imbalance can cause rapid failure. Never run the engine with a loose intake pipe or missing air filter — unfiltered air will destroy the engine in minutes.

Tip: Air filter condition is the single most influential factor in turbocharger life. A restricted air filter creates high vacuum on the turbo inlet, which pulls oil past the compressor seal (oil consumption) and causes the turbo to work harder (higher temperatures and speeds). Replace air filters on schedule regardless of visual appearance. Use OEM or equivalent quality filters only.

Documentation: Itala ang air filter restriction indicator reading, air filter element condition/replacement, turbocharger shaft play measurements (radial at axial), turbo wheel condition, intake piping and hose condition, intercooler condition, boost leak test result (kung ginawa).

Mechanic → Lead Mechanic verification

8.34 Miscellaneous — General Vehicle Inspection

General Vehicle Walk-Around Inspection and Miscellaneous Items

Purpose: Perform a general vehicle walk-around covering items not covered in other specific sections: windshield wipers, washer fluid, mirrors, seat belts, horn, door hardware, fluid levels, and a general visual check for leaks and damage. This is the final go/no-go check before dispatch.

Frequency: Daily (pre-trip) — documented in full during each PM service

Tools & Materials: Flashlight, rag, windshield washer fluid, tire pressure gauge (if checking tire pressures — see Section 8.8)

Scope: All trucks — perform daily (pre-trip) and during scheduled PM

PPE Required: Safety glasses, high-visibility vest

Step-by-Step Procedure:

- 1. Windshield Wipers and Washer:** Check both windshield wiper blades for: torn or missing rubber, hardened or cracked rubber, bent wiper arm. Replace wiper blades if they streak, skip, or fail to clear the windshield in one pass. Fill the windshield washer reservoir with appropriate washer fluid (use winter-rated fluid with freeze protection in cold climates). Test the washer spray: nozzles should spray evenly on both sides of the windshield, the spray pattern should cover the driver's field of view. Adjust nozzles if misaligned (use a pin to adjust). Check headlight washer (if equipped): verify spray function.
- 2. Mirrors and Mountings:** Inspect all mirrors (side mirrors, rearview mirror, fender mirrors, wide-angle/convex mirrors). Check for: cracked or broken mirror glass, loose mirror heads (adjust by hand — should hold position), vibration at highway speeds (indicating loose ball joints or mounting brackets), damaged mirror arms or brackets. Clean all mirror surfaces. Adjust mirrors for proper field of view according to the driver's preference.
- 3. Seat Belts:** Inspect all seat belts (driver, passenger, and any rear seats/cab berth). Check: webbing for fraying, cuts, or fading (UV damage), retractor function — pull the belt fully out, then allow it to retract; it should retract fully and smoothly, latch operation — insert the latch plate into the buckle; it should click and lock; pressing the release button should release cleanly. Check the locking mechanism: pull the belt quickly — it should lock immediately (emergency locking retractor). Check the buckle bolt and seat belt anchor bolt for tightness. Replace any seat belt with frayed webbing, weak retractor, or damaged buckle.
- 4. Horn:** Test the horn: press the horn pad — the horn should produce a clear, loud sound (minimum 90 dB for commercial vehicles). If the sound is weak, air horns may need compressor system attention. Electric horns: check for a single tone (dual horns should produce a harmonic pair). If the horn does not sound, check the fuse, relay, and horn power/ground connections.
- 5. Door Hardware:** Inspect all cab doors: door hinges — check for worn hinge pins (sagging door), check hinge bolts for tightness. Door latches — open and close each door; the latch should engage securely and release cleanly. Door handles — both interior and exterior handles should operate smoothly. Door seals — inspect rubber weatherstripping for compression set, tears, or missing sections. Replace damaged seals to prevent water ingress,

dust, and wind noise. Lubricate door hinges and latch mechanisms with white lithium grease or silicone spray (not WD-40). Inspect the door lock mechanism for proper operation.

- 6. Fluid Level Check — All Systems:** Verify all fluid levels are within the normal operating range: engine oil (see Section 8.1), coolant level in the recovery tank (see Section 8.32), power steering fluid (see Section 8.31), brake fluid or air brake system (see Sections 8.5-8.7), windshield washer fluid, transmission oil (see Section 8.24), hydraulic fluid (if equipped, see Section 8.26), battery electrolyte level (if serviceable battery — see Section 8.18). Top off any low fluids with the correct grade/type. Note any unexplained fluid loss for investigation.
- 7. Walk-Around for Leaks:** Park on a clean, dry surface. Allow the engine to idle for 2-3 minutes, then turn off. Walk completely around the truck and inspect the ground underneath for: engine oil drips (front engine area), coolant drips (front center — water pump, radiator, hoses), transmission oil drips (under transmission area), differential oil drips (under rear axle area), fuel leaks (anywhere along the fuel system), brake fluid leaks (at master cylinder, calipers, wheel cylinders), power steering fluid drips (front steering area), air leaks (hissing sound — with engine running). If any puddle is larger than a few drops, the leak should be investigated and repaired.
- 8. General Visual Checklist:** Complete these items during the walk-around: check that the fuel cap is secured, verify that all compartment hatches are closed and latched (tanker trucks), verify that drain valves are closed, check that all access panels and tool box lids are closed and latched, inspect the fifth wheel (if tractor) for wear on the coupling surface and secure locking mechanism, inspect the landing gear (if trailer) for smooth operation and secure locking, check that no tools, rags, or materials are left on the truck prior to dispatch.

Acceptance Criteria: Wiper blades clean windshield without streaking in one pass. Washer reservoir full, spray pattern correct. All mirrors intact, clean, and adjustable. Seat belts: webbing intact, retractors function properly, buckles latch and release cleanly. Horn produces clear loud sound. Doors open and close smoothly, latches securely. All fluid levels within normal range. No fluid leaks larger than a few drops on the ground. All hatches, valves, and panels closed and secured. No tools or materials left on the truck.

⚠ SAFETY: Pre-trip inspection is the last line of defense against road accidents and breakdowns. Never dispatch a truck with known defects in wipers, mirrors, seat belts, or significant fluid leaks. In many jurisdictions, driving with defective wipers, mirrors, or horn is a traffic violation. A cracked frame or missing wheel lug nut is an immediate out-of-service condition.

Tip: Create a laminated pre-trip checklist card that drivers keep in the cab. The card should cover all items in this section plus the Sections 8.8 (tires) and 8.20 (lighting). Drivers should initial each item and flag any defects to the shop immediately. The best PM program catches defects before the truck leaves the yard, not on the side of the highway.

Documentation: Itala ang lahat ng fluid levels at top-off, wiper blade condition, mirror condition, seat belt function sa lahat ng positions, horn operation, door hardware condition, leak check results (no leaks o specific leak locations found at corrected), pre-trip inspection completed.

Mechanic or Driver

10. Chapter 9: PM Documentation & Record-Keeping

9.1 Importance of Documentation

Proper documentation of every PM task is essential for:

- **Legal compliance:** Records of PM completion prove that the truck was maintained according to regulatory requirements.
- **Depot access:** Many petroleum depots require proof of PM compliance before allowing access.
- **Trend analysis:** PM history helps identify recurring issues and plan more effective maintenance strategies.
- **Resale value:** Complete maintenance records increase the resale value of the truck.
- **Warranty compliance:** Many manufacturer warranties require proof of scheduled maintenance.

9.2 What Should Be Documented

After each PM procedure, ensure the following is recorded in the Job Order line item or the PM Schedule note:

- Date and time of completion.
- Mileage and engine hours at the time of completion.
- Parts used (with part numbers and quantities).
- Actual labor hours spent on the task.
- Observations: any unexpected findings, measurements taken, or follow-up actions needed.
- Mechanic's name who performed the work.
- Lead mechanic or supervisor verification (for critical systems: brakes, steering, suspension, chassis).

9.3 When PM is Performed Outside the Job Order System

If a PM task is performed outside the Job Order system (e.g., at a third-party shop, roadside service, or by the driver), the PM Schedule can be updated manually:

1. Navigate to the truck detail page in the system.
2. Find the PM Schedule section.
3. Locate the task that was performed and click the schedule number (the underlined ID).
4. Update `last_completed_at`, `last_mileage_km`, and `last_engine_hours` to reflect the service that was done.
5. Click **"Save"**. The system will automatically recalculate the next due date.

Important: Manual updates should include a note in the schedule record explaining what was done and why it was not processed through a Job Order. This maintains an accurate audit trail.

9.4 Using the ☒ **Mark Complete** Button

For items on the PM Schedules list, the ☒ **Mark Complete** button provides a streamlined workflow. This creates a `ServiceLogEntry` directly without requiring a Job Order. Use this for:

- Simple PM tasks that do not involve repair work.
- Driver-performed inspections and checks.
- Visual-only tasks such as walk-around inspections.

When using ☒ **Mark Complete**, record: mileage and engine hours, labor hours spent, any parts used (name, quantity, unit cost), and notes or observations.

Volume 4 — Repair Operations

11. Chapter 10: Initial Assessment & Triage

10.1 Receiving a Truck for Repair

When a truck arrives at the workshop with a reported problem, follow this standard intake procedure:

1. **Driver Report:** Interview the driver or the person reporting the issue. Ask the five key questions:
 - What is the problem? (be specific — what does it do, or not do?)
 - When did it start? (sudden or gradual onset?)
 - Under what conditions does it happen? (speed, load, temperature, gear, turning?)
 - Are there any warning lights? (which ones, when did they come on?)
 - Any recent repairs or changes? (what was touched last?)
2. **Visual Walk-Around:** Walk around the truck and look for obvious signs: fluid leaks (color, location, amount), damaged components, loose or hanging parts, abnormal tire wear, broken lights, damaged hoses.
3. **Fluid Check:** Check all fluid levels and conditions — engine oil, coolant, transmission oil, power steering, brake fluid (if applicable), windshield washer. Note any fluid that is low, contaminated, or has an unusual odor.
4. **Functional Check:** With the engine running (if safe), verify: warning lights on instrument panel, unusual noises (knocking, squealing, hissing, grinding), unusual smells (burning oil, fuel, rubber, electrical), brake function (pedal feel, air pressure buildup and leak-down rate), steering response (free play, pulling).
5. **Diagnostic Scan:** Connect a diagnostic scan tool (if available) to check for stored fault codes. Record all active and historic codes.
6. **Triage Decision:** Based on the initial assessment, classify the repair:
 - **Emergency / Safety-Critical:** Truck cannot be operated safely. Must be repaired before any movement. Examples: brake failure, steering failure, significant fuel/oil leak, tire blowout, no lights at night.
 - **Routine:** Truck can operate but should be scheduled for repair within the week. Examples: minor oil leak, worn belt, one burned-out marker light, small air leak.
 - **Deferred:** Issue does not affect safety or operation but should be monitored and addressed at the next scheduled PM. Examples: minor cosmetic damage, slight oil seepage from gasket, worn but not critical tire.

10.2 Repair vs. Replace Decision

For any failed or worn component, evaluate the following factors before deciding whether to repair or replace:

Factor	Favor Repair	Favor Replace
Cost	Repair cost < 50% of replacement cost	Repair cost ≥ 50% of replacement cost
Age / Wear	Component is relatively new (<50% of expected life used)	Component is near or past its expected service life
Availability	Replacement part not immediately available but repair is possible	Replacement part is in stock or available quickly
Reliability	Repair restores component to like-new condition (e.g., rebuild)	Repair is a temporary fix; failure likely to recur soon
Labor	Repair labor is significantly less than replacement labor	Replacement requires similar or less labor than disassembly/repair
Safety	Repair meets or exceeds safety standards	Repaired component does not meet original safety margin
Warranty	Repair is covered under existing warranty	New part carries fresh warranty

Rule of thumb: When in doubt on safety-critical components (brakes, steering, suspension, tires, wheel bearings), replace rather than repair. The cost difference is small compared to the consequences of a field failure.

10.3 Creating a Job Order from Assessment

1. Navigate to **Sidebar** → **Job Orders** → **New Job Order**.
2. Select the **Truck** from the dropdown.
3. Enter a descriptive **Title** — be specific about the issue. Examples:
 - ✗ "Truck problem" — too vague.
 - ☑ "Engine overheating — coolant leak at upper radiator hose" — actionable.
 - ☑ "Brake air leak — driver-side rear chamber hissing at 100 psi" — specific.
4. Select the **Type**: PM / Repair / Inspection / Contractor Service.
5. Set **Priority**: Low / Medium / High / Emergency. Use Emergency only for safety-critical issues that require immediate attention (truck is down and blocking operations).
6. **Assign To**: Select the mechanic most qualified for the job (by specialization if possible). For contractor service, select the **Contractor** instead of an individual mechanic.
7. Click **"Save"**.
8. On the Job Order detail page, add **Line Items** for each specific task required (see Chapter 11 for line item details).

12. Chapter 11: Job Order Creation & Assignment

11.1 Understanding the 3-State Workflow

Every Job Order passes through three states:

OPEN → IN PROGRESS → CLOSED

State	Meaning	Who Can Change It	Actions Permitted
OPEN	JO has been created but no work has started. Equivalent to "pending assignment."	Admin/Staff	Edit, add line items, assign mechanic, change priority, delete (if no work done). Mechanic can view but cannot change status.
IN PROGRESS	Work is actively being performed. Mechanic has accepted the assignment.	Mechanic (changes line item status), Admin/Staff (all actions)	Line items can be marked as Done, In Progress, or Not Applicable. Parts can be added. JO can be edited.
CLOSED	All work is complete. JO is finalized and cannot be changed.	Admin/Staff (initiates close)	Read-only. Service log entries created automatically. PM schedules updated automatically.

11.2 Creating a Job Order — Step by Step

1. Navigate to **Sidebar → Job Orders**.
2. Click **"New Job Order"**.
3. Complete the **Job Order Form**:
 - **Truck**: Select from the fleet list. The search bar filters by plate number or unit number.
 - **Title**: Brief but descriptive (max 200 characters). Begin with a verb: "Replace," "Inspect," "Repair," "Adjust," "Investigate."
 - **Description**: Optional. Add more context, driver's complaint, troubleshooting notes.
 - **Job Type**: One of: PM, Repair, Inspection, Contractor Service. The type affects reporting but does not restrict the line items you can add.
 - **Priority**: Defaults to Medium. Use High or Emergency judiciously — if everything is high priority, nothing is.
 - **Assigned To**: For PM/Repair/Inspection: select a specific mechanic. For Contractor Service: select the contractor company.
 - **PMS Line Items (optional)**: If this JO was triggered by one or more PM schedule items being due, you can select them from the multi-select list. This links the JO to those specific PM schedules so they update when the JO closes.
4. Click **"Save"**.
5. You are redirected to the **Job Order Detail Page**. The JO now has an OPEN status.

11.3 Assigning Work to a Mechanic

Work is assigned during JO creation (by selecting the mechanic). To reassign after creation:

1. Open the Job Order detail page.
2. Click **"Edit"**.
3. Change the **Assigned To** field to the new mechanic.
4. Click **"Save"**.
5. The new mechanic will see the JO in their **My Assignments** list.
6. **Communication:** After assigning or reassigning, inform the mechanic verbally or through the communication channel so they know to check the system. The system does not send automatic notifications.

11.4 Adding Line Items (Tasks) to a Job Order

Line items are the individual tasks that make up the job. A single JO can have one or many line items.

1. From the JO Detail page, click the **"Add"** button in the Line Items section.
2. Fill in each field:
 - **Task Template:** Optional. If this task corresponds to an existing PM template, select it. This links the line item to the PM schedule for auto-update on close.
 - **Category:** Optional. The PM category this task falls under (used for reporting).
 - **Description:** What needs to be done. Be specific — include quantities, locations, or special instructions. Example: "Replace upper radiator hose — Cummins ISX engine, driver side" rather than just "Fix hose."
 - **Estimated Hours:** How long you expect this task to take. Used for workload planning and KPI comparison.
 - **Notes:** Special instructions, safety precautions, or part numbers.
3. Click **"Add"**.
4. Repeat for each task needed.
5. Line items can be reordered or deleted (if not yet started) using the icons in the line item list.

13. Chapter 12: Work Execution & Parts Tracking

12.1 Mechanic Workflow — Starting a Job

1. Log into the system.
2. Navigate to **Sidebar** → **My Assignments**.
3. You will see all Job Orders assigned to you, sorted by priority (Emergency → High → Medium → Low) and then by creation date (oldest first).
4. Click the JO title to open the detail page and see all line items.
5. Review the line items and any supporting information (notes, attached images if applicable).
6. Set the JO status to **"In Progress"** by clicking the status button at the top of the page. This signals to the admin/Staff that work has begun.

12.2 Completing Line Items

1. As you complete each task, update its status in the system.
2. For each line item, click the ☒ (**green checkmark**) icon to mark it as **"Done."**
3. A small form opens where you must enter:
 - **Actual Hours:** The real time spent on this task. Be honest — this data feeds the KPI reports and helps with future estimating. If a job took longer than estimated, note why in the notes field.
 - **Notes:** Any observations, challenges, or things to watch for next time. This is valuable institutional knowledge for the next mechanic who works on this truck.
4. Click **"Save."**
5. If you are unable to complete a task, you may set it to **"Not Applicable"** (if circumstances changed) or leave it **"Open"** (if the task is waiting for parts or clarification).

12.3 Recording Parts Used

Parts tracking is essential for cost analysis. Every part used on a Job Order must be recorded in the system.

1. On the JO detail page, find the line item that used the part.
2. Click the  (**box**) icon next to that line item to expand the parts section.
3. Click **"Add Part"**.
4. Complete the form:
 - **Part Name:** Accurate description. Use the manufacturer's part name and number if available. Example: "Oil Filter — Fleetguard LF16001" (not just "oil filter").
 - **Quantity:** Number of units used. If a part is sold in a kit, record "1" for the kit.
 - **Unit Cost:** The cost per unit in the local currency. **Important:** Be accurate — these costs feed KPI reports, cost trend analysis, and predictive analytics. If the exact cost is unknown at the time of adding, record the best estimate and update when the invoice arrives.
5. Click **"Save."**
6. The part appears in the expanded section with the calculated total cost (quantity × unit cost).
7. To add more parts, click **"Add Part"** again.

8. To edit or delete a part, click the pencil or trash icon next to the part entry.

Best Practice: Record parts as you install them — do not wait until the end of the job. This prevents forgotten parts and keeps the cost running total visible to the admin. Include the source of the part if relevant (e.g., "from parts stock" vs. "purchased at AutoSupply Co.").

12.4 Multi-Mechanic / Team Work

If a Job Order requires multiple mechanics (e.g., one mechanic on brakes, another on tires), the current system structure assigns the JO to one primary mechanic. Coordination options:

- **Option A:** Create separate JOs for each mechanic's portion, cross-referenced in the description.
- **Option B:** Assign the JO to the lead mechanic. They update the line items as the team completes them. Each line item can record the actual hours of the person who did that task in the notes field.

The admin/staff supervisor should coordinate the work assignment to avoid confusion.

14. Chapter 13: Quality Control & Pre-Return Inspection

13.1 Purpose of Quality Control

Before any truck is returned to service after a repair or PM, a quality control inspection must be performed. This ensures:

- All work was completed correctly and safely.
- No tools, parts, or materials were left in or on the truck.
- All fasteners are properly tightened.
- All fluids are at correct levels.
- No new issues were introduced during the repair.
- The truck is safe to operate on public roads.

13.2 QC Inspection Checklist

Responsible Role: Lead Mechanic or Supervisor (not the same person who performed the work)

1. **Walk-around inspection:** Walk completely around the truck. Look for: tools or rags left on fenders, steps, or cab roof, loose parts or fasteners, fluid leaks (any new drips from areas that were worked on), component alignment (hoses, wiring, brackets not rubbing or contacting), all access panels and covers reinstalled and secured.
2. **Under-hood / Under-truck check:** Verify all connections are tight (hose clamps, electrical connectors, bolts). Check that all fluid caps/dipsticks are installed and secure. Remove any rags, gloves, or trash left in the engine compartment or cab. Confirm that all removed guards and shields are reinstalled.
3. **Functional test:** Start engine and let idle for 2-3 minutes. Check for: unusual noises from worked-on areas, warning lights (all should be off after initial bulb check), abnormal smoke from exhaust, fluid leaks at connections that were loosened. Test the systems that were repaired: brake application and release (if brakes were touched), steering lock-to-lock (if steering components were touched), light operation (if electrical was touched), horn, windshield wipers.
4. **Road test (if applicable):** Drive the truck in a safe area. Check for: proper operation of the repaired system (e.g., brakes stop straight, transmission shifts smoothly, engine runs without missing), abnormal noises, vibrations, or smells, gauges reading normally (temperature, oil pressure, air pressure).
5. **Documentation check:** Verify all line items are marked Done (or Not Applicable). Verify all parts used have been entered, with correct quantities and costs. Verify the actual hours entered are reasonable and match the work performed. Ensure the Job Order notes field contains any important observations or follow-up recommendations.

⚠ **Critical:** If the QC inspection reveals any deficiency, the truck must NOT be returned to service until the issue is corrected. The mechanic who performed the work must re-address the deficiency and the QC must be repeated. No exceptions — even for "minor" issues.

13.3 Common QC Failures & How to Prevent Them

QC Failure	Root Cause	Prevention
Fluid leak after oil change	Drain plug not torqued, filter not tightened, old gasket not replaced	Always replace drain plug gasket. Hand-tighten filter + 3/4 turn. Run engine and inspect before returning to service.
Wheel lug nuts loose after 50 km	Lugs not re-torqued after initial tightening	Always re-torque wheel lug nuts after 50 km of driving. Mark with torque seal to verify visually.
Electrical connector left unplugged	Haste, distraction during reassembly	Use a systematic approach: disconnect ONE connector at a time during diagnosis, reconnect immediately after testing. Use the "connect all connectors" check as the last step before closing the hood.
Wiring or hose routed incorrectly (chafing risk)	Reassembly without reference photos	Take a photo before disassembly for complex routing. Always use zip ties to secure wiring away from moving parts and heat sources.
Tools left in engine bay	No tool inventory check	Perform a tool count before and after each job. Use shadow boards (tool outlines) in the toolbox to see missing tools at a glance.

15. Chapter 14: Job Order Close-out & Service Logging

14.1 When to Close a Job Order

A Job Order should be closed only when **all** of the following are true:

1. All line items are marked Done or Not Applicable.
2. All parts used have been entered into the system.
3. All actual hours have been recorded for each line item.
4. QC inspection has been passed.
5. Road test (if needed) has been completed successfully.
6. The truck is ready to return to service.

14.2 Closing a Job Order — Step by Step

Responsible Role: Admin, Staff, or Super Admin only (not the mechanic performing the work)

1. Open the Job Order detail page.
2. Review all line items and parts entries for completeness and accuracy.
3. Click the **"Close Job Order"** button.
4. A confirmation dialog appears explaining that closing is irreversible — read and confirm.
5. The close form opens showing:
 - **Current Mileage:** Pre-filled from the truck's current odometer reading. Verify and adjust if needed (the truck may have been driven since the reading was last updated).
 - **Current Engine Hours:** Pre-filled from the truck's current hour meter. Verify.
 - **Notes:** Optional closing notes — summarize the work performed, any follow-up recommendations, or parts that need to be ordered for the next service.
6. Click **"Close Job Order"** to finalize.

14.3 What Happens Automatically on Close

When a Job Order is closed, the system performs the following actions automatically:

1. **Status changes** to CLOSED. The JO becomes read-only.
2. **Truck's mileage and engine hours** are updated from the values entered on the close form.
3. **Any open line items** are automatically marked as Done (with a note "Auto-completed on JO close").
4. **Service Log Entries** are created in the immutable service ledger for:
 - Each line item that was completed.
 - The JO creation and close events.
5. **PM Schedule update:** Any line item that was linked to a PM template will update the corresponding PM schedule with:
 - `last_completed_at` = close timestamp.
 - `last_mileage_km` = mileage at close.
 - `last_engine_hours` = hours at close.

- The next due date is recalculated automatically.

14.4 Service Ledger — Immutable Audit Trail

The Service Ledger is a permanent, unchangeable record of every action performed. It serves as:

- **Legal record** of maintenance performed on each truck.
- **Audit trail** for compliance with depot access requirements.
- **Historical reference** for recurring issues on a specific truck.
- **Mechanic accountability** — each entry is tied to the person who performed the work.

Accessing the Service Ledger:

- **Full Ledger:** Sidebar → Service Ledger (Admin+). Shows all entries, filterable by truck and action type.
- **Per-Truck Ledger:** Open any truck detail page → Click "Full Ledger" button in the Service History section.

Important: The Service Ledger cannot be edited or deleted — not even by the Super Admin through the Django Admin interface. This immutability is a deliberate design feature to ensure the integrity of the maintenance record. If an error is discovered in a closed JO, a new JO must be created to correct it, and the original record remains as-is.

14.5 Reopening a Closed Job Order (Not Possible)

Once a Job Order is closed, it cannot be reopened. If additional work is needed on the same truck:

1. Create a **new Job Order** for the new work.
2. In the description, reference the original JO number: "Continuation of work from JO-2026-0042 — unresolved brake pull issue."
3. This maintains a clean audit trail and prevents data integrity issues.

14.6 Archived / Historical Job Orders

Closed Job Orders remain visible in the Job Orders list with a "Closed" status. Use the status filter to view only closed JOs when researching history. The system does not automatically archive or delete old JOs — data is retained indefinitely for historical reporting.

Volume 5 — Contractor Management

16. Chapter 15: Contractor Onboarding & Skills Registry

15.1 When to Use a Contractor

Contractors are external service providers used when:

- The required repair or service is outside the in-house mechanics' expertise (e.g., calibration, specialized welding, electrical diagnostics for OEM systems).
- In-house capacity is fully utilized and the job cannot be deferred.
- The job requires specialized equipment not available in the workshop (e.g., hydrostatic testing, ultrasonic thickness gauging, dynamometer testing).
- Warranty or insurance requirements mandate OEM-certified technicians.

15.2 Registering a New Contractor in the System

Responsible Role: Admin or Super Admin

1. Navigate to **Sidebar** → **Contractors**.
2. Click **"Add Contractor"**.
3. Complete the form:
 - **Company Name:** Legal business name as registered with government or as stated on the invoice.
 - **Contact Person:** Name of the person you communicate with for scheduling and invoicing.
 - **Mobile Number:** Direct line to the contact person.
 - **Email Address:** For sending Job Orders, correspondence, and invoices.
 - **Address:** Physical office or workshop address.
 - **Skills:** Comma-separated list of services the contractor provides. Be specific: "Welding, Calibration, Electrical Diagnostics, Vapor Recovery Testing, Tank Hydrostatic Testing, Hydraulic Repairs."
 - **Active:** Check this box to allow the contractor to be assigned Job Orders. Uncheck to temporarily or permanently disable contractor assignments without deleting the record.
4. Click **"Save"**.

15.3 Contractor User Account

Each contractor company can have a user account in the system with the **CONTRACTOR** role. This account:

- Can only see Job Orders assigned to their company.
- Has the same workflow capabilities as a mechanic (mark line items Done, record parts, add hours).

- Does not have access to any other part of the system (truck list read-only, no PM schedules, no reports).
- Is redirected to **My Assignments** on login for immediate access to work orders.

To create a contractor user account, see Chapter 2 (User Account Lifecycle) — select Role = CONTRACTOR.

17. Chapter 16: Contractor Service Execution

16.1 Creating a Contractor Job Order

1. Navigate to **Sidebar** → **Job Orders** → **New Job Order**.
2. Select the **Truck**.
3. Set **Type** = "Contractor Service."
4. In the **Assigned To** field, select the **Contractor Company** (not an individual mechanic).
5. Complete the remaining fields (title, description, priority).
6. Click **"Save"**.
7. Add line items for each specific task the contractor must perform.

16.2 Contractor On-Site Procedure

1. **Check-in:** When the contractor arrives at the workshop, they must:
 - Report to the supervisor or admin on duty.
 - Present any required credentials or certifications for the job.
 - Review the Job Order scope with the supervisor before starting work.
 - Be briefed on workshop safety rules, emergency exits, and the location of fire extinguishers and first aid kits.
2. **During work:**
 - The contractor updates the Job Order line items in the system (or the supervisor updates if the contractor does not have system access).
 - The supervisor may perform spot checks on the work in progress.
 - Any additional work found during the job must be approved by the supervisor before proceeding (verbal approval is acceptable; documentation follows).
3. **Check-out:**
 - Work is inspected and approved by the supervisor (see QC inspection in Chapter 13).
 - The contractor's service report (paper or digital) is submitted to the supervisor.
 - The supervisor signs off on the Job Order.

16.3 Invoicing & Cost Tracking

- All parts and materials used by the contractor are recorded in the Job Order line items (see Section 12.3).
- The contractor's labor cost is recorded as the contractor's quoted rate in the line item notes or as a separate part entry for "labor charge."
- Upon completion, the contractor submits an invoice referencing the Job Order number.
- The supervisor verifies the invoice against the Job Order (line items, parts, hours) and approves for payment.

18. Chapter 17: Contractor Performance Evaluation

17.1 Performance Metrics

The KPI Reports section tracks contractor performance automatically. Key metrics:

Metric	Definition	Target
Total Jobs	Number of Job Orders assigned to this contractor	Varies by contractor
Completed Jobs	Number of JOs closed (completed successfully)	As assigned
Open Jobs	JOs currently in progress	Minimize — should be <10% of total
Completion Rate	Completed jobs / Total jobs × 100%	≥ 90%
Average Turnaround Time	Days from creation to close	Varies by job type — track trends

17.2 Evaluating Contractor Quality

In addition to quantitative metrics, evaluate contractors on:

- **Work quality:** Does the work pass QC inspection on the first attempt? Are there recurring issues that trace back to contractor work?
- **Reliability:** Does the contractor show up on the scheduled date and time? Do they complete work within the estimated time?
- **Documentation:** Does the contractor provide complete service reports, photos (if applicable), and accurate invoices?
- **Safety compliance:** Does the contractor follow workshop safety rules? Do they wear proper PPE? Do they bring their own safety equipment?
- **Communication:** Does the contractor communicate delays, parts issues, or additional findings promptly?

Evaluation notes can be recorded in the contractor's record using the **Notes** field available in the Contractor detail page (accessible from Sidebar → Contractors).

Volume 6 — Training & On-the-Job Training (OJT)

19. Chapter 18: Trainee Onboarding & System Access

18.1 Overview of the Training Program

The OJT/Training module tracks the progress of apprentice mechanics and trainees as they learn the trade. The system supports three aspects of training:

- **Attendance tracking:** Daily check-in and check-out to monitor presence.
- **Task ratings:** Supervisors rate OJTs on specific PM/repair tasks they perform (1-5 scale).
- **Weekly reviews:** Supervisors provide structured evaluations with strengths, areas for improvement, and overall scores.

18.2 Setting Up a New Trainee

Responsible Role: Admin or Super Admin

Step 1 — Create User Account:

1. Navigate to **Sidebar** → **User Management**.
2. Click **"Add User"**.
3. Set **Role** = "OJT / Trainee."
4. Complete all required fields (username, password, names, etc.).
5. Click **"Save"**.

Step 2 — Assign to Training:

1. Navigate to **Sidebar** → **Assign Training**.
2. Select the OJT you just created from the dropdown.
3. Select their **Supervisor** — this must be a user with Staff, Admin, or Super Admin role. The supervisor will be responsible for rating and reviewing the OJT.
4. Set the **Start Date** — typically the OJT's first day at the workshop.
5. Click **"Assign"**.

The OJT can now log in and access the Training Dashboard. The supervisor can see the OJT in their supervisor dashboard.

18.3 Supervisor Selection Guidelines

The supervisor plays a critical role in the OJT's development. Choose a supervisor who:

- Has at least 2 years of experience in truck maintenance.
- Is patient and communicates well.
- Can dedicate time each week to fill out ratings and reviews.
- Is proficient in using the FleetPulse system.

A supervisor may oversee multiple OJTs, but should not have more than 3 OJTs assigned at one time to maintain quality of mentorship.

20. Chapter 19: Daily Attendance Procedure

19.1 How Attendance Works

Attendance is recorded daily. Each OJT checks in when they arrive and checks out when they leave. The system records timestamps and prevents duplicate check-ins on the same day.

19.2 OJT — Checking In

1. Log into the system (redirected to Training Dashboard automatically).
2. On the Attendance card, click "**Check In**".
3. The current time is recorded as `time_in`.
4. A confirmation message appears: "Checked in at [time]."

19.3 OJT — Checking Out

1. Before leaving for the day, ensure all tasks or training activities for the day are complete.
2. On the Attendance card, click "**Check Out**".
3. The current time is recorded as `time_out`.
4. A confirmation message appears: "Checked out at [time]."

19.4 Attendance Rules

- An OJT can only check in **once per day**. If they forget to check in, they should notify their supervisor to record the attendance manually.
- Checking out is optional but recommended. An OJT who forgets to check out can still check out the next day (the system records the previous day's check-out time).
- If an OJT is absent, no attendance record is created for that day. The supervisor may note the absence in the weekly review.

19.5 Attendance Reports (Supervisor Access)

Supervisors can view an OJT's attendance history from the Training Dashboard or by navigating to the specific OJT's profile page. The attendance log shows each day's check-in and check-out times, with total hours calculated.

21. Chapter 20: Task Rating & Skills Assessment

20.1 Purpose of Task Ratings

Task ratings track an OJT's progress on specific maintenance tasks. Each rating records:

- What task was performed (linked to PM task template or custom task name).
- Who rated them (the supervisor).
- A numerical score (1-5).
- Comments or observations.

20.2 Rating Scale

Score	Label	Description
1	Novice	Requires constant supervision. Cannot perform task independently. Needs significant guidance on safety, tool selection, and procedure.
2	Beginner	Can perform task with step-by-step guidance. Knows the general process but needs coaching on technique and efficiency.
3	Competent	Can perform task independently with minimal supervision. Follows the correct procedure. Completes work within reasonable time. Understands safety requirements.
4	Proficient	Performs task efficiently and consistently. Can identify and correct common issues without assistance. Understands underlying principles. Good diagnostic ability.
5	Expert	Can perform task flawlessly. Can teach others. Able to handle complex or unusual variations. Understands the system-level impact of the task.

20.3 Creating a Task Rating (Supervisor)

1. Navigate to the Training Dashboard or the specific OJT's profile.
2. Click **"New Rating"**.
3. Complete the form:
 - **OJT:** Select the trainee you are rating.
 - **Rating:** Select 1-5 based on the scale above.
 - **PM Task:** Optional. If the OJT performed a specific PM task, select it from the dropdown (populated from the 154 PM templates). This links the rating to a standardized task for progress tracking.
 - **Task Name (Custom):** If the task is not a standard PM template, type a custom description. Example: "Assisted with transmission removal," "Practiced diagnostic scan tool usage."
 - **Comments:** Provide specific, actionable feedback. Mention what was done well AND what needs improvement. Avoid vague comments like "Good job" — instead say "Performed brake lining inspection correctly, measured all positions, identified the right-rear as needing replacement. Needs to work on speed — took 45 minutes vs. the standard 20 minutes."

4. Click **"Save Rating"**.
5. The OJT can view the rating on their Training Dashboard.

20.4 Rating Guidelines for Supervisors

- **Rate frequently:** Aim for at least 3-5 ratings per OJT per week. Frequent, small feedback is more effective than infrequent, lengthy evaluations.
- **Rate promptly:** Complete the rating while the task is still fresh in your mind — ideally within 2 hours of the OJT completing the task.
- **Be objective:** Base the rating on the OJT's demonstrated performance on that specific task, not on your general impression of them. A competent mechanic may still be a novice at a specific new task.
- **Include positive and constructive feedback:** Even a score of 3 (Competent) should note what was done correctly plus one area for improvement.
- **Use the system:** Ratings feed into the OJT's average score and provide a quantitative record of skill development over time.

22. Chapter 21: Weekly Review & Progress Tracking

21.1 Purpose of Weekly Reviews

Weekly reviews provide a structured, holistic evaluation of the OJT's progress. Unlike task ratings (which are task-specific), the weekly review covers overall performance, attitude, safety awareness, and professional development.

21.2 Creating a Weekly Review (Supervisor)

1. Navigate to the Training Dashboard or the specific OJT's profile.
2. Click **"New Review"**.
3. Complete the form:
 - **OJT:** Select the trainee.
 - **Week Start / Week End:** The Monday-to-Sunday (or applicable work week) period being reviewed.
 - **Overall Score:** 1-5 scale. This is a holistic score for the week, considering: task performance, safety behavior, punctuality, teamwork, attitude, and initiative. See the rating scale in Chapter 20.2 for guidance.
 - **Strengths:** What did the OJT do well this week? Be specific: "Demonstrated strong initiative by organizing the tool cart without being asked. Showed good understanding of brake system inspection procedure."
 - **Areas for Improvement:** What specific skills or behaviors need development? "Needs to improve speed on oil changes — target is under 30 minutes including documentation. Also needs to remember to replace the drain plug gasket every time — missed it twice this week."
 - **Supervisor Notes:** Any additional comments, concerns, or recommendations. This may include: recommendations for additional training, observations about attitude or teamwork, safety incidents or near-misses, notable achievements.
4. Click **"Save Review"**.

21.3 Review Cadence

Period	Review Frequency	Focus
Weeks 1-4 (First Month)	Weekly	Safety awareness, tool familiarity, basic skills, workshop orientation, punctuality, attitude
Weeks 5-12 (Months 2-3)	Every 2 weeks	Skill development on specific PM procedures, diagnostic thinking, speed vs. quality balance
Weeks 13+ (Ongoing)	Monthly	Independence level, ability to handle complex tasks, readiness for unsupervised work

21.4 OJT Dashboard — What the Trainee Sees

When an OJT logs in, they see a dashboard with the following sections:

Section	Content	Purpose
Attendance Card	Check In / Check Out buttons. Shows today's time_in and time_out if recorded.	Quick attendance logging
My Ratings	Average score across all task ratings, total number of ratings received.	See overall performance trend
Training Info	Start date, supervisor name, status (active/inactive).	Know who to go to for feedback
Recent Ratings	Last 10 task ratings with scores and comments.	Review specific feedback
Weekly Reviews	Last 5 weekly review summaries with scores and key points.	Track long-term progress

21.5 Supervisor Dashboard — What the Supervisor Sees

Section	Content	Purpose
My OJTs	List of active trainees with links to their profiles.	Quick access to each OJT
Recent Ratings Given	Last 20 task ratings, showing OJT name, task, score, and date.	Review recent feedback given
Weekly Reviews	Recent reviews with buttons to create new reviews.	Track review completion

21.6 Interpreting OJT Progress

Use the trend in average scores over time to assess development:

- **Upward trend** (scores increasing over weeks): OJT is learning and developing as expected.
- **Flat trend at 3-4** (scores consistently in the Competent-Proficient range): OJT is making solid progress. Look for opportunities to introduce more challenging tasks.
- **Flat trend at 1-2** (scores staying at Novice-Beginner after 4+ weeks): This is a red flag. Investigate: Is the OJT receiving adequate instruction? Are the tasks too advanced for their current level? Is there an attitude or motivation issue? Consider adjusting the training approach or having a direct conversation with the OJT.
- **Downward trend** (scores decreasing): Investigate immediately. Possible causes: boredom or lack of challenge, personal issues affecting performance, inadequate supervision, or a mismatch between the OJT and the trade.

Note on OJT Role & System Access: OJT/Trainee accounts have the most restricted access in the system. They can only see the Training Dashboard, truck list (read-only), and their own ratings and reviews. They

cannot create or modify any operational records. This is intentional — their role is to learn, not to perform unsupervised work.

Volume 7 — Performance & Analytics

23. Chapter 22: Mechanic KPI Reports

22.1 Purpose of KPI Tracking

Key Performance Indicators (KPIs) provide objective data on individual and team performance. The system generates KPI reports automatically from the data entered during Job Order execution (line items, hours, parts). This chapter explains each KPI metric and how to interpret it.

22.2 Mechanic / Staff KPI Table

Navigate to **Sidebar** → **KPI Reports** to view the Mechanic KPI table. The table shows the following columns for each mechanic:

Column	Source Data	What It Measures
Mechanic	User profile — name and specialization	Who the data belongs to
Jobs Completed	Count of closed JOs assigned to this mechanic	Workload volume — how many jobs the mechanic has finished
Total Labor Hours	Sum of actual hours recorded on all line items	Total time spent on all completed work
Average Hours per Job	Total Labor Hours ÷ Jobs Completed	Average job size — larger numbers may indicate complex work or slower performance
Total Parts Cost	Sum of all part costs (qty × unit cost) across all JOs	Total cost of materials used by this mechanic
Average Parts Cost per Job	Total Parts Cost ÷ Jobs Completed	Typical parts spend per job — useful for benchmarking

22.3 Interpreting Mechanic KPI Data

When evaluating performance, consider these questions:

- **Volume:** Is the mechanic completing a reasonable number of jobs compared to their peers? A mechanic with significantly fewer closed JOs may be: spending too much time per job (inefficiency), assigned more complex jobs (check the job types), or frequently waiting for parts or instructions.
- **Efficiency:** Compare Estimated Hours (from the line items) vs. Actual Hours (recorded when the line item was marked Done). A mechanic who consistently exceeds estimated hours may need: more training on the specific task, better tools or equipment, or more realistic estimation practices.

- **Cost:** Compare parts cost across mechanics doing similar work. A mechanic with consistently higher parts costs may be: replacing parts that could be repaired, not diagnosing root causes (replacing symptoms instead), or working on trucks with more severe wear (not the mechanic's fault — investigate further).
- **Quality:** While not directly tracked in KPI numbers, look for: rework (same truck back in the shop with the same issue within 30 days — indicates incomplete diagnosis or repair), QC failures (work that was flagged during QC inspection).

Best Practice: KPI data is most useful for identifying trends over time, not for evaluating a single point in time. Review KPIs monthly and look for patterns. A single month of low numbers is not a concern. Three months of declining numbers warrants a conversation.

22.4 Contractor Performance View

The KPI page also shows a Contractor Performance table with:

- **Company:** Contractor company name.
- **Total Jobs:** All JOs assigned to this contractor.
- **Completed:** Closed JOs.
- **Open:** JOs currently in progress.
- **Completion Rate:** Completed ÷ Total as a percentage.

Use this data to identify contractors who consistently complete work on time and those who may need follow-up on open or overdue assignments.

24. Chapter 23: Predictive Analytics Interpretation

23.1 Overview

The Predictive Analytics dashboard provides a visual overview of fleet health using five interactive charts. All charts are generated from live data — every page load queries the current database state. This is not a machine-learning model; it is a real-time data visualization tool for informed decision-making.

Navigate to **Sidebar** → **Predictive Analytics** (Admin+ access).

23.2 The Five Charts

23.2.1 Failure Frequency (Bar Chart)

What it shows: The top 10 trucks with the most Repair-type Job Orders (breakdowns).

How to read it: Taller bars = more breakdowns. A truck that appears in this chart consistently month-over-month may need: a comprehensive diagnostic evaluation to find the root cause of repeated failures, consideration for retirement if repair costs exceed the truck's value, or a review of its PM schedule — are the right PM tasks assigned at the right intervals?

Action items:

- Trucks in the top 3 for three consecutive months: schedule a full diagnostic review.
- New entries in the top 10: investigate what changed (new routes, new driver, recent repair?).
- A truck that drops out of the top 10: identify what changed positively and apply that lesson to other trucks.

23.2.2 PM Compliance (Donut Chart)

What it shows: A donut chart breaking down all PM schedules across the fleet by status: OK (green), Due (yellow), Overdue (red), No Data (gray).

How to read it:

- **Target:** $\geq 85\%$ of all PM schedules should be in OK (green) status.
- If Due (yellow) exceeds 10%: PM scheduling needs attention — consider increasing shop capacity or prioritizing PMs over repairs for a period.
- If Overdue (red) exceeds 5%: This is a serious concern. Overdue PMs indicate maintenance is being deferred, which increases the risk of breakdowns. Investigate systemic causes: not enough mechanics, parts availability issues, or lack of prioritization from operations.
- If No Data (gray) is significant: Some PM schedules have never been completed. This may indicate new templates that were not synced, or trucks that have not been through a full PM cycle.

Action items:

- Review Overdue PMs weekly — assign JOs for each overdue item.
- Target: reduce overdue to 0% within one month by dedicating one day per week to "overdue catch-up."

23.2.3 Cost Trends (Line Chart)

What it shows: Monthly breakdown of parts cost and labor hours cost over the last 6 months.

How to read it:

- **Upward trend:** Costs are increasing. Investigate: Are parts prices rising? Is the fleet aging and requiring more repairs? Are new trucks being added? Is there a seasonal pattern (more breakdowns during rainy season or summer)?
- **Downward trend:** Positive sign — the fleet may be getting healthier, PM may be working, or newer trucks are reducing repair needs.
- **Spike in a single month:** What happened? Major accident? Multiple breakdowns on the same truck? Large parts purchase (e.g., replaced all tires on multiple trucks)? Identify the event that caused the spike to understand if it was one-time or recurring.

Action items:

- Investigate any month-over-month increase exceeding 15%.
- Compare parts vs. labor ratio: a rising parts ratio may indicate more heavy repairs (replacing components) compared to labor-only work (adjustments, inspections).

23.2.4 Overdue Alerts (Bar Chart)

What it shows: The trucks with the highest number of overdue PM schedule items.

How to read it: Each bar represents one truck. The taller the bar, the more overdue items that truck has accumulated. A truck with many overdue items is at high risk of breakdown and may also have difficulty accessing depots (some depots check PM compliance before allowing loading).

Action items:

- For any truck with 5+ overdue items: schedule a comprehensive PM catch-up within the week.
- For trucks that repeatedly appear on this chart: is the PM schedule realistic for this truck's utilization rate? Consider adjusting the interval or adding shop capacity.

23.2.5 Top Replaced Parts (Bar Chart)

What it shows: The 10 most frequently replaced parts across all Job Orders, with total cost.

How to read it: This chart helps with inventory planning. If a specific part appears consistently at the top, ensure you have adequate stock on hand to avoid delays waiting for parts.

Action items:

- For the top 5 parts: maintain a minimum stock level (safety stock) in the parts inventory.
- If a part suddenly appears in the top 10 (not previously common): investigate — is there a quality issue with the current batch? Is there a fleet-wide issue (e.g., a bad batch of tires or filters)?
- Use this data to negotiate bulk pricing with suppliers for frequently replaced parts.

23.3 Using Predictive Analytics for Decision-Making**Monthly Review Process:**

1. Open the Predictive Analytics dashboard.
2. Review all five charts — this takes approximately 5 minutes.
3. Note any significant changes from the previous month's review.
4. Identify the top 3 issues to address (e.g., "Truck UNIT-004 has 8 overdue PM items and appears in the top 3 for failure frequency — prioritize it").

5. Assign action items to specific people with deadlines.
6. Record the key observations and decisions in the meeting minutes or system notes.

Quarterly Fleet Health Review:

- Export the data from KPI Reports and Predictive Analytics for a broader review.
- Compare quarter-over-quarter trends in all five chart areas.
- Evaluate if the fleet PM program is improving overall fleet reliability (are breakdowns decreasing? Is PM compliance increasing?).
- Adjust PM intervals, task assignments, or shop procedures based on the data.

Volume 8 — Safety & Compliance

25. Chapter 24: General Workshop Safety & PPE

24.1 The Safety Culture Principle

Safety is not a set of rules — it is a mindset. Every mechanic, driver, and supervisor in this organization is expected to:

- **Stop any task** that feels unsafe, without fear of reprisal.
- **Speak up** when they see an unsafe act or condition involving anyone.
- **Lead by example** — wear proper PPE even for "quick" tasks.
- **Report every near-miss** — a near-miss today prevents an accident tomorrow.

⚠ **Golden Rule:** If you cannot do a job safely, do not do it at all. No job is so urgent that it cannot be done safely.

24.2 General Workshop Safety Rules

The following rules apply at all times within the workshop, depot, or any location where truck maintenance is performed:

1. **Housekeeping:** Keep all work areas clean and organized. Clean up spills immediately. Return tools to their designated storage after use. Do not leave tools, parts, or debris on the floor — they are tripping hazards. Dispose of rags contaminated with oil, fuel, or solvent in approved fire-safe containers.
2. **Spill Containment:** Any spill of oil, fuel, coolant, or other fluids must be contained immediately using absorbent pads or granular absorbent. Report spills over 5 liters to the supervisor. Do not wash spills down drains — use proper disposal procedures.
3. **Ventilation:** Ensure adequate ventilation when working with solvents, paints, welding equipment, or running engines indoors. Carbon monoxide from running engines is lethal in enclosed spaces. Use exhaust extraction hoses for all running engines.
4. **Fire Prevention:** No smoking in the workshop. Store flammable liquids (fuel, solvents, paints) in approved fire-rated cabinets. Keep ignition sources away from battery charging areas (hydrogen gas). Know the location of all fire extinguishers and emergency exits.
5. **Compressed Air Safety:** Never use compressed air to clean clothing, skin, or hair — air can enter the bloodstream through small cuts and cause fatal embolisms. Never point an air nozzle at another person. Use pressure regulators — do not exceed rated tool pressure.
6. **Confined Space:** Tank compartments, pits, and storage tanks are confined spaces. NEVER enter a confined space without: a confined space permit, atmospheric testing (oxygen, LEL, toxic gas), continuous ventilation, a safety harness and retrieval line, and a trained attendant outside the space.

7. **Hot Work:** Welding, grinding, cutting, or any activity that produces sparks requires a hot work permit. Clear the area of flammable materials within 10 meters. Have a fire watcher with an extinguisher present. Monitor for 30 minutes after work is completed.
8. **Drugs and Alcohol:** Working under the influence of drugs or alcohol is strictly prohibited. Prescription medications that cause drowsiness must be reported to the supervisor before starting work.
9. **Personal Conduct:** Horseplay, running, and practical jokes in the workshop are prohibited. Use designated walkways. Do not jump off loading docks or truck beds.
10. **Reporting:** All injuries, no matter how minor, must be reported to the supervisor and recorded in the incident log. All near-misses must be reported and investigated.

24.3 Personal Protective Equipment (PPE) — Complete Reference

PPE is the last line of defense after engineering controls and safe work practices. The following table specifies the minimum PPE required for each task category.

Body Part	PPE Item	When Required	Specification
Head	Hard hat / Safety helmet	Any work under a raised vehicle, on top of tank, in pit, or where overhead hazards exist (tools being passed, crane operations)	ANSI Z89.1 Type I or II. Replace if cracked or after any impact.
Eyes	Safety glasses (primary)	ALL work in the workshop at all times. Grinding, welding, cutting, battery work, compressed air use, chemical handling.	ANSI Z87.1 rated. Z87+ for high-impact. Side shields required. Prescription safety glasses acceptable.
Eyes (high hazard)	Goggles or face shield	Grinding (over safety glasses), battery acid handling, chemical mixing, welding (with proper shade), pressurized fluid systems	Face shield must be used OVER primary safety glasses — not instead of.
Hearing	Earplugs or earmuffs	Running engines, air tools (impact wrenches, grinders), pneumatic hammer, compressed air blow-off, any area with noise above 85 dB	NRR 25+ for earplugs, NRR 20+ for earmuffs. Double protection (plugs + muffs) for >100 dB.
Hands	General work gloves	Material handling, rigging, sweeping, yard work	Leather or knit with latex coating. Inspect for tears before use.
Hands	Mechanic's gloves (nitrile-coated)	General mechanical work where dexterity is needed	Cut-resistant level A2-A5. Not for chemical handling.
Hands	Chemical-resistant gloves	Handling fuels, solvents, acids, caustics, battery acid, coolant concentrate	Nitrile (for petroleum), Neoprene (for acids), PVC (for caustics). Check manufacturer compatibility chart.

Body Part	PPE Item	When Required	Specification
Hands	Welding gloves	Welding, cutting, torch work	Leather gauntlet style. Must be dry — wet leather conducts electricity.
Hands	High-voltage gloves	Work on hybrid/electric vehicle high-voltage systems	Class 00 (500V) or Class 0 (1000V). Must be dielectric tested within last 6 months.
Feet	Steel-toe boots	ALL work in the workshop, yard, or depot at all times.	ASTM F2413-18 rated. Minimum 75 mm heel height. Oil-resistant sole. Lace-up type (not pull-on) for secure fit.
Feet	Metatarsal guard boots	Handling heavy parts, drum/rotor replacement, rigging heavy equipment	Integrated metatarsal guard or add-on guard over laces.
Body	Coveralls / Shop coat	All mechanical work	Cotton or cotton-poly blend (NOT 100% synthetic — melts in fire). Close-fitting at cuffs. No loose flaps or dangling drawstrings.
Body	High-visibility vest	Any work near moving vehicles, in yard, at depot loading bays, night work	ANSI 107 Class 2 or 3. Fluorescent yellow-green or orange with reflective stripes.
Body	Welding jacket / apron	Welding, cutting, torch work	Flame-resistant treated leather or cotton. No polyester or nylon.
Respiratory	N95 dust mask	Brake dust (due diligence — see danger note), dry sweeping, grinding (non-toxic materials)	NIOSH N95 approved. Disposable — discard after use or when moist.
Respiratory	Half-face respirator with organic vapor cartridge	Painting, solvent use, fuel system work, confined space standby (NOT for entry)	NIOSH approved. Cartridge type: organic vapor (OV) + P100 particulate. Fit-tested required — facial hair breaks the seal.
Respiratory	Full-face respirator	Any task generating hazardous vapors where eye protection is also needed	NIOSH approved with appropriate cartridge. Must be fit-tested.
Fall Protection	Full-body harness + lanyard	Work above 2 meters on tank top, truck roof, or scaffold without guardrails	ANSI Z359.1 or equivalent. Lanyard with shock absorber. Inspect before each use. Never use a body belt for fall arrest.

⚠ **Critical — Brake Dust Hazard:** Brake dust may contain asbestos, even in modern brake linings where it is banned. **NEVER** use compressed air to clean brake drums or backing plates. Use a HEPA vacuum or wet

cleaning method approved for asbestos-containing materials. Wear a properly fitted N100 or P100 respirator when working on brakes.

24.4 PPE Inspection & Care

PPE Item	Inspect Before Each Use	Replace When
Hard hat	Cracks, dents, suspension system (straps, ratchet)	After any impact, or every 5 years (date on inside of shell)
Safety glasses	Scratches (that impair vision), cracks, bent frames	Scratched or cracked lenses
Hearing protection	Earplugs: cleanliness, deformation. Earmuffs: cracks in cup, torn cushions, weak headband tension	Earplugs: daily (disposable). Earmuffs: worn cushions or cracked cups
Gloves	Holes, tears, chemical permeation (swelling, discoloration), stiffness	Any defect found. Chemical gloves: discard after 1 hour total exposure to solvents
Steel-toe boots	Sole separation, worn tread (smooth), exposed toe cap, cracked leather, torn stitching	Tread worn through to midsole, or toe cap exposed, or any sole separation
Respirator	Seal integrity (stretch and release rubber — cracks?), exhalation valve flap, strap elasticity, cartridge expiration date	Cartridges: open date + 6 months (sealed) or after use (discard after each shift for OV). Facepiece: cracked or distorted rubber
Fall harness	Webbing cuts, fraying, UV damage (faded or powdery), buckle deformation, D-ring cracks, stitch integrity (label must be legible)	After any fall, or every 5 years (date on label)

24.5 Proper Tool Handling & Safety

24.5.1 Hand Tools — General Rules

1. **Use the right tool for the job.** Never use a wrench as a hammer, a screwdriver as a pry bar, or pliers as a wrench. Using the wrong tool damages both the tool and the fastener, and increases injury risk.
2. **Inspect before use.** Check wrenches for cracked jaws, sockets for worn drive ends, screwdrivers for mushroomed tips, hammers for loose heads. Any damaged tool must be removed from service and tagged for repair or disposal.
3. **Pull — do not push.** Whenever possible, pull a wrench toward your body rather than pushing away. If you must push, use the palm of your hand with an open hand (not a clenched fist) to avoid smashed fingers if the wrench slips.
4. **Never extend a wrench with a pipe or cheater bar** unless the wrench is specifically rated for such use. Using a cheater bar on a standard wrench can cause the jaw to snap catastrophically.
5. **Keep cutting tools sharp.** Dull tools require more force and are more likely to slip. Store sharp tools with blades covered or in a designated drawer.
6. **Do not carry tools in pockets.** Use a tool belt or tool cart. Sharp or pointed tools in pockets can cause injury when bending or sitting.

24.5.2 Specific Hand Tool Safety

Tool	Safe Use Rules	Do NOT
Wrenches (open-end, box-end, combo)	Pull toward you when possible. Ensure the jaw fully engages the fastener. Use box-end for breaking loose tight fasteners; use open-end for speed on loose fasteners.	Do not use wrenches with spread jaws (more than 10% wider than nominal). Do not strike wrenches with a hammer (use a strike-face wrench or impact socket).
Socket wrenches / Ratchets	Always use the correct size socket (SAE or metric — never mix). Ensure the socket fully seats on the fastener. Use impact sockets with impact tools only (chrome sockets can shatter).	Do not use ratchets for breakaway torque beyond their rated capacity. Do not use a ratchet with a cheater bar. Do not use extensions at extreme angles.
Torque wrenches	Always "click" or "release" at the set torque — do not continue pulling. Store at the lowest setting (zero out the spring). Calibrate annually or after 5,000 cycles. Use a smooth, steady pull — not jerky.	Do NOT use a torque wrench to break loose fasteners. Do NOT drop or impact a torque wrench (damages internal mechanism). Never exceed the maximum rated torque of the tool.
Screwdrivers	Match the tip to the screw head exactly — undersized tips will cam out and damage the head. Use the correct type (Phillips, flat, Pozidriv, Torx, etc.).	Never use a screwdriver as a pry bar, chisel, punch, or scraper. Never hold a workpiece in your hand while driving a screw — use a vise or clamp.
Hammers	Use the correct hammer for the material: ball-peen for metal, dead-blow for striking without rebound, brass or soft-face for delicate work. Strike the work surface squarely — glancing blows are dangerous.	Do not use a hammer with a loose head or cracked handle. Do not strike one hammer with another. Do not use a hammer as a wedge or pry bar.
Pliers / Vise-Grips	Use lineman's or combination pliers for gripping and twisting. Use locking pliers (Vise-Grips) only for temporary holding — not as a permanent substitute for the correct wrench. Use diagonal cutters for wire cutting only.	Do not use pliers on nuts or bolts if a wrench is available — pliers will round the corners. Do not use wire cutters on hardened wire (springs, cables) — use dedicated cable cutters.
Files / Rasps	Always use a file with a handle — the tang can puncture your palm if the file slips. Apply pressure on the forward stroke only; lift on the return stroke to avoid dulling the teeth.	Never use a file as a pry bar. Never use a file without a handle. Do not use files on hardened steel (they will skip and may fracture).
Chisels / Punches	Keep the striking face dressed (not mushroomed). Wear safety glasses under a face shield. Use a hammer of appropriate weight for the chisel size. Hold the chisel at the correct angle (60° for cutting, 30° for shearing).	Do not use a chisel with a mushroomed head — chips can break off and cause eye injury. Redress the head on a grinder before use.

24.5.3 Power Tool Safety

Tool Type	Safe Use Rules
Impact Wrench (Pneumatic / Electric)	Use only impact-rated sockets (black oxide finish). Do NOT use chrome sockets — they shatter. Set torque regulator to the minimum needed for the task before applying full power. Hold the tool with both hands. Always wear safety glasses and hearing protection. Lubricate pneumatic tools daily with inline oiler.
Angle Grinder	Use ONLY grinding/cutting wheels rated for the tool's maximum RPM (check the label on the wheel). Inspect the wheel for cracks before mounting — tap-test with a non-metallic object: a good wheel rings, a cracked wheel sounds dull. Do not use a wheel that has been dropped. Always use the guard in position between your body and the wheel. Wear face shield over safety glasses. Use two hands. Allow the tool to reach full speed before contacting the work. Never use a cutting wheel for grinding (side-loading causes wheel breakage). Disconnect power before changing wheels.
Drill (Electric / Pneumatic)	Secure the workpiece with clamps — never hold it in your hand. Use sharp drill bits at the correct speed (slow for large bits, fast for small bits). Apply steady, moderate pressure — too much pressure can bind the bit and cause the drill to spin violently. Use cutting oil for metal drilling. Remove chips frequently. Disconnect power before changing bits or clearing jams.
Welding Equipment	Inspect cables for cracked insulation and loose connections — damaged cables cause voltage drop and fire hazard. Wear proper welding helmet with correct shade lens (shade 10-12 for arc welding, shade 5-8 for gas welding). Wear dry welding gloves. Ensure adequate ventilation — welding fumes are toxic. Keep a fire extinguisher within reach. Clear all flammable materials within 10 meters. Never weld on a closed container (tank, drum, pipe) — it can explode. Have a fire watcher present.
Air Ratchet / Die Grinder	Check the hose connection for secure locking (use a whip check on connections). Lubricate before use. Wear hearing protection — air tools are typically 90-100 dB. Control the tool with both hands at all times. Do not exceed the manufacturer's maximum pressure rating (typically 90 psi).
Hydraulic Jack / Press	Inspect hydraulic hoses for cuts or bulges before use — a hose failure under pressure can whip violently. Do not exceed the rated capacity of the jack. Always use jack stands rated for the vehicle weight BEFORE working under the vehicle. Never work under a vehicle supported only by a jack. Use a block of wood between the jack saddle and the vehicle frame to distribute load and prevent slipping. Inspect for hydraulic fluid leaks — a leaking jack can drop without warning.

24.5.4 Vehicle Support Safety (Jacks & Stands)

⚠ **CRITICAL — LIFTING SAFETY:** The single most deadly mistake in truck maintenance is working under a vehicle supported only by a hydraulic jack. A jack can fail suddenly due to seal blowout, hydraulic leak, or mechanical failure. **ALWAYS use properly rated jack stands.** Never use cinder blocks, concrete blocks, or stacked lumber as vehicle supports — they can crumble without warning.

Proper Lifting Procedure:

1. Park on solid, level ground. Engage parking brake. Chock both sides of the wheels that remain on the ground.

2. Identify correct lifting points on the frame rails or axle (refer to manufacturer's manual). Never lift on the differential, steering components, or body panels.
3. Position the jack saddle with a block of wood (to prevent metal-on-metal contact and slipping).
4. Lift the vehicle slowly and steadily. Watch for any shifting or instability.
5. Once at desired height, place jack stands under the frame at the designated lift points. Use stands of the same type, in pairs, at the same height setting.
6. Adjust the ratchet or pin mechanism — ensure it is fully engaged and cannot collapse.
7. **Gently lower the vehicle onto the stands.** Do not lower with a bang. Lower until the full weight is on the stands and the jack is free.
8. Test the stability by pushing the vehicle laterally at shoulder height. If it rocks, adjust stands or reposition.
9. Only then may you work under the vehicle. The jack may be left in place as a secondary safety measure but must NOT bear the main load.

Jack Stand Ratings: Use stands rated for at least 1.5x the weight on that end of the vehicle. For a typical loaded tanker weighing 20,000 kg on the rear tandem: use stands rated at 30,000 lbs (13,600 kg) per pair minimum — two stands per axle.

24.6 Chemical Handling & Hazard Communication

24.6.1 Understanding Chemical Labels (GHS / HMIS)

Every chemical in the workshop must have a legible label. The Globally Harmonized System (GHS) uses nine pictograms to communicate hazards:

Pictogram	Meaning	Examples in Workshop
 (flame)	Flammable	Gasoline, diesel, solvents, thinners, brake cleaner
 (skull & crossbones)	Acute toxicity (fatal or toxic)	Coolant concentrate (ethylene glycol), some degreasers, battery acid
 (corrosion)	Corrosive — causes skin burns, eye damage	Battery acid (sulfuric acid), caustic degreasers, wheel cleaner
 (exclamation mark)	Irritant, skin sensitizer, or less severe toxicity	Used engine oil (skin irritant/carcinogen), brake dust
 (environment)	Aquatic toxicity — harmful to the environment	Used oil, coolant, fuel, any chemical that should not enter drains
 (exploding bomb)	Explosive or self-reactive	Aerosol cans (when heated), compressed gas cylinders
 (health hazard)	Respiratory sensitizer, carcinogen, organ damage	Diesel exhaust, welding fumes, silica dust, used engine oil
 (flame over circle)	Oxidizer — may cause or intensify fire	Oxygen cylinders, some pool chemicals (rare in workshop)

Pictogram	Meaning	Examples in Workshop
 (gas cylinder)	Gas under pressure	Oxygen, acetylene, MIG welding gas cylinders, air tanks

24.6.2 Safe Chemical Handling Rules

1. **Read the Safety Data Sheet (SDS)** before using any chemical you are unfamiliar with. SDS must be available in the workshop (physical binder or digital access).
2. **Use the correct chemical for the job.** Do not substitute — using the wrong solvent can damage components, create toxic fumes, or cause a fire.
3. **Never mix chemicals.** Mixing bleach with ammonia produces deadly chlorine gas. Mixing acids with solvents can cause explosions. If in doubt, do not mix.
4. **Use proper dispensing equipment.** Use pumps, funnels, or spouts — never pour from large containers without control. Use a bonding wire when transferring flammable liquids between metal containers (prevents static spark).
5. **Keep containers closed** when not in use. Open containers evaporate, creating fire hazards and exposing workers to vapors.
6. **Waste disposal:** Used oil, coolant, solvents, and oily rags are hazardous waste. Dispose of each in designated, labeled containers. Never pour chemicals down drains, onto the ground, or into trash bins.
7. **Spill response:** Know the location of the spill kit. For chemical spills: contain with absorbent pads or booms. For fuel spills: eliminate all ignition sources. For acid spills: neutralize with baking soda before absorbing. Report spills to supervisor — documentation may be required for regulatory compliance.

24.6.3 Common Workshop Chemicals & Specific Hazards

Chemical	Primary Hazard	Safe Handling Notes
Gasoline / Diesel	Flammable (gasoline flash point: -43°C), toxic vapors, carcinogenic (diesel exhaust)	No smoking. Use only approved safety cans. Bond and ground containers during transfer. Store in flammable liquids cabinet. Limit stored quantity.
Engine Oil (new)	Mild irritant, slip hazard (spills)	Wear gloves. Clean up spills immediately with absorbent.
Engine Oil (used)	Skin carcinogen (contains PAHs from combustion), contains heavy metals	Wear nitrile gloves. Avoid skin contact. Do not use for lubrication or rust protection. Dispose in used oil tank. Used oil is a reportable hazardous waste.
Coolant (ethylene glycol)	Toxic (sweet taste attracts animals — lethal dose ~100 mL for humans), SLIGHTLY flammable	Store in labeled containers, away from children and animals. Clean spills with absorbent — glycol is toxic to pets. Used coolant is hazardous waste.
Brake Cleaner	Flammable (aerosol), irritant, solvent vapors	Use in well-ventilated area. No smoking. Do not spray near open flame or hot surfaces. Aerosol cans may explode if overheated.

Chemical	Primary Hazard	Safe Handling Notes
Battery Acid (sulfuric acid 30-50%)	Corrosive — severe burns on contact, produces hydrogen gas (explosive) when charging	Wear acid-resistant gloves, goggles, and apron. Add acid to water, never water to acid (violent boiling/eruption). Neutralize spills with baking soda. Charge batteries in well-ventilated area (hydrogen gas).
Parts Washer Solvent (Stoddard solvent / kerosene-type)	Flammable, skin irritant, vapor inhalation hazard	Use with ventilation. Wear nitrile gloves. Never use for cleaning hands. Do not use near hot surfaces or electrical equipment.
Acetylene (welding)	Extremely flammable, can decompose explosively above 15 psi	Never use above 15 psi outlet pressure. Keep cylinders upright and chained. Close valve when not in use. Check for leaks with soapy water (never flame). Flashback arrestors required at regulator and torch.
Compressed Oxygen (welding)	Strong oxidizer — greatly intensifies combustion. Oil + oxygen = spontaneous fire/explosion	NEVER use oil or grease on oxygen fittings or valves. Keep oxygen cylinders away from fuel cylinders (minimum 6 meters or fire-rated wall). Open cylinder valve slowly. Use only oxygen-rated regulators and hoses (green).

24.7 Lockout / Tagout (LOTO) Procedure

Lockout/Tagout is used to ensure equipment is isolated from all energy sources before maintenance begins. The following energy sources exist in the workshop and on trucks:

- **Electrical:** Battery power (12/24V), shore power, charger connections
- **Pneumatic:** Compressed air in brake system, shop air lines
- **Hydraulic:** Power steering, hydraulic lift gates, water brake systems
- **Mechanical (stored energy):** Spring brake chambers, suspension springs, pressurized tanks
- **Thermal:** Hot engine, exhaust, cooling system under pressure
- **Chemical:** Fuel in tanks/lines, battery acid, refrigerants

Six-Step LOTO Procedure:

1. **NOTIFY** all affected personnel that equipment will be locked out and why.
2. **SHUT DOWN** the equipment using normal shutdown procedures (turn off engine, etc.).
3. **ISOLATE** all energy sources: disconnect battery (negative first), close fuel valves, drain air pressure, release spring brakes, etc.
4. **LOCK & TAG** each isolation point with your personal padlock and a tag stating: your name, date, reason, and expected return time. Tags must be weather-resistant and attached with a non-reusable zip tie or wire.
5. **VERIFY** zero energy: attempt to start the engine (it should not start), test for voltage with a multimeter, verify air pressure is zero, manually release stored energy (e.g., pry against a spring with caution), check that the cooling system is depressurized.
6. **PROCEED** with work. Only the person who applied the lock may remove it. To remove: complete the work, ensure all tools are cleared, ensure all guards are replaced, notify affected personnel, remove locks and tags.

⚠ **Exception — "Hot Work" LOTO:** Welding or grinding on a fuel tanker requires additional steps: purge the tank compartment (gas-free certificate), open all manlids for ventilation, continuously monitor with a combustible gas detector (LEL), have a fire watcher present with extinguisher, and secure a hot work permit from the supervisor.

24.8 Fire Safety

24.8.1 Fire Classes & Extinguisher Types

Class	Fuel Type	Example	Extinguisher Type
A	Ordinary combustibles	Paper, wood, cloth, trash, rubber	Water, ABC dry chemical, foam
B	Flammable liquids	Gasoline, diesel, oil, solvents, paint	ABC dry chemical, CO ₂ , foam, Purple-K
C	Energized electrical equipment	Motors, wiring, batteries, chargers, control panels	ABC dry chemical, CO ₂ (non-conductive agents only — NEVER water)
D	Combustible metals	Magnesium, aluminum powder, titanium (rare in workshop)	Class D extinguisher (specialized dry powder)
K	Cooking oils/fats	Kitchen grease fires (cafeteria — not typical workshop)	Wet chemical (class K)

Workshop Recommendation: All workshop extinguishers should be ABC-rated multi-purpose dry chemical (Type ABC, 5-10 lb capacity). One extinguisher per 200 square meters, with a travel distance of no more than 15 meters to the nearest extinguisher. Check monthly (gauge in green, tamper seal intact).

24.8.2 PASS Extinguisher Operation

1. **P**ull the pin — break the tamper seal.
2. **A**im at the base of the fire — not at the flames.
3. **S**queeze the handle — slowly and evenly.
4. **S**weep side to side — covering the entire base of the fire.

If the fire does not extinguish after one extinguisher, **evacuate immediately** and call emergency services (local fire department). Do not attempt to fight a fire that is spreading rapidly, involves gas cylinders, or is located in a confined space.

24.8.3 Fire Prevention in the Workshop

- No smoking anywhere inside the workshop or within 10 meters of fuel storage areas.
- Store flammable liquids in approved flammable storage cabinets. Cabinet doors must be self-closing and latched.
- Oily rags: store in metal containers with self-closing lids. Empty daily.
- Charging batteries: charge in a designated area with ventilation (hydrogen gas is explosive). Do not charge near flammables.
- Electrical panels: keep clear — at least 1 meter of unobstructed space. Do not store anything against electrical panels.

- No extension cords as permanent wiring. Use power bars with built-in circuit breakers.
- Hot work (welding, grinding, cutting): clear combustibles, have extinguisher ready, post a fire watch for 30 minutes after work completes.
- Fuel transfer: bond and ground containers. No smoking. Engine off. Have absorbent pads and a fire extinguisher positioned nearby.

24.9 Safe Lifting & Ergonomics

24.9.1 Manual Lifting — Proper Technique

1. **Size up the load:** Before lifting, push or tilt the object to gauge its weight. If it is too heavy (over 23 kg / 50 lb for one person), get mechanical assistance (forklift, hoist, or second person).
2. **Stance:** Feet shoulder-width apart, one foot slightly ahead of the other for stability.
3. **Bend your knees — not your back.** Keep the natural curve of your spine (do not round your back).
4. **Grip firmly:** Use full palm grip, not just fingertips. Wear gloves if the object has sharp edges.
5. **Lift with your legs:** Straighten your legs while keeping your back straight. Keep the load close to your body.
6. **Do not twist:** Turn by moving your feet, not by rotating your torso while carrying the load.
7. **Set down:** Reverse the process — bend your knees, keep back straight, lower slowly.

24.9.2 Mechanical Assistance Guidelines

Weight Range	Method	Notes
Up to 23 kg (50 lb)	Manual lifting (one person)	Use proper technique. Ask for help if awkward shape or size.
23-45 kg (50-100 lb)	Two-person lift	One person calls the count: "Lift in 3, 2, 1 — lift." Both lift and lower together.
45-90 kg (100-200 lb)	Engine hoist / cherry picker / transmission jack	Never attempt to lift these weights manually even with two people — risk of back injury if one person slips.
Over 90 kg (200 lb)	Forklift, overhead crane, or hoist	Use rated lifting equipment. Inspect slings and chains before use. Follow rigging procedures.

24.9.3 Common Ergonomic Risk Factors

Risk Factor	Examples	Solution
Overhead work	Working on top of engine, overhead brake lines, roof lighting	Use a step stool or ladder. Take breaks — overhead work is 5x more fatiguing than chest-level work. Limit to 15-minute intervals.
Kneeling / squatting	Wheel work, under-body inspection, suspension work	Use a mechanics stool (rolling) or knee pads. Take standing breaks every 10 minutes.
Awkward wrist positions	Wrenching in tight spaces, overhead fasteners	Use offset wrenches, flex-head ratchets, or universal joint sockets. Avoid bending wrists at extreme angles.

Risk Factor	Examples	Solution
Repetitive motion	Hammering, using an impact wrench continuously, grinding	Rotate tasks every 30 minutes. Use vibration-dampening gloves. Take micro-breaks.
Prolonged standing	Workbench tasks, welding	Use anti-fatigue mats. Wear comfortable boots with good arch support. Use a sit-stand stool if appropriate.

26. Chapter 25: Emergency Response Procedures

25.1 Emergency Contact Numbers

The following numbers must be posted prominently in the workshop, near the phone or communications station:

Emergency	Number	Notes
Local Fire Department	___-___	Fill in local number
Local Ambulance / EMT	___-___	Fill in local number
Local Police / Highway Patrol	___-___	Fill in local number
Nearest Hospital	___-___	Fill in name + number
Company Safety Officer	___-___	Fill in name + mobile
Fleet Supervisor (on-call)	___-___	Fill in name + mobile
Poison Control	___-___	Fill in local number

25.2 Fire Emergency

1. **SHOUT "FIRE! FIRE! FIRE!"** to alert everyone in the area.
2. **EVACUATE** the immediate area. Close doors behind you if safe to do so (contains the fire).
3. **ACTIVATE** the fire alarm if the workshop has one.
4. **ATTEMPT TO EXTINGUISH** only if ALL of the following apply:
 - The fire is small (trash can size or smaller).
 - You have a clear escape route behind you.
 - You have been trained on extinguisher use (PASS method).
 - You are using the correct extinguisher for the fire class.
 - No gas cylinders, fuel tanks, or hazardous materials are involved.
5. **If fire grows, or if you cannot extinguish it:** evacuate, call the fire department, and account for all personnel at the designated assembly point.
6. **DO NOT** re-enter the building for any reason until fire officials declare it safe.

25.3 Fuel / Chemical Spill Emergency

1. **ASSESS** the spill: is it a small drip (can be absorbed with a rag) or a major release (flowing liquid, large pool)?
2. **Small spill:** absorb with rags or absorbent pads. Place absorbed material in a hazardous waste container. Clean area with degreaser.
3. **Major spill:**
 - **Protect drains:** Place absorbent booms or drain covers around drains immediately.

- **Eliminate ignition sources:** No smoking, no electrical equipment operation, turn off any running engines in the area.
- **Contain:** Use absorbent booms to surround the spill and prevent spreading. Use granular absorbent to soak up liquid within the containment area.
- **Notify:** Call the supervisor and safety officer immediately. For fuel spills over 25 liters, regulatory reporting may be required.
- **Clean up:** Collect all contaminated absorbent materials into hazardous waste drums. Label and seal drums for proper disposal.
- **Document:** Complete a spill report with: date, time, location, product spilled, estimated volume, cause, and corrective actions taken.

25.4 Injury / Medical Emergency

1. **Call for help:** Shout for assistance or call emergency services. Do not move the injured person unless they are in immediate danger (fire, chemical exposure, collapsing structure).
2. **First Aid — life-threatening conditions (ABC):**
 - **A - Airway:** Check if the person is conscious. Tilt head back, lift chin to open airway.
 - **B - Breathing:** Look, listen, and feel for breathing for 10 seconds. If not breathing normally, begin CPR (30 chest compressions at 100-120/min, then 2 rescue breaths).
 - **C - Circulation:** Control severe bleeding with direct pressure using a clean cloth or bandage. Raise the injured area above the heart if possible. Apply a tourniquet only for life-threatening limb bleeding that cannot be controlled with direct pressure.
3. **Chemical exposure:**
 - **Eyes:** Flush with clean water from the eyewash station for at least 15 minutes. Hold eyelids open. Remove contact lenses if present.
 - **Skin:** Remove contaminated clothing. Rinse affected area with water for at least 15 minutes. Use soap and water for non-corrosive chemicals.
 - **Inhalation:** Move the person to fresh air immediately. If not breathing, begin CPR.
 - **Ingestion:** Call Poison Control. Do NOT induce vomiting unless instructed by Poison Control. Do not give anything by mouth to an unconscious person.
4. **Report:** Every injury, no matter how minor, must be reported to the supervisor and recorded in the company's incident log. This is not optional — even a small cut can become infected, and reporting protects both the worker and the company.

25.5 First Aid Kit Contents

Each workshop and each truck must carry a stocked first aid kit. Minimum contents:

Item	Workshop Kit	Truck Kit
Adhesive bandages (assorted sizes)	50	15
Sterile gauze pads (4×4 inch)	20	6
Medical tape (roll)	2	1

Item	Workshop Kit	Truck Kit
Triangular bandage (sling)	2	1
Rolled gauze (2-inch and 4-inch)	2 each	1 each
Antiseptic wipes	20	10
Antibiotic ointment (single-use)	10	5
Disposable nitrile gloves	10 pairs	4 pairs
CPR mask / face shield	2	1
Trauma shears	2	1
Tweezers / splinter forceps	2	1
Instant cold pack	4	1
Emergency blanket	2	1
Burn cream / burn dressing	1 tube / 2 dressings	1 dressing
Eye wash (sterile single-use)	4 bottles	2 bottles
First aid guide / card	1	1

25.6 Incident Reporting & Investigation

All incidents (injuries, near-misses, property damage, environmental spills) must be reported within 2 hours to the supervisor, and formally documented within 24 hours.

Incident Report Form — Minimum Required Information:

1. Date, time, and exact location of the incident.
2. Names of all persons involved and witnesses.
3. Description of what happened (who, what, when, where, how).
4. Equipment involved (truck number, tools, chemicals used).
5. Injuries sustained (nature and severity).
6. First aid or medical treatment provided.
7. Root cause (what conditions or actions led to this incident?).
8. Corrective actions (what will be done to prevent recurrence?).
9. Signature of person reporting and supervisor.

No-Blame Culture: The purpose of incident reporting is **prevention**, not punishment. Workers who report incidents honestly will not face disciplinary action. Retaliation against anyone who reports a safety concern is a serious violation of company policy.

27. Chapter 26: HAZMAT & Dangerous Goods Compliance

26.1 Applicable Regulations

All petroleum transportation activities are subject to national and international dangerous goods regulations. The operator and driver are jointly responsible for compliance. Non-compliance can result in fines, vehicle impoundment, and denial of depot access.

Key regulations for petroleum tanker operations:

- **Dangerous Goods Regulations (DGR):** National implementation of UN Model Regulations for transport of dangerous goods by road.
- **Depot Safety Rules:** Each petroleum depot has its own safety requirements — these may vary and must be reviewed.
- **Environmental Regulations:** Governing spill prevention, discharge reporting, and waste disposal.

26.2 Classification of Petroleum Products

Product	UN Number	Class	Division	Packing Group
Gasoline (Petrol)	UN 1203	3 (Flammable liquid)	—	II
Diesel / Fuel Oil	UN 1993	3 (Flammable liquid)	—	III
Kerosene	UN 1223	3 (Flammable liquid)	—	III
Used Engine Oil	UN 1993	3 (Flammable liquid)	—	III
Gas Oil / Heating Oil	UN 1202	3 (Flammable liquid)	—	III

26.3 Placarding Requirements

Each truck transporting dangerous goods must display placards on all four sides (front, rear, both sides):

- **Hazard Class Placard:** Class 3 (Flammable Liquid) — red diamond with flame symbol. Minimum size: 250 mm × 250 mm.
- **UN Number Placard:** Orange panel (or integrated into hazard placard) showing the appropriate UN number for the product being carried.
- **Multi-compartment trucks:** If carrying different products in different compartments, placard for the highest hazard product. Some jurisdictions require separate placarding per side if different products are loaded.
- **Empty (not gas-free) tanks:** Must still display the placard for the last product carried. Only gas-free and certified empty tanks may remove placards.

26.4 Transport Documentation

Each trip must carry the following documents in the cab, accessible to the driver and emergency responders:

1. **Dangerous Goods Transport Document (DG Document / Waybill):** Including: shipper name and address, consignee name and address, description of goods (UN number, proper shipping name, class, packing group, quantity in liters), date of transport.
2. **Shipping Manifest or Delivery Receipt:** For each compartment showing product type and quantity loaded.
3. **Truck Registration / OR/CR:** Current registration documents.
4. **Insurance Certificate:** Comprehensive insurance including third-party liability.
5. **Driver's License:** Appropriate professional driver's license with HAZMAT endorsement (where required).
6. **Driver HAZMAT Training Certificate:** Proof of current dangerous goods awareness training.

26.5 Emergency Response Information

The driver must carry, and be familiar with, the **Emergency Response Guide** for the products being carried. At minimum:

- **ERG Guide 128** (Flammable Liquids — Non-Polar / Water-Immiscible) for petroleum products.
- **Emergency actions:** For any spill or leak: stop engine, eliminate ignition sources, evacuate area, call emergency services, and provide the product information (UN number) and truck details.
- **Spill containment:** Use on-board spill kit (absorbent pads, booms, drain covers). Do not wash product into drains or stormwater systems.
- **First aid:** For inhalation (move to fresh air), skin contact (remove clothing, wash with soap and water), eye contact (flush with water 15 minutes), ingestion (do NOT induce vomiting — call poison control).

26.6 Customer / Receiving Site Safety

1. Follow all site-specific safety rules (posted at the entrance).
2. Verify the product and quantity before unloading.
3. Connect grounding cable to the designated grounding point before opening any hatch or valve.
4. Use only approved hoses and fittings for the product being unloaded.
5. Chock wheels before disconnecting any hoses.
6. Stay with the vehicle during the entire unloading process — never leave a truck unattended while it is connected for loading or unloading.
7. Report any product loss, overfill, or leak immediately to the site supervisor and your dispatcher.

Appendices

28. Appendix A — Complete 154-Task PM Checklist

This appendix lists all 154 PM task templates organized by category. Use this checklist for printing and posting in the workshop, or as a quick reference during pre-trip and PM inspections. The checklist includes the interval type and value for each task so you know when each item is due.

Legend: M = Mileage (km), H = Engine Hours, C = Calendar (days), V = Visual (manual check)

Engine PM

Task Name	Interval	Type	Status
Change engine oil	5,000	M	☐
Replace oil filter	5,000	M	☐
Check coolant level and condition	10,000	M	☐
Inspect radiator hoses and clamps	10,000	M	☐
Inspect fan belt — condition and tension	10,000	M	☐
Check coolant freeze point (refractometer)	30	C	☐
Flush and replace coolant	730	C	☐
Inspect radiator fins for blockage	10,000	M	☐
Replace fuel filter (primary)	10,000	M	☐
Replace fuel filter (secondary)	20,000	M	☐
Check air filter restriction indicator	5,000	M	☐
Replace air filter element	30,000	M	☐
Inspect turbocharger — boost and noise	20,000	M	☐
Check engine mounting bolts	30,000	M	☐

Brakes

Task Name	Interval	Type	Status
Inspect brake lining thickness — all wheels	10,000	M	☐
Inspect brake drums — cracks, scoring, out-of-round	30,000	M	☐
Inspect brake discs — thickness, runout, hard spots	30,000	M	☐
Check air brake system — pressure buildup and leak rate	1	C	☐
Test low pressure warning device	1	C	☐
Drain air tanks — moisture and contaminants	1	C	☐
Check slack adjuster stroke — all wheels	5,000	M	☐
Inspect brake chamber and pushrod	10,000	M	☐

Task Name	Interval	Type	Status
Inspect air hoses and tubing	10,000	M	☐
Check brake valve function	20,000	M	☐
Inspect compressor and governor operation	20,000	M	☐
Test spring brake (parking brake) application	10,000	M	☐
Inspect ABS sensors and wiring	30,000	M	☐

Tires

Task Name	Interval	Type	Status
Visual tire inspection — cuts, bulges, damage	1	C	☐
Check tire pressure — all positions	7	C	☐
Measure tread depth — all positions	7	C	☐
Inspect wheels and rims — cracks, bent flanges	10,000	M	☐
Check lug nut torque — all wheels	5,000	M	☐
Inspect spare tire — pressure and condition	30	C	☐
Rotate tires (if applicable)	30,000	M	☐

Tank & Valves

Task Name	Interval	Type	Status
Inspect overfill prevention sensor — each compartment	30	C	☐
Test overfill prevention — manual actuation	90	C	☐
Inspect manlid gasket — each compartment	30	C	☐
Check manlid clamping mechanism	30	C	☐
Inspect API bottom-loading adapter o-rings	7	C	☐
Test API adapter internal valve	90	C	☐
Inspect camlock couplings — ears, pins, gasket	30	C	☐
Inspect internal safety valve (foot valve) operation	90	C	☐
Check emergency shutoff linkage and cable	30	C	☐

Task Name	Interval	Type	Status
Inspect discharge valves — butterfly/ball	90	C	☐
Check tank compartment markings and labels	30	C	☐
Inspect tank barrel for dents, cracks, or leaks	90	C	☐
Hydrostatic tank test (certified)	365	C	☐

Vapor Recovery

Task Name	Interval	Type	Status
Inspect vapor recovery hose — cuts, bulges	30	C	☐
Check vapor adaptor poppet valve operation	30	C	☐
Inspect vapor plumbed lines and connections	90	C	☐
Check P/V vent — free movement, clean screen	90	C	☐
Vapor system pressure test (1 psi hold)	180	C	☐
Inspect vapor recovery adaptor o-ring	30	C	☐

Grounding & Static

Task Name	Interval	Type	Status
Inspect grounding cable — jacket, strands	7	C	☐
Check grounding clamp — clean contact surface, spring	7	C	☐
Continuity test — clamp to chassis (≤ 5 ohms)	30	C	☐
Inspect chassis ground connection — clean, tight	90	C	☐

Hoses & Couplings

Task Name	Interval	Type	Status
Inspect product transfer hoses — full length	7	C	☐
Check hose end couplings — crimp, threads	30	C	☐
Pressure test transfer hoses (annual)	365	C	☐
Inspect hose storage — clean, no sharp edges	30	C	☐

Task Name	Interval	Type	Status
Check product piping — corrosion, leaks, supports	90	C	☐
Inspect flexible connections at tank ends	90	C	☐

Safety Equipment

Task Name	Interval	Type	Status
Check fire extinguisher — gauge in green, seal intact	30	C	☐
Inspect reflective triangles — 3 present, reflectors OK	30	C	☐
Check first aid kit — stocked, items current	30	C	☐
Inspect wheel chocks — 2 present, good condition	30	C	☐
Inspect reflective tape — all sides, ≥50% reflectivity	30	C	☐
Check HAZMAT placards — correct, clean, legible	7	C	☐
Test horn — minimum 90 dB	30	C	☐
Test backup alarm	30	C	☐
Inspect fall arrest system / top rail	90	C	☐
Check compartment labels — legible, secure	30	C	☐

Electrical & Lighting

Task Name	Interval	Type	Status
Inspect battery — case, terminals, hold-down	30	C	☐
Check battery resting voltage	30	C	☐
Alternator output test — voltage at idle and 1500 RPM	30	C	☐
Inspect battery cables — fraying, corrosion	30	C	☐
Check starter circuit voltage drop	90	C	☐
Inspect starter — mounting, solenoid connections	90	C	☐
Inspect wiring harness — chafing, loose connectors	90	C	☐
Check engine-to-chassis ground strap	90	C	☐
Test all exterior lights — head, tail, brake, turn, marker	1	C	☐
Inspect light lenses — cracked, moisture, hazing	30	C	☐

Task Name	Interval	Type	Status
Check instrument panel — all warning lights	1	C	☐
Inspect wiring for lighting — connectors, corrosion	90	C	☐

Drivetrain

Task Name	Interval	Type	Status
Inspect U-joints — rotational play, grease	10,000	M	☐
Check center / carrier bearing	20,000	M	☐
Check differential oil level and condition	10,000	M	☐
Inspect differential for leaks — pinion, axle seals	10,000	M	☐
Replace differential oil	60,000	M	☐
Inspect wheel bearings — end play, noise	30,000	M	☐
Check driveshaft bolts and straps	10,000	M	☐
Grease driveline fittings	5,000	M	☐

Water Brake System

Task Name	Interval	Type	Status
Check water brake fluid level	30	C	☐
Inspect water brake pump — leaks, noise	90	C	☐
Check water brake hoses — chafing, bulges	90	C	☐
Inspect heat exchanger — debris blockages	90	C	☐
Test water brake function — engage/disengage	7	C	☐
Check control linkage — full range, smooth	90	C	☐
Replace water brake fluid	365	C	☐

HVAC

Task Name	Interval	Type	Status
Check air conditioning — vent temperature	90	C	☐
Inspect AC compressor — belt, clutch operation	90	C	☐

Task Name	Interval	Type	Status
Check heater operation	30	C	☐
Inspect cabin air filter — replace if dirty	90	C	☐

Hydraulics

Task Name	Interval	Type	Status
Check hydraulic fluid level (if applicable)	30	C	☐
Inspect hydraulic hoses — leaks, chafing	90	C	☐
Check hydraulic pump operation	90	C	☐
Inspect hydraulic cylinder — rod seal leaks	90	C	☐

Suspension & Steering

Task Name	Interval	Type	Status
Inspect leaf springs — cracks, broken leaves	10,000	M	☐
Check shock absorbers — leaks, mounting	20,000	M	☐
Inspect steering linkage — ball joints, tie rod ends	10,000	M	☐
Inspect tie rod ends — play, dust boots, threads	10,000	M	☐
Inspect ball joints — play, boots, grease fittings	10,000	M	☐
Check steering king pin & bushing — play, lubrication	10,000	M	☐
Lubricate steering king pins & tie rod ends	5,000	M	☐
Check power steering fluid level	5,000	M	☐
Inspect power steering hoses — leaks, chafing	20,000	M	☐
Check steering box — mounting bolts, leaks	30,000	M	☐
Measure steering free play	10,000	M	☐

Transmission

Task Name	Interval	Type	Status
Check transmission oil level	10,000	M	☐
Replace transmission oil and filter	60,000	M	☐

Task Name	Interval	Type	Status
Check clutch pedal free play (manual)	10,000	M	☐
Inspect transmission mounts	30,000	M	☐

Exhaust

Task Name	Interval	Type	Status
Inspect exhaust system — leaks, mounting	10,000	M	☐
Check exhaust hangers and brackets	20,000	M	☐
Inspect muffler — corrosion, damage	30,000	M	☐

Body & Chassis

Task Name	Interval	Type	Status
Inspect chassis frame — cracks, loose bolts	30,000	M	☐
Check body mounting bolts and brackets	30,000	M	☐
Inspect cab mounts — rubber cushions	30,000	M	☐
Check steps and grab handles — secure	10,000	M	☐

Compliance & Regulatory

Task Name	Interval	Type	Status
Verify OR/CR registration — in truck, not expired	30	C	☐
Check insurance certificate — in truck, current	30	C	☐
Verify DOST / PSTC calibration stickers	180	C	☐
Check speed limiter seal intact	90	C	☐
Verify HAZMAT registration and permits	30	C	☐
Check tollway RFID tags — mounted, functional	90	C	☐
Verify depot access documentation	30	C	☐
Check compliance / data plate — present, legible	180	C	☐
Inspect HAZMAT UN marking panels	30	C	☐

Cooling System (Auxiliary)

Task Name	Interval	Type	Status
Check auxiliary cooling fan operation	10,000	M	☐
Inspect coolant recovery tank — level, cracks	5,000	M	☐
Check radiator cap — gasket, pressure rating	10,000	M	☐

Miscellaneous

Task Name	Interval	Type	Status
General visual inspection — walk around all systems	1	C	☐
Check windshield wipers and washer fluid	30	C	☐
Inspect mirrors and mountings	30	C	☐
Check seat belts — all positions	30	C	☐
Lubricate door hinges and locks	90	C	☐
Check all fluid levels — top off as needed	7	C	☐
Inspect for fuel, oil, coolant, or air leaks	7	C	☐

29. Appendix B — Form Templates

B.1 Job Order Form Fields

The system uses the following Job Order form layout. For paper-based backup (when system is unavailable), use this template:

JOB ORDER — JO-YYYY-NNNN	
Truck: [Plate Number / Unit Number]	Date: [YYYY-MM-DD]
Title: [Brief description]	Priority: [Low / Med / High / Emerg]
Type: [PM / Repair / Inspection / Contractor]	Assigned To: [Name]
Description: [Detailed description of the issue]	

Line Items:

#	Task	Est. Hrs	Actual Hrs	Status	Notes
1	[Task description]	[]	[]	☐ Open / ☐ Done / ☐ N/A	[]
2	[]	[]	[]	☐ Open / ☐ Done / ☐ N/A	[]

Parts Used:

Part Name	Qty	Unit Cost	Total
[]	[]	[]	[]

Close-out:

Mileage at close: _____ km Engine Hours: _____ hrs

Sign-off: Mechanic: _____ Lead Mechanic/Supervisor: _____

B.2 Service Log Entry Format

Field	Value
Date	System-generated (timestamp)
Truck	Link to truck record
Job Order	Link to JO record (if applicable)
Action	Created / Done / Closed / Updated
Description	Text description of the action

Field	Value
Mileage / Hours	Recorded at time of action
Parts Cost	Total cost of parts (if applicable)
Performed By	User name

B.3 PM Schedule Status Report

Generated from the PMS Schedules page. CSV export includes:

Column	Content
Truck	Plate number
Category	PM category name
Task	Task template name
Interval	e.g., "5,000 km"
Last Done	Date of last completion
Last Mileage	Mileage at last completion
Last Hours	Engine hours at last completion
Status	OK / Due / Overdue / Visual / No Data

30. Appendix C — Troubleshooting Quick Reference

C.1 Engine Won't Start

Symptom	Likely Cause	Check / Action
Starter doesn't turn — no sound	Dead battery, loose/corroded connection, neutral safety switch	Check battery voltage, clean terminals, try in neutral
Starter clicks but doesn't turn	Low battery, bad solenoid, seized engine	Jump start or charge, check engine oil, try to bar engine over by hand
Starter turns slowly	Weak battery, high resistance connections, bad starter	Voltage drop test, clean grounds, load test battery
Engine cranks but no start	Fuel, air, or compression issue	Check fuel level, fuel filter, air filter, glow plugs (cold start)
White smoke when cranking, no start	Fuel not reaching cylinders, low compression	Prime fuel system, check for air in fuel lines
Starts then dies immediately	Fuel restriction, air in fuel, injection pump issue	Check fuel supply, bleed air from system

C.2 Engine Performance Issues

Symptom	Likely Cause	Check / Action
Lack of power / sluggish	Clogged air filter, bad fuel, turbo issue, fuel restriction	Check air filter restriction gauge, inspect turbo, replace fuel filters
Excessive black smoke	Over-fueling, restricted air intake, bad injectors	Check air filter, check boost pressure, injector balance test
Excessive white smoke (warm engine)	Coolant in combustion chamber (blown head gasket), injector timing	Check coolant level and for bubbles in radiator, compression test
Excessive blue smoke	Oil burning — worn piston rings, valve guides, turbo seals	Check oil consumption, compression test, inspect turbo for oil leaks
Engine overheating	Low coolant, stuck thermostat, clogged radiator, bad fan clutch	Check coolant level, feel radiator (should be hot across), check fan operation
Engine running hot but not overheating	Wrong thermostat, gauge error, air in cooling system	Verify with infrared thermometer, burp cooling system
Knocking noise from engine	Bad injector, rod bearing wear, valve train issue	Isolate cylinder with stethoscope, injector balance test, check oil pressure

C.3 Brake System

Symptom	Likely Cause	Check / Action
Low air pressure / won't build	Compressor issue, governor issue, major air leak	Check compressor drive, governor cut-out setting, listen for leaks
Air pressure drops quickly when parked	Leak in brake system	Soap test all fittings, chambers, valves. Check gladhands.
Brake pedal feels spongy	Air in hydraulic system (if equipped), low fluid, worn linings	Bleed brakes, check fluid, check lining thickness
Truck pulls to one side when braking	Uneven brake adjustment, contaminated lining, seized caliper	Check pushrod stroke both sides, inspect linings, check caliper slides
Brake squeal or squeak	Worn linings, glazed drums, lack of anti-squeal compound	Inspect lining and drum condition, apply anti-squeal to pad backing
Brake grab / jerky braking	Grease/oil on lining, out-of-round drum, bad brake hose (collapsing)	Inspect lining for contamination, measure drum roundness

C.4 Electrical System

Symptom	Likely Cause	Check / Action
Battery goes dead overnight	Parasitic draw, bad alternator diode	Parasitic draw test, alternator diode test
Headlights dim at idle, bright at RPM	Weak alternator, bad voltage regulator, loose belt	Alternator output test, belt tension check
Warning lights flickering	Loose connection, bad alternator	Check chassis grounds, alternator output
Turn signal blinks fast (hyper-flash)	Burned-out bulb on that circuit	Check and replace bulb. If LEDs, may need resistor.

C.5 Air Conditioning

Symptom	Likely Cause	Check / Action
No cold air — compressor not engaging	Low refrigerant, bad relay, blown fuse, clutch gap too large	Check fuse/relay, short-cycle clutch to test, check pressure
Warm air — compressor cycles briefly	Low refrigerant (leak) or bad expansion valve	Check pressures, UV dye test for leaks
Weak airflow	Blower motor resistor, clogged cabin filter	Check blower speeds, replace cabin filter

31. Appendix D — Glossary of Terms

Term	Definition
API Adapter	American Petroleum Institute standard bottom-loading adapter — the connection point for loading arms at a terminal.
Bendix Drive	The mechanism on a starter motor that engages the pinion gear with the engine's flywheel ring gear during cranking.
Camlock Coupling	A quick-connect hose coupling using pivoting cam arms that lock over a mating flange. Standard for top-loading transfer hoses.
CFR / Code of Federal Regulations	U.S. federal regulations (often referenced for HAZMAT and trucking compliance standards).
CR / Certificate of Registration	The official government document proving vehicle registration (LTO in the Philippines).
DOT / Department of Transportation	Government agency regulating commercial vehicle safety standards. DOT-compliant means meeting minimum safety requirements.
Dead-blow Hammer	A hammer filled with shot or sand that delivers force without rebound — prevents marring surfaces and reduces user fatigue.
Dielectric Grease	Non-conductive silicone grease used on electrical connections to prevent corrosion and moisture ingress.
Differential	The gear assembly that allows the left and right wheels to rotate at different speeds while receiving power from the driveshaft (necessary for turning).
DOST / Department of Science and Technology	Philippine government agency that certifies tank compartment calibration and weighing equipment.
Drain Plug Gasket	The crush washer or O-ring that seals the oil drain plug to the oil pan. Should be replaced at every oil change.
ERC 20	Emergency Response Guidebook — the guide for first responders to hazardous material incidents.
GHS / Globally Harmonized System	International standard for chemical hazard classification and labeling (pictograms, signal words, hazard statements).
Gladhand	The quick-connect coupling for air brake hoses between tractor and trailer (or between truck sections).
Governor (Air Brake)	The device that controls when the air compressor loads (builds pressure) and unloads (idles), maintaining system pressure between cut-in and cut-out.
GVW / Gross Vehicle Weight	The total weight of the truck including chassis, body, payload, fuel, and occupants.

Term	Definition
HAZMAT / Hazardous Materials	Dangerous goods that require special handling, placarding, and documentation during transport (e.g., petroleum products).
HEPA / High-Efficiency Particulate Air	A filtration standard that captures at least 99.97% of particles 0.3 microns — used for vacuuming brake dust.
Interval Value	The numeric value in the PM schedule that determines when a task becomes due, combined with the interval unit (km, hrs, days).
JO / Job Order	A digital work order in the system that records: truck, tasks, parts used, labor hours, and completion status.
KPI / Key Performance Indicator	A measurable value that tracks performance — in this system: jobs completed, hours worked, parts cost, etc.
LEL / Lower Explosive Limit	The minimum concentration of a flammable gas or vapor in air that can ignite. Used in confined space and tank entry monitoring.
LOTO / Lockout-Tagout	A safety procedure that ensures equipment is isolated from all energy sources and locked to prevent accidental re-energization during maintenance.
LTO / Land Transportation Office	Philippine government agency responsible for vehicle registration and driver licensing.
Manlid	The hinged cover on top of each tank compartment through which product is loaded (top-loaded trucks) or inspected.
NIOSH / National Institute for Occupational Safety and Health	U.S. agency that certifies respirators and other safety equipment (N95, N100, P100 ratings).
OR / Official Receipt	The proof of payment for vehicle registration renewal (LTO), carried together with the CR.
Overfill Prevention	A safety system that automatically stops product flow when a tank compartment reaches its maximum fill level.
P/V Vent / Pressure-Vacuum Vent	A valve on top of the tank that allows air to enter (when product is being pumped out) and allows vapors to exit (if pressure exceeds set point), preventing tank collapse or overpressure.
PASS Method	Fire extinguisher operation sequence: Pull the pin, Aim at the base, Squeeze the handle, Sweep side to side.
PM / Preventive Maintenance	Scheduled, proactive maintenance performed to prevent equipment failure before it occurs.
PMVI / Private Motor Vehicle Inspection	Annual mandatory vehicle roadworthiness inspection in the Philippines (LTO-accredited).
PPE / Personal Protective Equipment	Safety equipment worn to protect against workplace hazards: safety glasses, gloves, steel-toe boots, etc.

Term	Definition
Pushrod Stroke	The distance the brake chamber pushrod travels when brakes are applied. Measured to determine if the slack adjuster needs adjustment.
QC / Quality Control	The inspection process performed before returning a truck to service after any repair or PM.
SAE / Society of Automotive Engineers	Standards organization that defines oil viscosity grades (e.g., SAE 15W-40) and fastener specifications.
SDS / Safety Data Sheet	A document that lists the properties, hazards, handling, and emergency procedures for a chemical product. Must be available for every chemical in the workshop.
Slack Adjuster	A lever arm on the brake camshaft that converts the pushrod's linear motion into rotary motion to apply the brake shoes. Can be manual or automatic.
Spring Brake	A parking/emergency brake that uses a powerful spring to apply the brakes mechanically when air pressure is released (fall-safe design).
Torque Seal	A colored paint or lacquer applied to a tightened fastener to visually indicate that it has been torqued and has not moved.
U-Joint / Universal Joint	A flexible coupling that allows the driveshaft to transmit power at varying angles (necessary for suspension movement).
UN Number	A four-digit number that identifies a hazardous substance (e.g., UN 1203 = Gasoline, UN 1993 = Diesel/Fuel Oil). Used on placards and transport documents.
Vapor Recovery	A system that captures fuel vapors displaced during loading and returns them to the terminal, preventing vapor release to the atmosphere.
Water Brake (Retarder)	A supplemental braking system that uses hydraulic resistance to slow the vehicle, preserving the service brakes for emergency stops. Common on heavy trucks in mountainous terrain.
WeasyPrint	A Python library that converts HTML+CSS into PDF documents. Used to generate this SOP manual as a downloadable PDF.

End of Appendix D